How Big Can Mechanical Rebalancing Get?*

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Abstract

A large share of institutional capital rebalances mechanically: today's prices set tomorrow's trades through volatility targets, margins, value-at-risk limits, and collateral haircuts. We ask how large such a sector can grow before its own rule-induced trading makes the market non viable, meaning a profitable round trip exists. The test needs only the mechanical rebalancing rule and a market-impact estimate, so a supervisor can run it from public information. For volatility-managed portfolios the sector's safe size is below 0.17 days of average daily volume. At a size of one day of volume, a profitable round trip exists 83% of the time.

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1 Introduction

Today's transaction prices determine tomorrow's mandated trades, target exposures, or binding requirements. Volatility targets, risk-control overlays, leverage limits, collateral haircuts, margin schedules, and target-weight momentum, and reversal mandates all share this structure, and now govern a large and growing share of institutional capital. When markets are not perfectly elastic (Gabaix and Koijen, 2021) and the rule-following sector is large enough to move prices, a feedback loop forms: a trade moves prices, the price change shifts the statistic the rule reads, and the rule turns it into predictable order flow that moves prices again. This feedback loop can run on its own. For instance, in February 2018, the disclosed rebalancing rules of exchange-traded volatility products turned a one-day spike in the VIX into mandated end-of-day futures purchases that amplified the spike and erased most of the products' value (Sushko and Turner, 2018; Augustin et al., 2021); in the autumn of 2022, price-based collateral calls on leveraged gilt positions forced sales that raised yields and triggered further calls, until the Bank of England intervened (Pinter, 2023; Chen and Kemp, 2023). Neither episode needed a manipulator, but the same feedback can be exploited deliberately: a strategic trader can generate forced flows and unwind against the induced rebalancing. The question is therefore how large can the rule-following sector grow before its own price-triggered rebalancing makes the market non viable, in the sense that the induced feedback admits a profitable round trip?

We provide a novel test that answers this question. The test rests on a no-arbitrage restriction that the classical benchmark of Huberman and Stanzl (2004) omits. A market can satisfy the Huberman and Stanzl (2004) condition that every admissible round trip has nonnegative expected execution cost, yet still admit profitable round trips, because current trades now move future constraints and hence future forced flow. Theorem 1 in Section 4 supplies the missing restriction and turns it into a capacity test for any disclosed price-based rule. Given a market-impact estimate and the rule mapping sampled prices into next-period requirements or target exposures, the test returns the largest size of the mechanically rebalancing sector consistent with non-manipulability. It is an eigenvalue test on an impact matrix augmented to internalize the rule's own flow. Because it needs only the disclosed rule and a price impact estimate, a supervisor can run it using public information alone. We then apply the test to volatility-managed indices used in structured products and indexed annuities in Section 7, a leading case of mechanical rebalancing. From the published volatility-control rule and an impact estimate, we compute a size bound for the linked sector and vulnerability curves that show, for each sector size, how often the no-manipulation condition fails. We use the two rule families that span published risk-control index methodologies. Template A scales a single parent index by a capped inverse-volatility weight; Template B scales a fixed multi-asset basket by one volatility-control factor. Within each we vary liquidity, target volatility, implementation lags, volatility dynamics, cross-impact, and return correlations. At a manipulation horizon of $T = 126$ days and a baseline impact of 12 bps per 1% of average daily volume (ADV), admissible scale is small: safe capacity is below 0.17 days of ADV[1], about 0.14% of the underlying market; when the sector size is equal to one day of volume, the no-manipulation condition fails 83.4% of the time under Template A,

[1]These capacities are stated in full-day volume; because the rule rebalances in the closing window, which carries only a fraction of daily volume (Bogousslavsky and Muravyev, 2023), the safe scale is even smaller, roughly in proportion to that fraction.

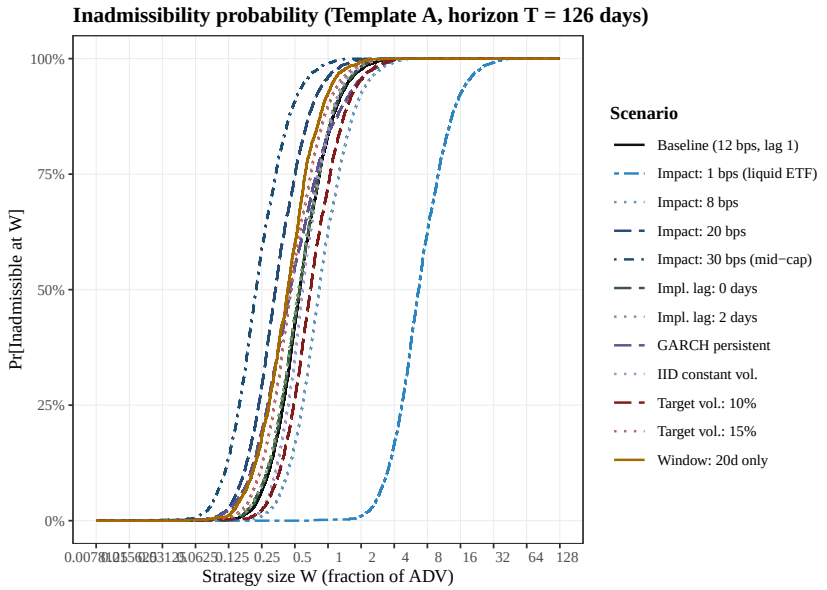


Figure 1 Vulnerability curves for a volatility-controlled rule over horizon $T = 126$ days.
Each stress state z is a joint draw of the rule inputs (recent returns that determine the realized-volatility estimate and regime) and a local liquidity environment (impact operator $\mathcal{I}$). For each z we compute the statewise capacity $W_{\max}(z) = \sup\{W \geq 0 : \lambda_{\min}^{\mathcal{R}_T}(\widehat{H}(z; W)) \geq 0\}$, the largest sector scale W that passes the restricted-eigenvalue screen over horizon T. The figure plots the vulnerability curve $\Pr_z[W > W_{\max}(z)]$ against W, where W is the sector's linked notional expressed in ADV units. This is the $T = 126$ panel of Figure 2.

and 95.7%, 97.5%, and 85.0% under Template B for $N = 2, 4, 8$ assets.[2]

Figure 1 plots the vulnerability curves, that is, the fraction of states in which the test fails as a function of the linked notional W. We span liquidity from 1 bp per 1% of volume traded, a deep underlying ETF, to 30 bps, representing an illiquid name.[3] Illiquidity is a primary driver of vulnerability, but the feedback can be present even in liquid markets when the mechanical sector is not too small. Under the baseline calibration, the test fails in 50% of states by $W \approx 0.56$ days of ADV. At the deepest liquidity, 1 bp, it still fails in roughly 40% of states by $W \approx 6$ days of ADV, which the median ETF ADV-to-AUM ratio of Khomyn et al. (2024) places at about 5% of the underlying ETF. Other scenarios, including alternative implementation lags, volatility dynamics, target volatilities, and windows, cluster close to the baseline.

In Section 5 we characterize the round trip that maximizes the manipulator's profits. The trader moves the sampled risk statistic that governs future constraints, then profits by unwinding against the predictable rebalancing the tighter constraint induces. The payoffs are driven by how sensitive the sampled statistic is to current prices, and how strongly mandated demand responds to that statis-

[2]To anchor the ADV scale, take a liquid ETF with \$5 billion ADV and \$600 billion in net assets, consistent with the median ETF secondary-market dollar ADV of about 0.9% of AUM per day reported by Khomyn et al. (2024). Throughout, W denotes the sector's total linked notional expressed in days of ADV, so $W = 1$ corresponds to a linked notional of \$5 billion, about 0.83% of the underlying ETF; with a typical volatility-control weight of 0.80, the associated risky exposure is about \$4 billion (0.67% of the ETF).

[3]Risk-control and target-volatility rules are licensed not only on broad equity indices but also on emerging-market and commodity indices, and, through dealer quantitative-investment-strategy desks, on single-name and worst-of baskets, for which impact per unit of volume is larger.

tic. Under convex risk measures, including volatility-control rules, the trigger is often inventory-light, because the attacker can raise measured risk with offsetting trades inside the sampling window rather than by carrying a large directional position. Because the risk measure is computed over a rolling window, the induced tightening and its reversal are mechanically timed, and therefore predictable. This differs from corners, squeezes, and price-level manipulation: the profit comes from moving a disclosed statistic and harvesting the rebalancing it induces, not from impairing the rebalancing sector's net worth. Solving the optimal-attack dynamic program at vulnerable states, the median annualized return on average capital is 11% at one day of volume and 61% at five days, with annualized Sharpe ratios from 2.4 at one day of volume to 1.4 at five days, and mean peak inventory of only 0.05 to 0.06 days of volume. We find that the attacker captures a shrinking share of the damage: the transfer she extracts falls from 60% of the rebalancing sector's excess cost at one day of volume to 1% at five, with the remainder borne as a higher aggregate execution cost, a deadweight loss to the extent that impact reflects the real cost of liquidity provision. This is because the extra trading costs induced by manipulation grow significantly with the size of the rebalancing sector. We also consider the multi-asset case in Section 6, where optimal exploitation reduces to the two-dimensional span of two portfolios: the most efficient direction for moving risk, and the direction of liquidation.

Another implication of rules based on convex risk measures is that fragility can be larger in calm markets, because the risk input is then easiest to move. For target-volatility rules, manipulability is highest when measured volatility is low. Scaling by inverse volatility makes exposure most sensitive to the volatility estimate, and a given rise in realized variance has the largest percentage effect on measured volatility from a low base. Calm markets are therefore precisely where a small spike-and-revert strategy can move measured volatility enough to induce material, predictable rebalancing. This is a rule-based version of the paradox of financial instability (Borio and Drehmann, 2009; Borio, 2011). It is related to, but distinct from, the leverage cycle (Brunnermeier and Pedersen, 2009; Geanakoplos, 2010; Adrian and Shin, 2010), where amplification operates through balance-sheet capacity as collateral values fall or funding conditions worsen. Here amplification arises because the requirement is a deterministic function of a sampled risk statistic computed from recent prices, and trading can move that statistic with no change in fundamentals, collateral values, or intermediary capital. It also differs from the self-inflation of illiquid funds (van der Beck et al., 2025), where return-chasing flows inflate a fund and leave overvaluation that later reverses: our forced flow is instead set by a disclosed rule, which makes it predictable and possible to generate deliberately, so it admits a profitable round trip, not mispricing alone. Our mechanism includes van der Beck et al. (2025)'s setting as a special case, but applies far more broadly to any rebalancing strategy that depends on past returns.

1.1 Contributions to the literature

Our first contribution is to extend the no-dynamic-arbitrage benchmark of Huberman and Stanzl (2004), Gatheral (2010), and Schneider and Lillo (2019) to mechanical rebalancing.[4] In that bench-

[4]In the benchmark, nonlinear permanent impact admits quasi-arbitrage (Huberman and Stanzl, 2004), and concave transient impact is arbitrage-free only for restricted combinations of curvature and kernel decay (Gatheral, 2010). Linear generators sit inside this benchmark. In the Internet Appendix A.1 we show that a linear kernel's permanent component

mark, trades move execution prices but not tomorrow's constraints or order flow. Mechanical rebalancing breaks that exogeneity: transaction prices enter next-period requirements and generate predictable flow, so classical impact restrictions no longer suffice for viability. Frey and Stremme (1997) show that mechanical hedging demand feeds back into prices and raises volatility; we ask instead whether such feedback overturns the no-arbitrage benchmark, and at what sector scale. Theorem 1 supplies the missing condition by augmenting the impact-cost matrix to internalize rule-induced flow. Dynamic arbitrage is present exactly when that matrix fails to be positive semidefinite on round trips, which yields a tractable eigenvalue test. Because the augmented matrix is built from the impact operator and the disclosed rule alone, the condition characterizes viability of the market with a rebalancing sector of a given scale, independent of any attacker: it bounds how large the mechanical sector can be before the price system ceases to be arbitrage-free.

Our second contribution is to turn this restriction into an implementable diagnostic. Given a disclosed price-based rule and a market-impact calibration, we compute a conservative bound on the scale of the rebalancing sector consistent with market viability, together with vulnerability curves that map sector size into the probability that the market fails the test. In that sense the paper is close to work that turns observable inputs into supervision-relevant objects such as systemic-risk measures, capital shortfalls, and vulnerability diagnostics (Acharya et al., 2012; Adrian and Brunnermeier, 2016; Acharya et al., 2017; Brownlees and Engle, 2017; Greenwood et al., 2015).

Our third contribution is to characterize optimal strategic trading when forced flow is both predictable and manipulable. Order-anticipation models take the investor's need to trade as a primitive: an exogenous liquidity shock the attacker foresees but cannot create (Carlin et al., 2007; Sannikov and Skrzypacz, 2016; Fardeau, 2021). Where the attacker does induce the imbalance, she induces it by impairing the victim's solvency or capital position (Brunnermeier and Pedersen, 2005; Attari et al., 2005). Here it is generated by a disclosed rule that maps sampled prices into future requirements, so the attacker creates it by moving a public statistic, leaving fundamentals, collateral values, and the rebalancing sector's net worth unchanged. A related empirical literature measures the price impact of predictable, anticipated demand: Treasury auction supply (Lou et al., 2013), index rebalancing (Li, 2021; Pegoraro et al., 2026), rebalancing to target portfolio weights (Harvey et al., 2025), and predictable trades more generally (Bessembinder et al., 2016). Here the flow is not only predictable[5] but can be generated: the attacker's own trades move the statistic that mandates it, so the optimal strategy creates and then harvests the imbalance rather than front-running one that would have arrived anyway. Mechanically, the paper is closest to contract-based and benchmark manipulation, where trading moves a reference statistic that governs payoffs or behavior (Kumar and Seppi, 1992; Hillion and Suominen, 2004; Dutt and Harris, 2005; Onur and Reiffen, 2018; Zhang, 2022).

Finally, this paper is closely related to Nicolai (2026), which shows that no price-based, risk-sensitive rule can simultaneously guarantee liquidity continuity and round-trip manipulation-proofness, and studies the mechanism design of such rules. We take the institutional environment and the me-

is a pure cost and never causes manipulation, which can arise only from the transient shape (Proposition 5); we also show that a linear generator with decay reproduces the concave, approximately square-root impact of metaorders documented empirically (Tóth et al., 2011), which linear propagator models generate (Proposition 6). Section 2 and Internet Appendix B elaborate further on square-root models.

[5]The aggregate noise of Chinco and Fos (2021) arises from many funds rebalancing on idiosyncratic triggers; here the sector responds to a common statistic that the attacker moves, so its induced flow is correlated.

chanical trading rule as given, and ask a different question: for a given rule and impact calibration, when does the induced feedback admit dynamic arbitrage, and how large can the rebalancing sector grow before that happens? This yields a computable admissibility test, conservative capacity bounds, and vulnerability curves.

Roadmap. Section 2 sets up the baseline impact environment and reviews the classical no-manipulation condition, in both the single- and multi-asset cases. Section 3 models mechanical rebalancing as a deterministic price-based constraint, and Section 4 derives the corresponding augmented no-arbitrage condition (Theorem 1). Sections 5 and 6 characterize the round trip that exploits the vulnerability, in the single- and multi-asset cases, and establish the two-portfolio reduction in the multi-asset setting (Theorem 2). Section 7 implements the stress test for volatility-controlled indices, and Section 8 concludes.

2 Model setup

2.1 Single asset

Fix a finite horizon T with trading dates $t = 0, 1, \dots, T-1$. Random variables are defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq -1}, \mathbb{P})$, where $\mathcal{F}_t$ is public information just after date-t outcomes realize and $\mathcal{F}_{-1}$ is the information before the first trade, so a time-t trade u_t is chosen using $\mathcal{F}_{t-1}$.

Definition 1 (Inventory dynamics and admissible strategies)**.** *A single-asset strategy is a predictable process $u = (u_t)_{t=0}^{T-1}$ with $u_t \in \mathbb{R}$, meaning u_t is $\mathcal{F}_{t-1}$-measurable for each $t = 0, \dots, T-1$. We interpret $u_t > 0$ as a purchase and $u_t < 0$ as a sale. Inventory evolves according to*

$$x_{t+1} = x_t + u_t, \qquad x_0 = 0. \tag{1}$$

A strategy is admissible *if it is predictable and uniformly bounded: there exists $\bar{u} > 0$ such that $|u_t| \leq \bar{u}$ for all t and all $\omega \in \Omega$.*

Definition 2 (Round trips)**.** *An admissible strategy u is a round trip if it starts and ends flat,*

$$x_T = 0 \quad a.s. \tag{2}$$

Equivalently, $\sum_{t=0}^{T-1} u_t = 0$ a.s.

2.1.1 Prices

We distinguish the unaffected price S_t, which evolves absent the strategy's trades, from the execution price P_t, at which the strategy transacts and which incorporates market impact. Let $(S_t)_{t=0}^{T-1}$ be an exogenous process adapted to $(\mathcal{F}_t)_{t\geq -1}$.

Assumption 1 (Unaffected price)**.** *$(S_t)_{t=0}^{T-1}$ is an $\mathcal{F}_t$-martingale and $\mathbb{E}[|S_t|] < \infty$ for all t.*

Assumption 1 rules out predictable drift in S_t. Any expected profits must therefore come from market impact and from the deterministic institutional feedback introduced below, not from forecasting S_t.

Execution prices typically move against the trader: buys execute above S_t and sells below S_t. We capture this wedge with a standard linear impact specification.

Assumption 2 (Linear price impact)**.** *Fix coefficients $G_0, G_1, \ldots, G_{T-1} \in \mathbb{R}$ and an impact slope $\eta > 0$. Given a strategy u, the execution price at time t is*

$$P_t(u) = S_t + \eta \sum_{s=0}^{t} G_{t-s} u_s. \tag{3}$$

We treat Assumption 2 as reduced form. The wedge $P_t(u) - S_t$ may reflect asymmetric information, the inventory and risk-bearing costs of liquidity suppliers, latent order-book depth and resilience, and the limited balance-sheet or funding capacity of dealers and arbitrageurs; we do not model these primitives separately or impose a specific market equilibrium. The term $\eta G_{t-s} u_s$ is the contribution of the time-s trade to the time-t execution price, so past flows move current prices, and the sequence (G_ℓ) governs persistence: smaller G_ℓ at longer lags means faster decay. Purely temporary impact, $G_0 = 1$ and $G_\ell = 0$ for $\ell \geq 1$, gives $P_t(u) = S_t + \eta u_t$.[6]

2.1.2 Execution cost and dynamic arbitrage

Trading at execution prices generates the realized cash outflow

$$\sum_{t=0}^{T-1} u_t P_t(u).$$

For a round trip this is the entire payoff, since the strategy starts and ends flat. Define realized trading profit by

$$\Pi(u) = -\sum_{t=0}^{T-1} u_t P_t(u).$$

Under the martingale benchmark for the unaffected price, a round trip cannot generate expected profits from S_t alone.

Lemma 1. *Under Assumption 1, for any admissible round trip u,*

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t S_t\right] = 0. \tag{4}$$

[6] Linearity in flow is consistent with the concave, approximately square-root impact of metaorders documented empirically (Tóth et al., 2011). The square-root law describes the cumulative impact of a parent order executed over many periods, not the one-period response to flow. Linear propagators of the form in Assumption 2 can lead to concavity once kernel decay and autocorrelated order flow are accounted for; Proposition 6 in the Internet Appendix A.1 shows this for a power-law linear kernel.

Proof. Let x_t denote inventory. Since $u_t = x_{t+1} - x_t$,

$$\begin{aligned}\sum_{t=0}^{T-1} u_t S_t &= \sum_{t=0}^{T-1} (x_{t+1} - x_t) S_t \\ &= x_T S_{T-1} - x_0 S_0 + \sum_{t=1}^{T-1} x_t (S_{t-1} - S_t). \end{aligned} \tag{5}$$

For a round trip, $x_0 = 0$ and $x_T = 0$ a.s., so the boundary terms vanish. Predictability implies x_t is $\mathcal{F}_{t-1}$-measurable. Taking expectations and using $\mathbb{E}[S_t \mid \mathcal{F}_{t-1}] = S_{t-1}$ yields, for each $t \geq 1$,

$$\mathbb{E}[x_t (S_{t-1} - S_t)] = \mathbb{E}[x_t \, \mathbb{E}[S_{t-1} - S_t \mid \mathcal{F}_{t-1}]] = 0.$$

Summing over t gives the claim. □

We start from the classical finite-horizon notion of price manipulation in Huberman and Stanzl (2004) (and its transient-impact refinements in Gatheral (2010)). Throughout we impose this no-manipulation benchmark, so any profitable round trip below cannot come from the classic mechanism of moving prices through impact and then unwinding. The gains we study instead run through the institutional feedback that maps transacted prices into binding requirements and induced forced flow.[7]

Definition 3 (Finite-time dynamic arbitrage)**.** *Under Assumptions 1–2, an admissible round trip u is a price manipulation strategy if its expected execution cost is negative:*

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t P_t(u)\right] < 0. \tag{6}$$

Equivalently, $\mathbb{E}[\Pi(u)] > 0$ for $\Pi(u) = -\sum_{t=0}^{T-1} u_t P_t(u)$. The impact model is manipulation free (no finite-time dynamic arbitrage) if no admissible round trip is a price manipulation strategy.

2.1.3 A discrete-time no-manipulation condition

Stack trades into $u = (u_0, \ldots, u_{T-1})^\top \in \mathbb{R}^T$. Define the lower-triangular impact matrix $K \in \mathbb{R}^{T \times T}$ with elements

$$K_{t,s} = G_{t-s} \, \mathbf{1}_{\{t \geq s\}}, \qquad t, s \in \{0, 1, \ldots, T-1\}, \tag{7}$$

so that, for each t,

$$(Ku)_t = \sum_{s=0}^{t} G_{t-s} u_s.$$

Under Assumption 2, the execution price is $P_t(u) = S_t + \eta (Ku)_t$, and total execution cost is

$$\sum_{t=0}^{T-1} u_t P_t(u) = \sum_{t=0}^{T-1} u_t S_t + \eta \, u^\top K u.$$

[7] Admissibility (Definition 1) excludes doubling-type schemes with unbounded positions or arbitrarily large interim losses.

For a round trip, Lemma 1 implies $\mathbb{E}[\sum_t u_t S_t] = 0$, so expected cost is determined by the impact term. It is convenient to define the following symmetric matrix

$$H = \eta\,(K + K^\top). \tag{8}$$

Since $u^\top K u$ is a scalar, $u^\top K u = u^\top K^\top u$, hence

$$\eta\, u^\top K u = \frac{1}{2}\, u^\top H u.$$

Our notion of dynamic arbitrage restricts attention to the subspace of round trips:

$$\mathcal{R}_T = \left\{ u \in \mathbb{R}^T : \sum_{t=0}^{T-1} u_t = 0 \right\}. \tag{9}$$

The standard viability condition requires H to be positive semidefinite on $\mathcal{R}_T$.

Proposition 1. *Suppose Assumptions 1–2 hold. Then, for any admissible round trip u,*

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t P_t(u)\right] = \frac{1}{2}\,\mathbb{E}\Big[u^\top H u\Big]. \tag{10}$$

Consequently, the following are equivalent:

(i) The impact model is manipulation free in the sense of Definition 3.

(ii) The symmetric matrix H is positive semidefinite on $\mathcal{R}_T$, that is, $u^\top H u \geq 0$ for all $u \in \mathcal{R}_T$.

Proof. Fix an admissible round trip u. By Lemma 1,

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t P_t(u)\right] = \mathbb{E}\left[\sum_{t=0}^{T-1} u_t\big(P_t(u) - S_t\big)\right].$$

Assumption 2 gives $P_t(u) - S_t = \eta(Ku)_t$, so

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t P_t(u)\right] = \eta\,\mathbb{E}[u^\top K u] = \frac{1}{2}\,\mathbb{E}\Big[u^\top \eta(K + K^\top) u\Big] = \frac{1}{2}\,\mathbb{E}[u^\top H u],$$

which proves (10).

(ii)⇒(i): If $u^\top H u \geq 0$ for all $u \in \mathcal{R}_T$, then for any admissible round trip the realized trade vector satisfies $u(\omega) \in \mathcal{R}_T$ a.s., hence $u(\omega)^\top H u(\omega) \geq 0$ a.s. Taking expectations and using (10) yields $\mathbb{E}[\sum_t u_t P_t(u)] \geq 0$, so no admissible round trip is a manipulation strategy.

(i)⇒(ii): If H is not positive semidefinite on $\mathcal{R}_T$, then there exists $\tilde{u} \in \mathcal{R}_T$ with $\tilde{u}^\top H \tilde{u} < 0$. Consider the deterministic strategy $u = \tilde{u}$. It is a round trip by construction, and it is admissible for any bound $\bar{u} \geq \max_t |\tilde{u}_t|$. Then (10) gives

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t P_t(u)\right] = \frac{1}{2}\,\tilde{u}^\top H \tilde{u} < 0,$$

so u is a manipulation strategy, contradicting (i). □

Proposition 1 is the standard viability condition for the linear-impact class: on the round-trip set $\mathcal{R}_T$, impact behaves like a true trading cost if and only if H is positive semidefinite. It is the transient-impact no-manipulation condition of Gatheral et al. (2012), of which our H is the discrete counterpart, and refines the classical benchmark of Huberman and Stanzl (2004) for permanent impact. We impose this baseline below. Linearity is part of the benchmark rather than an added restriction: nonlinear permanent impact admits quasi-arbitrage (Huberman and Stanzl, 2004), and concave transient impact is consistent with the absence of dynamic arbitrage only for restricted combinations of decay (Gatheral, 2010). A concave generator would generically admit profitable round trips even without a constrained sector, so the linear class is the environment least favorable to the manipulation we study.

2.1.4 Example: temporary linear impact

Under purely temporary impact ($G_0 = 1$, $G_\ell = 0$ for $\ell \geq 1$), only the contemporaneous trade affects the execution price and $K = I_T$, so

$$H = \eta(K + K^\top) = 2\eta\, I_T.$$

Then $u^\top H u = 2\eta \|u\|^2 \geq 0$ for all u, so by Proposition 1 the model is manipulation free.

2.2 More than one asset

Fix N assets. At each trading date $t = 0, 1, \ldots, T-1$, the trader chooses a trade vector $u_t \in \mathbb{R}^N$. Inventory $x_t \in \mathbb{R}^N$ evolves as

$$x_{t+1} = x_t + u_t, \qquad x_0 = 0.$$

A vector round trip starts and ends flat,

$$x_T = 0 \text{ a.s.} \qquad \text{equivalently} \qquad \sum_{t=0}^{T-1} u_t = 0$$

with equalities understood componentwise. As before, $S_t \in \mathbb{R}^N$ denotes the unaffected price vector and $P_t(u) \in \mathbb{R}^N$ the execution price vector induced by the strategy.

Assumption 3 (Multi-asset unaffected prices). *$(S_t)_{t=0}^{T-1}$ is an $(\mathcal{F}_t)$-martingale in $\mathbb{R}^N$ and $\mathbb{E}[\|S_t\|] < \infty$ for all t.*

Lemma 2 (Multi-asset martingale benchmark). *Under Assumption 3, let $u = (u_t)_{t=0}^{T-1}$ be a predictable $\mathbb{R}^N$-valued process and uniformly bounded: there exists $\bar{u} > 0$ such that $\|u_t\| \leq \bar{u}$ for all t and all ω. If $\sum_{t=0}^{T-1} u_t = 0$, then*

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t^\top S_t\right] = 0. \tag{11}$$

Proof. Let $x_t = \sum_{s=0}^{t-1} u_s$ be inventory, so $u_t = x_{t+1} - x_t$. We have

$$\sum_{t=0}^{T-1} u_t^\top S_t = x_T^\top S_{T-1} - x_0^\top S_0 + \sum_{t=1}^{T-1} x_t^\top (S_{t-1} - S_t).$$

On a round trip $x_0 = x_T = 0$ a.s. Predictability implies x_t is $\mathcal{F}_{t-1}$-measurable, and the martingale property yields $\mathbb{E}[S_t \mid \mathcal{F}_{t-1}] = S_{t-1}$, hence for each $t \geq 1$,

$$\mathbb{E}\Big[x_t^\top (S_{t-1} - S_t)\Big] = \mathbb{E}\Big[x_t^\top \, \mathbb{E}[S_{t-1} - S_t \mid \mathcal{F}_{t-1}]\Big] = 0.$$

Summing over t gives the claim. □

Assumption 4 (Linear cross-impact). *Fix matrices $G_0, G_1, \ldots, G_{T-1} \in \mathbb{R}^{N\times N}$. Given a strategy u, the execution price vector at time t is*

$$P_t(u) = S_t + \sum_{s=0}^{t} G_{t-s}\, u_s. \tag{12}$$

The matrix G_{t-s} maps the time-s trade vector into its contribution to the time-t execution price; diagonal entries are own-asset impact, off-diagonal entries cross-impact. Purely temporary cross-impact, $G_0 = A \in \mathbb{R}^{N\times N}$ and $G_\ell = 0$ for $\ell \geq 1$, gives $P_t(u) = S_t + Au_t$.

2.2.1 Execution cost and manipulation

The realized execution cost of a multi-asset strategy is $\sum_{t=0}^{T-1} u_t^\top P_t(u)$. On a round trip this equals minus realized profit. Stack trades over time into $U \in \mathbb{R}^{NT}$, $U = (u_0^\top, \ldots, u_{T-1}^\top)^\top$. Define the block lower-triangular matrix $\mathbf{K} \in \mathbb{R}^{NT\times NT}$ by its $N \times N$ blocks

$$\mathbf{K}_{t,s} = G_{t-s}\, \mathbf{1}_{\{t\geq s\}}, \qquad t, s \in \{0, 1, \ldots, T-1\}. \tag{13}$$

Then the t-th block of $\mathbf{K}U$ is

$$(\mathbf{K}U)_t = \sum_{s=0}^{t} G_{t-s} u_s,$$

so Assumption 4 implies $P_t(u) = S_t + (\mathbf{K}U)_t$. As before, define the symmetric matrix

$$\mathbf{H} = \mathbf{K} + \mathbf{K}^\top.$$

Collect vector round trips in

$$\mathbf{R}_T = \left\{ U \in \mathbb{R}^{NT} : \sum_{t=0}^{T-1} u_t = 0 \right\}, \tag{14}$$

where the constraint is a vector equality in $\mathbb{R}^N$.

Proposition 2. *Suppose Assumptions 3 and 4 hold. Then, for any admissible round trip u,*

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t^\top P_t(u)\right] = \frac{1}{2}\,\mathbb{E}\Big[U^\top \mathbf{H} U\Big]. \tag{15}$$

Consequently, the model is manipulation free if and only if $\mathbf{H}$ *is positive semidefinite on* $\mathbf{R}_T$.

Proof. Fix an admissible round trip u. Lemma 2 gives $\mathbb{E}[\sum_{t=0}^{T-1} u_t^\top S_t] = 0$, hence

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t^\top P_t(u)\right] = \mathbb{E}\left[\sum_{t=0}^{T-1} u_t^\top \big(P_t(u) - S_t\big)\right].$$

By Assumption 4, $P_t(u) - S_t = \sum_{s=0}^{t} G_{t-s} u_s$, so

$$\sum_{t=0}^{T-1} u_t^\top \big(P_t(u) - S_t\big) = \sum_{t=0}^{T-1} \sum_{s=0}^{t} u_t^\top G_{t-s} u_s = U^\top \mathbf{K} U.$$

Since $U^\top \mathbf{K} U$ is a scalar, $U^\top \mathbf{K} U = U^\top \mathbf{K}^\top U$, hence

$$U^\top \mathbf{K} U = \frac{1}{2} U^\top (\mathbf{K} + \mathbf{K}^\top) U = \frac{1}{2} U^\top \mathbf{H} U.$$

Taking expectations yields (15). The model admits a manipulation strategy if and only if there exists $U \in \mathbf{R}_T$ with $U^\top \mathbf{H} U < 0$, which is equivalent to failure of positive semidefiniteness of $\mathbf{H}$ on $\mathbf{R}_T$. □

With multiple assets, round-trip profits can arise from cross-impact even when each asset in isolation is manipulation free. If $\mathbf{H}$ fails to be positive semidefinite on $\mathbf{R}_T$, then some vector round trip has negative expected execution cost: trading in one asset moves execution prices in another, altering the cost of subsequent legs and allowing the strategy to close out flat at a gain. Proposition 2 gives the finite-horizon condition that rules out such cross-impact loops within the linear class. It is the multi-asset counterpart of the single-asset condition in (10), and the discrete analogue of the cross-impact no-arbitrage restrictions of Alfonsi et al. (2016) and Schneider and Lillo (2019).

2.2.2 Example: two-asset arbitrage

To illustrate, take a pair of tightly linked instruments with strong cross-impact: a liquid index future (asset 1) and a less liquid cash instrument tracking the same index (asset 2), such as an ETF or basket proxy. Buying the future aggressively leads liquidity providers to hedge by buying the cash instrument, transmitting order flow from the future into the cash execution price; the reverse channel can be weaker when the cash instrument is more fragmented or less informative. Such lead-lag patterns are well documented in futures versus spot settings (Kawaller et al., 1987; Chan, 1992). Model this with purely temporary cross-impact, $P_t(u) = S_t + A u_t$, and

$$A = \begin{pmatrix} \lambda_F & \epsilon \\ \Lambda & \lambda_C \end{pmatrix}, \qquad \lambda_F > 0,\ \lambda_C > 0,\ \Lambda > \epsilon \geq 0.$$

Trading asset 1 shifts asset 2's execution price with slope Λ, and asset 2 shifts asset 1's with a small slope ϵ, while λ_F, λ_C are own-impact. Expected execution cost depends on

$$H = \frac{A + A^\top}{2} = \begin{pmatrix} \lambda_F & (\epsilon + \Lambda)/2 \\ (\epsilon + \Lambda)/2 & \lambda_C \end{pmatrix}.$$

A profitable round trip exists if and only if H is not positive semidefinite, equivalently if there exists $q \neq 0$ with $q^\top H q < 0$. In two dimensions this is equivalent to

$$\det(H) = \lambda_F \lambda_C - \left(\frac{\epsilon + \Lambda}{2}\right)^2 < 0.$$

Cross-impact can dominate own-impact, creating a direction in which quadratic impact cost is negative. Only the symmetric part H enters round-trip cost, so what drives the arbitrage is the size of cross-impact relative to own-impact, not its asymmetry: a symmetric matrix with the same average off-diagonal $(\epsilon + \Lambda)/2$ would admit the same profitable round trip. The simplest direction of attack is the spread $v = (1, -1)^\top$ (buy asset 1, sell asset 2). Along v,

$$v^\top A v = v^\top H v = \lambda_F + \lambda_C - (\epsilon + \Lambda).$$

If $\epsilon + \Lambda > \lambda_F + \lambda_C$, then trading the spread has negative quadratic cost. With $T = 2$, consider the round trip $u_0 = v$ and $u_1 = -v$, which opens the spread at $t = 0$ and fully reverses it at $t = 1$. The strategy starts flat and ends flat, and

$$\mathbb{E}\left[\sum_{t=0}^{1} u_t^\top P_t(u)\right] = \mathbb{E}\left[\sum_{t=0}^{1} u_t^\top A u_t\right] = 2\, v^\top A v.$$

For instance, if $\lambda_F = \lambda_C = 0.50$, $\epsilon = 0.05$, and $\Lambda = 1.50$, then $v^\top A v = 0.50 + 0.50 - (0.05 + 1.50) = -0.55$, so expected execution cost equals $2(-0.55) = -1.10 < 0$ and expected profit is strictly positive. The opening trade in the liquid instrument pushes the execution price of the other leg through hedging-induced cross-impact, and, because temporary impact carries no memory, the reversal books the same favorable wedge; together the two legs more than offset the own-impact paid on both legs. Proposition 2 rules this out by requiring $\mathbf{H}$ p.s.d. on $\mathbf{R}_T$.

3 Price-based constraints and mechanical rebalancing

This section formalizes mechanical rebalancing as a price-based constraint. The structure is widespread: volatility targeting, risk parity, drawdown and VaR limits, broker-dealer haircuts and credit limits, central-counterparty initial margins, risk-control overlays embedded in structured products, derivatives-venue margin and liquidation thresholds in crypto, lending-protocol collateral revaluations, and passive momentum, reversal, or equal-weighted strategies all update binding requirements from recent transaction prices. Past transaction prices then determine tomorrow's feasible positions, and with price impact a feedback loop forms: prices affect tomorrow's constraint; a binding constraint forces trades; forced trades move prices. We model the rule and the induced sector demand, impose the classical no-arbitrage benchmark, and identify the new channel through which dynamic arbitrage appears even when the standard no-manipulation condition holds.

Fix trading dates $t = 0, \ldots, T-1$ and a window length $m \geq 0$, and let P_t denote transaction prices (scalar in the single-asset case, vector in the multi-asset case). At each date t, the rule maps the most

recent $m+1$ transaction prices into a scalar risk input,

$$\Gamma_t = \Gamma(P_{t-m}, \ldots, P_t), \tag{16}$$

and sets the next-date requirement

$$M_{t+1} = g(\Gamma_t), \tag{17}$$

where g is nondecreasing. The constrained sector holds positions X_t subject to a requirement-based feasibility constraint. A representative example is

$$|X_t(i)|\, M_t \leq E(i) \quad \text{for all constrained agents } i, \tag{18}$$

where $E(i)$ denotes equity, as in Brunnermeier and Pedersen (2009) and Gârleanu and Pedersen (2011). When the constraint binds, an increase in M_t forces a reduction in $|X_t(i)|$. Aggregating across constrained agents yields an induced sector demand rule $X_t = X(M_t)$. The mechanism needs only that constrained demand is sensitive to past prices, $dX_t/dP_{t-s} \neq 0$. Fix a binding requirement M_0 and define the local sensitivity $B = -X'(M_0) > 0$. Rebalancing generates forced order flow $v_t = X_t - X_{t-1}$. Let u_t denote the strategic trader's order, and define total order flow

$$q_t = u_t + v_t.$$

The fully constrained benchmark is $X = W/M$, with W total sector equity (hence $B = W/M_0^2$).

Assumption 5 (Temporary linear impact with total order flow)**.** *Transaction prices satisfy*

$$P_t = S_t + \mathcal{A} q_t, \tag{19}$$

where $\mathcal{A} = \eta > 0$ in the single-asset case, and $\mathcal{A} = A \in \mathbb{R}^{N \times N}$ in the multi-asset case, with A symmetric and $z^\top A z \geq 0$ for all $z \in \mathbb{R}^N$.

Assumption 5 imposes the classical no-manipulation benchmark: absent the price-based constraint, no admissible round trip has positive expected profit. Any profitable round trip in the rest of the paper therefore runs through a distinct, institutional channel: transaction prices enter the rule (16)–(17), which moves a binding requirement and induces forced flow v_t that feeds back into prices. The mechanism is pinned down by four primitives: the statistic Γ_t, the schedule g, the sector demand mapping $X(M)$ (equivalently, the mapping from M_t into v_t), and the impact map from total flow q_t into transaction prices P_t. The rest of the paper takes these as given, characterizes when their composition admits profitable round trips, and derives stress tests.

3.1 Profit decomposition and the source of arbitrage

Consider the single-asset case. For an order sequence $u = (u_t)_{t=0}^{T-1}$, realized trading profit is

$$\Pi(u) = -\sum_{t=0}^{T-1} u_t P_t. \tag{20}$$

For a round trip, Lemma 1 implies

$$\mathbb{E}[\Pi(u)] = -\mathbb{E}\left[\sum_{t=0}^{T-1} u_t \left(P_t - S_t\right)\right].$$

Under Assumption 5, $P_t - S_t = \eta q_t = \eta(u_t + v_t)$. Substituting gives

$$\mathbb{E}[\Pi(u)] = -\eta\,\mathbb{E}\left[\sum_{t=0}^{T-1} u_t^2\right] - \eta\,\mathbb{E}\left[\sum_{t=0}^{T-1} u_t v_t\right]. \tag{21}$$

The term $-\eta \sum_t u_t^2$ is the direct impact cost. The term $-\eta \sum_t u_t v_t$ is the only source of expected profit: positive exactly when u_t tends to have the opposite sign of forced flow v_t. For instance, during forced liquidation $v_t < 0$, a buy order $u_t > 0$ makes $u_t v_t < 0$ and increases expected profit. The source of arbitrage is that v_t is endogenous and rule-driven. Forced flow is the sector's rebalancing,

$$v_t = X_t - X_{t-1}, \qquad X_t = X(M_t).$$

Because transaction prices satisfy $P_t = S_t + \eta(u_t + v_t)$, the strategic order u_t affects the realized prices sampled by the rule, hence future requirements M_{t+1} and therefore future forced flow $v_{t+1}, v_{t+2}, \ldots$.

3.2 The excess cost of manipulation and its incidence

The decomposition (21) is the attacker's side; we now turn to the mechanical rebalancing (MR) sector's side. Define the sector's execution cost under the attack and the no-attack benchmark by

$$C^{MR}(u) = \eta \sum_{t=0}^{T-1} v_t\,(u_t + v_t), \qquad C^{MR}(0) = \eta \sum_{t=0}^{T-1} (v_t^0)^2, \tag{22}$$

and

$$\Delta C^{MR} = C^{MR}(u) - C^{MR}(0). \tag{23}$$

In the single-asset case,

$$\Delta C^{MR} = \eta \sum_{t=0}^{T-1} v_t^0 u_t + \eta \sum_{t=0}^{T-1} (v_t - v_t^0)(u_t + 2v_t^0) + \eta \sum_{t=0}^{T-1} (v_t - v_t^0)^2. \tag{24}$$

The three terms are, respectively, the direct price effect, the flow-distortion cross-term, and excess self-impact. The first term in (24) is the effect of the attacker's flow on the execution price of the unchanged baseline flow v_t^0. It equals minus the attacker's interaction profit at v_t^0. The remaining two terms arise only because the attack changes the rule-induced flow from v_t^0 to v_t. The transfer to the attacker and the change in aggregate execution cost are identified separately below. Indeed,

$$\Delta C^{MR} + \eta \sum_{t=0}^{T-1} u_t(u_t + v_t) = \eta \sum_{t=0}^{T-1} \Big[(u_t + v_t)^2 - (v_t^0)^2\Big]. \tag{25}$$

Taking expectations and using (21) shows that the wedge between the sector's expected excess cost and the attacker's expected profit is exactly the expected change in total quadratic execution cost relative to the no-attack benchmark. The attacker captures her profit as a transfer; the remaining wedge is the attack-induced rise in aggregate execution cost, borne beyond that transfer and received by the liquidity-supplying side. Whether this residual is a deadweight social loss or a transfer to liquidity suppliers depends on the interpretation of the impact coefficient η: it is a social loss to the extent that impact compensates the real costs of liquidity provision (risk-bearing and balance-sheet capacity), and a transfer to those suppliers to the extent that it is a rent. Either way it is not captured by the attacker, and Section 7.4 shows it grows with the scale of the rebalancing sector.

4 An augmented no-arbitrage condition

We derive the no-arbitrage condition that applies once mechanical rebalancing feeds back into prices. The starting point is the standard finite-horizon benchmark for market impact: absent any mechanical feedback, every admissible round trip has nonnegative expected execution cost (Propositions 1–2). The question is whether a deterministic price-based constraint can nevertheless generate profitable round trips. Theorem 1, the main result of this section, gives a sharp answer: round-trip manipulation is absent if and only if a symmetric cost matrix that internalizes rule-induced flow is positive semidefinite on the round-trip subspace. We work in a local linearization around a configuration in which the requirement is binding. The augmented cost matrix is built from six primitives: the impact map $\mathcal{I}$, the Jacobian J of the risk statistic, the schedule slope $s = g'(\Gamma_0)$, the requirement-to-position sensitivity B, and timing operators L and D that encode implementation lags. Any mechanical-rebalancing rule fixes a particular instance of these primitives, so the test applies uniformly across settings. Failure of the linearized test implies a truly profitable, sufficiently small round trip in the exact nonlinear model (Proposition 8 in Internet Appendix A.5).

4.1 The mechanism

Before the general treatment, a two-period example shows how a rule can overturn the classical benchmark. Consider an impact matrix that is manipulation free in the classical sense, two dates $t = 0, 1$, temporary linear impact under total flow,

$$P_t = S_t + \eta q_t, \qquad q_t = u_t + v_t, \qquad \eta > 0,$$

a martingale unaffected price ($S_t = S$), and the round-trip subspace $\mathcal{R}_2 = \{u \in \mathbb{R}^2 : u_0 + u_1 = 0\}$. Without feedback $\Pi(u) = -\eta(u_0^2 + u_1^2) \leq 0$ for every $u \in \mathcal{R}_2$: the classical benchmark holds. Now let the date-0 trade force flow at date 1, $v_0 = 0$ and $v_1 = ku_0$ with $k > 0$. For $u = (a, -a)$,

$$\Pi(u) = -\eta\big(u_0 q_0 + u_1 q_1\big) = \eta(k-2)a^2,$$

which is positive when $k > 2$: the round trip pays whenever the induced forced flow exceeds twice the size of the trade that triggers it. The rest of this section turns this mechanism into a test for any

disclosed rule and horizon; we return at the end to verify that the example is its two-period instance.

4.2 A linearized model

Fix a horizon T. We study a small trade-induced deviation, holding the unaffected price path S fixed. For any object Y, let δY denote its deviation from the reference value, so $\delta S = 0$. Let $u \in \mathbb{R}^T$ be the strategic trader's deviation from the current strategy and let $v \in \mathbb{R}^T$ be the resulting deviation in forced flow from the mechanically rebalancing sector. Total flow is

$$q = u + v.$$

The transaction price P carries a wedge over the unaffected price. In deviations,

$$\delta P = \delta S + \mathcal{I} q, \tag{26}$$

where $\mathcal{I}$ is the linear map from flow paths to the induced transaction-price wedge. It is convenient to denote this wedge by

$$\delta \Delta P = \delta P - \delta S = \mathcal{I} q.$$

Let $\delta\Gamma \in \mathbb{R}^T$ collect the statistic deviations over the horizon. Throughout this section the linearization is taken at a well-behaved reference configuration: the requirement binds, the statistic map $\Gamma(P)$ and the schedule g are differentiable there, so the Jacobian J and the slope $s = g'(\Gamma_0)$ exist, and the sector's baseline flow is zero. Where the rule has nondifferentiable components (caps, floors, or max/min operations across windows), the derivatives are understood at interior points. Because baseline flow is zero, deviations coincide with levels: $q = u + v$ and $P - S = \mathcal{I} q$. Assumption 6 in Internet Appendix A states these local regularity conditions precisely. The statistic is a deterministic function of the sampled transaction-price path, so its first-order approximation is

$$\delta\Gamma = J\, \delta \Delta P, \tag{27}$$

where J is the Jacobian of $\Gamma(P)$ at the reference configuration. Let $\delta M \in \mathbb{R}^T$ denote deviations in the posted requirements, aligned with the trading dates so that $(\delta M)_t$ corresponds to M_t. Under the timing $M_{t+1} = g(\Gamma_t)$, within-horizon deviations satisfy $(\delta M)_0 = 0$ and, for $t = 1, \ldots, T-1$,

$$(\delta M)_t = s\, (\delta\Gamma)_{t-1},$$

where $s = g'(\Gamma_0)$ is the schedule slope. Equivalently,

$$\delta M = s\, L\, \delta\Gamma,$$

where $L \in \mathbb{R}^{T \times T}$ is the one-step lag operator.

4.2.1 From requirements to forced flow

Let $B = -X'(M_0) > 0$ be the local sensitivity of constrained demand $X(M)$ at the binding point, equal to W/M_0^2 under the benchmark $X = W/M$. Linearizing around the binding point gives

$$\delta X = -B\,\delta M.$$

Forced flow is the change in constrained demand, $v_t = X_t - X_{t-1}$, so in deviations

$$v = -B\,D\,\delta M, \tag{28}$$

where $D \in \mathbb{R}^{T\times T}$ is the first-difference operator. The scalars s and B are evaluated at the binding reference, exact when the requirement is identical across dates; when it varies by date, the implementation replaces them by date-by-date diagonal matrices (Appendix A.4).

4.2.2 Closing the loop

Combining the linearizations yields the induced forced-flow response to a wedge-price deviation,

$$v = -B\,s\,D\,L\,J\,\delta\Delta P. \tag{29}$$

Using $\delta\Delta P = \mathcal{I}q$ and $q = u + v$,

$$q = u - BsDLJ\mathcal{I}q. \tag{30}$$

Define

$$\mathcal{K} = BsDLJ\mathcal{I}. \tag{31}$$

Then $(I + \mathcal{K})q = u$. $I + \mathcal{K}$ is invertible[8] and the fixed point selects a unique total-flow path for each strategic deviation u:

$$q = (I + \mathcal{K})^{-1}u, \qquad \delta\Delta P = \mathcal{I}(I + \mathcal{K})^{-1}u. \tag{32}$$

4.3 The augmented no-arbitrage condition

Fix an admissible round trip $u \in \mathcal{R}_T$. Under the martingale benchmark,

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t S_t\right] = 0$$

so expected profits depend only on the execution wedge $P - S$. In the linearized loop, (32) implies

$$P - S = \delta\Delta P = \mathcal{I}(I + \mathcal{K})^{-1}u.$$

[8]Lemma 5 in Internet Appendix A formalizes this argument. There, we also contrast lagged updating with instantaneous updating $M_t = g(\Gamma_t)$, where invertibility can fail. Small strategic trades can be strongly amplified into total flow, producing the quasi-arbitrage pathology highlighted there.

Define the effective impact

$$\widetilde{\mathcal{I}} = \mathcal{I}(I + \mathcal{K})^{-1}.$$

This matrix maps the proposed trade path into the entire wedge-price path, combining direct impact with the indirect component from induced forced flow. Expected execution cost is therefore

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t P_t\right] = \mathbb{E}\left[u^\top (P - S)\right] = \mathbb{E}\left[u^\top \widetilde{\mathcal{I}}\, u\right]. \tag{33}$$

As usual, define the augmented symmetric cost matrix

$$\widehat{H} = \widetilde{\mathcal{I}} + \widetilde{\mathcal{I}}^\top = \mathcal{I}(I + \mathcal{K})^{-1} + (I + \mathcal{K})^{-\top}\mathcal{I}^\top. \tag{34}$$

With $\Pi(u) = -\sum_{t=0}^{T-1} u_t P_t$, it follows that

$$\mathbb{E}[\Pi(u)] = -\mathbb{E}\left[u^\top \widetilde{\mathcal{I}}\, u\right] = -\frac{1}{2}\,\mathbb{E}\left[u^\top \widehat{H}\, u\right]. \tag{35}$$

A profitable round trip exists exactly when $\widehat{H}$ has a negative direction on $\mathcal{R}_T$, meaning that the rule-induced response tilts the induced flow so that the trader can unwind on favorable terms often enough to offset direct impact costs. The condition below is our key result. It takes only the disclosed rule and a market-impact estimate to derive a market viability test.

Theorem 1 (Augmented no-arbitrage condition)**.** *Consider the linearized feedback system* (26)–(32) *around a binding zero-baseline-flow reference state (Assumption 6), and assume $I + \mathcal{K}$ is invertible. A market is feedback manipulation free when no admissible round trip earns strictly positive expected profit in the linearized system (Internet Appendix A, Definition 4).*

(i) Single asset. The combined system is feedback manipulation free if and only if $\widehat{H}$ is positive semidefinite on $\mathcal{R}_T$.

(ii) Multi asset. Let $\mathcal{I} \in \mathbb{R}^{NT\times NT}$ map stacked total flow $q \in \mathbb{R}^{NT}$ into stacked wedge-price deviations $\delta\Delta P \in \mathbb{R}^{NT}$ via $\delta\Delta P = \mathcal{I}q$, and let $J \in \mathbb{R}^{T\times NT}$ map wedge-price paths into statistic deviations via $\delta\Gamma = J\,\delta\Delta P$. Under $M_{t+1} = g(\Gamma_t)$, write $\delta M = s\,L\,\delta\Gamma$ with $s = g'(\Gamma_0)$ and L the one-step lag operator. Suppose the mechanically rebalancing sector rebalances along a fixed liquidation direction $b \in \mathbb{R}^N$ with local sensitivity $B = -X'(M_0) > 0$, so the induced forced-flow deviations satisfy

$$v = -Bs\,(D \otimes I_N)(I_T \otimes b)\,L\,J\,\delta\Delta P.$$

Define

$$\mathcal{K} = Bs\,(D \otimes I_N)(I_T \otimes b)\,L\,J\,\mathcal{I} \in \mathbb{R}^{NT\times NT},$$

and

$$\widehat{\mathbf{H}} = \mathcal{I}(I + \mathcal{K})^{-1} + (I + \mathcal{K})^{-\top}\mathcal{I}^\top.$$

Then the combined system is feedback manipulation free if and only if $\widehat{\mathbf{H}}$ is positive semidefinite on round trips $u \in \mathbf{R}_T$.

Proof. Part (ii) follows from the same argument after stacking time-and-asset vectors, so it suffices to prove part (i). Fix an admissible round trip u. By Assumption 1 and the round-trip condition,

$$\mathbb{E}\left[\sum_{t=0}^{T-1} u_t S_t\right] = 0.$$

Using (32), we have $P - S = \widetilde{\mathcal{I}} u$, hence (35) gives

$$\mathbb{E}[\Pi(u)] = -\frac{1}{2}\,\mathbb{E}\left[u^\top \widehat{H}\, u\right].$$

If $\widehat{H}$ is positive semidefinite on $\mathcal{R}_T$, then $u(\omega) \in \mathcal{R}_T$ a.s. implies $u(\omega)^\top \widehat{H}\, u(\omega) \geq 0$ a.s., so $\mathbb{E}[\Pi(u)] \leq 0$ for every admissible round trip. Conversely, if $\widehat{H}$ is not positive semidefinite on $\mathcal{R}_T$, choose $\tilde{u} \in \mathcal{R}_T$ with $\tilde{u}^\top \widehat{H}\, \tilde{u} < 0$ and take the deterministic admissible round trip $u \equiv \tilde{u}$. Then $\mathbb{E}[\Pi(u)] = -\frac{1}{2}\, \tilde{u}^\top \widehat{H}\, \tilde{u} > 0$, so feedback manipulation exists. □

Internet Appendix A, Theorem 3 rewrites the condition as a restricted eigenvalue test, which is convenient for computation. The augmented matrix $\widehat{H}$ is built from the price-impact operator and the rule's feedback map alone, with no reference to any trader's objective, so the condition is a property of the market and the disclosed rule. When it fails, the market that embeds the rebalancing sector admits a profitable round trip and is no longer arbitrage-free; the largest scale of the sector consistent with the condition is a capacity of mechanical rebalancing, and a strategic trader only realizes and prices the arbitrage the failure already implies.

4.4 Back to the earlier example

The two-period example that opened this section is a special case of the linearized primitives above, with $v = -BsDLJ\,\delta\Delta P$ and $\delta\Delta P = \mathcal{I}q$. Let the statistic react only to the date-0 price, so J is nonzero only in its $(0,0)$ entry, and calibrate it to reproduce $v_1 = ku_0$; this gives $J_{0,0} = -k/(Bs\eta)$ and, with $\mathcal{I} = \eta I$,

$$\mathcal{K} = BsDLJ\mathcal{I} = \begin{pmatrix} 0 & 0 \\ -k & 0 \end{pmatrix}, \qquad (I + \mathcal{K})^{-1} = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}.$$

With effective impact $\widetilde{\mathcal{I}} = \mathcal{I}(I + \mathcal{K})^{-1}$, the augmented cost matrix $\widehat{H} = \widetilde{\mathcal{I}} + \widetilde{\mathcal{I}}^\top$ is

$$\widehat{H} = \eta \begin{pmatrix} 2 & k \\ k & 2 \end{pmatrix}, \qquad u^\top \widehat{H} u = -2\eta(k-2)a^2 \ \text{ for } u = (a, -a),$$

so $\widehat{H}$ is not p.s.d. exactly when $k > 2$, matching the direct profit above.

5 Single-asset optimal attack

Section 4 provides a diagnostic for when deterministic, price-based requirements overturn the classical no-dynamic-arbitrage benchmark in temporary-impact models. This section takes the rule and

impact as given and characterizes the finite-horizon round trip that exploits a breach of the capacity bound, in the single-asset environment. The solution separates two motives. First, the attacker profits from predictable, constraint-driven orders: she trades in the direction of the forced flow and reverses once it is done, as in models of order anticipation and predatory trading (Brunnermeier and Pedersen, 2005). Second, she can trade within the sampling window to move the rule's input and thereby reshape future forced flow. Predatory models also let the attacker induce the imbalance, but only through a solvency threshold, so she can trigger the liquidation or not; here the rule maps a sampled price statistic into a quantity of future demand, so she sets its magnitude. The dynamic program makes these incentives explicit.

5.1 The dynamic program

Throughout impose Assumption 5 and $X_t = W/M_t$. Fix a finite horizon T with trading dates $t = 0, 1, \ldots, T-1$. Initial conditions are $(y_0, P_{-1}, M_0, X_{-1}, X_0)$, where P_{-1} is the last pre-horizon transaction price, y_0 summarizes the remaining pre-horizon prices the statistic requires, and $x_0 = 0$ by Definition 1. For the optimal-attack dynamic programs in Sections 5–6, we work conditional on a deterministic unaffected-price path $(S_t)_{t=-1}^{T-1}$, so the state transition is deterministic given (s_t, u_t) and we omit conditional expectations. At the start of date t, before choosing u_t, the state is

$$s_t = (x_t,\ y_t,\ P_{t-1},\ M_t,\ X_t,\ X_{t-1}). \tag{36}$$

The attacker controls inventory x_t through trades. The variable y_t is the rule's internal memory: the minimal summary of past transaction prices, beyond the most recent price P_{t-1}, that the statistic Γ needs, with F its law of motion; in the volatility application of Section 7 it is the trailing-returns window $y_t = (r_{t-1}, \ldots, r_{t-m+1})$. The remaining components (M_t, X_t, X_{t-1}) pin down the sector's current and lagged holdings, which determine its mechanically induced net trade on date t. The rule is Markov in (y_t, P_{t-1}, P_t):

$$y_{t+1} = F(y_t, P_t), \qquad \Gamma_t = \Gamma(y_t, P_{t-1}, P_t), \qquad M_{t+1} = g(\Gamma_t). \tag{37}$$

The sector's date-t forced order flow is predetermined at the start of date t:

$$v_t = X_t - X_{t-1}. \tag{38}$$

The attacker observes v_t in the state and chooses u_t subject to a per-date bound $|u_t| \le \bar{u}$. To ensure feasibility of the terminal round-trip constraint $x_T = 0$, restrict actions to those that leave enough remaining capacity to unwind:

$$\mathcal{U}_t(x_t) = \{u \in \mathbb{R} :\ |u| \le \bar{u},\ |x_t + u| \le (T-1-t)\bar{u}\}\,. \tag{39}$$

Given u_t, total order flow is $q_t = u_t + v_t$ and transaction prices follow temporary impact:

$$P_t = S_t + \eta q_t = S_t + \eta(u_t + v_t). \tag{40}$$

The attacker's realized profit is $\Pi(u) = -\sum_{t=0}^{T-1} u_t P_t$. Because the strategy is a terminal round trip and S_t is a martingale, the unaffected-price component contributes zero to expected profit (Lemma 1). The dynamic program therefore maximizes the execution-wedge component of profit, whose expectation equals expected round-trip profit; conditional on the realized path (S_t), we write the per-period payoff as

$$\pi_t(s_t, u_t) = -u_t(P_t - S_t) = -\eta u_t(u_t + v_t), \tag{41}$$

which is the objective optimized by the dynamic program. The direct impact term $-\eta u_t^2$ is always negative, while the interaction term $-\eta u_t v_t$ is positive precisely when the attacker trades against the mechanically induced flow. Dynamic incentives arise because u_t changes both inventory, $x_{t+1} = x_t + u_t$, and the transaction price P_t, which enters (37) and can therefore move future requirements and future forced flow. Given s_t and u_t, the next state is

$$s_{t+1} = \mathcal{T}(s_t, u_t) = (x_t + u_t,\ F(y_t, P_t),\ P_t,\ g(\Gamma_t),\ W/g(\Gamma_t),\ X_t), \tag{42}$$

where P_t is given by (40) and Γ_t by (37). Define the value function

$$V_t(s) = \sup_{(u_\tau)_{\tau=t}^{T-1}} \sum_{\tau=t}^{T-1} \pi_\tau(s_\tau, u_\tau), \tag{43}$$

subject to $u_\tau \in \mathcal{U}_\tau(x_\tau)$, the transition (42), and the terminal constraint $x_T = 0$. The associated Bellman recursion is

$$V_T(s) = 0 \text{ if } x_T = 0, \qquad V_T(s) = -\infty \text{ otherwise}, \tag{44}$$

$$V_t(s) = \max_{u \in \mathcal{U}_t(x)} \left\{ \pi_t(s, u) + V_{t+1}(\mathcal{T}(s, u)) \right\}, \qquad t = T-1, \ldots, 0. \tag{45}$$

5.2 Trade decomposition: harvesting, inventory management, and triggering

The dynamic program highlights two distinct roles of a trade. First, it determines the current payoff through (41). Second, by moving P_t, it affects the rule and can therefore reshape future forced flow. A useful benchmark holds future incentives fixed and treats v_t as predetermined, isolating the purely mechanical intermediation problem. This is the single-period analogue of supplying liquidity against a given liquidation pressure in the predatory-trading literature.

It is useful to name the three forces that the interior optimum (49) below will isolate, and to relate them to the two motives identified above. The first, *harvesting* (the term $-v_t/2$), leans against the predetermined forced flow v_t and captures half its rent; this is the liquidity-supply motive of order anticipation and predatory trading. The third, *triggering* (the term $\nu_{t+1}/2$, with $\nu_{t+1} = \frac{d}{dP_t} V_{t+1}$), moves the current price P_t to reset the rule's future requirements and hence the forced flow the attacker will later harvest; this is the reshaping motive specific to price-based feedback. Between them sits *inventory management* (the term $\mu_{t+1}/(2\eta)$, with $\mu_{t+1} = \partial_x V_{t+1}$), the optimal-execution motive that tilts trading away from pure harvesting to discharge the terminal unwind $x_T = 0$. The remainder of the section derives this decomposition and the shadow values μ_{t+1}, ν_{t+1}, Λ_{t+1}, and χ_{t+1} that govern when each force operates.

Lemma 3 (Myopic harvesting). *Fix a date t and a state s_t with predetermined forced flow $v_t = X_t - X_{t-1}$. Under (41), the one-period payoff satisfies*

$$\pi_t(s_t, u) = -\eta\Big(u + \tfrac{v_t}{2}\Big)^2 + \frac{\eta}{4} v_t^2. \tag{46}$$

Consequently, holding the continuation value fixed, $\pi_t(s_t, u)$ is uniquely maximized at

$$u_t^{\text{myopic}} = -\frac{v_t}{2}, \qquad \max_{u \in \mathbb{R}} \pi_t(s_t, u) = \frac{\eta}{4} v_t^2. \tag{47}$$

If $v_t = 0$, then $\pi_t(s_t, u) \le 0$ for all u, with equality only at $u = 0$.

Proof. Completing the square in (41) gives (46). The square term is minimized at $u = -v_t/2$, yielding the unique maximizer and the maximal value in (47). If $v_t = 0$, then $\pi_t(s_t, u) = -\eta u^2 \le 0$, with equality only at $u = 0$. □

Lemma 3 provides a pure harvesting benchmark. When v_t is predetermined, gains arise only from trading against the forced order, and the myopic optimum absorbs exactly half of it. The attacker is compensated for providing immediacy against a predictable imbalance, while paying the usual temporary-impact cost on own orders.

Optimal attacks depart from the previous myopic harvesting benchmark because the trader must satisfy the terminal round-trip constraint and because u_t affects future requirements and thus future forced flow. The next result characterizes the interior optimum by rewriting the first-order condition in terms of two shadow values: the marginal value of inventory carried into $t+1$ and the marginal value of shifting the current price.

Proposition 3. *Fix $t \in \{0, \dots, T-1\}$ and a state s_t. Suppose the optimizer $u_t^\star$ in (45) is interior. Assume that V_{t+1} is differentiable at $s_{t+1}^\star = \mathcal{T}(s_t, u_t^\star)$ and that $\mathcal{T}(s_t, u)$ is differentiable at $u_t^\star$. Define*

$$\mu_{t+1} = \partial_x V_{t+1}(s_{t+1}^\star), \qquad \nu_{t+1} = \frac{d}{dP_t} V_{t+1}(s_{t+1}^\star), \qquad P_t = S_t + \eta(u_t^\star + v_t).$$

Then $u_t^\star$ satisfies the Euler equation

$$\eta(2u_t^\star + v_t) = \mu_{t+1} + \eta\,\nu_{t+1}, \tag{48}$$

or equivalently

$$u_t^\star = -\frac{v_t}{2} + \frac{1}{2\eta}\mu_{t+1} + \frac{1}{2}\nu_{t+1}. \tag{49}$$

Moreover, the resulting current-period payoff satisfies

$$\pi_t(s_t, u_t^\star) = \frac{\eta}{4} v_t^2 - \eta\Big(u_t^\star + \tfrac{v_t}{2}\Big)^2 = \frac{\eta}{4} v_t^2 - \frac{1}{4\eta}\big(\mu_{t+1} + \eta\nu_{t+1}\big)^2. \tag{50}$$

Proof. See Internet Appendix A.6.2. □

In (49), the inventory shadow value μ_{t+1} is the optimal-execution motive familiar from linear- and transient-impact models: even with future forced flow fixed, the attacker tilts away from myopic

harvesting to manage the unwind implied by $x_T = 0$. The price shadow value ν_{t+1} is specific to the feedback environment because P_t is an input to the requirement update; when a marginal change in the current price shifts future requirements and thus future forced flow, ν_{t+1} is the continuation-value benefit of moving P_t. The term $\frac{\eta}{4}v_t^2$ in equation (50) is the mechanical rent available from intermediating the predetermined imbalance, while the second term is the wedge cost of departing from the myopic liquidity-provision benchmark to manage inventory and to move the rule's input. Corollary 3 in Internet Appendix A.6.3 derives the resulting dichotomy formally: at a date with $v_t = 0$ the myopic payoff is nonpositive, so any nonzero trade is an investment in continuation value, with $u_t^\star = \mu_{t+1}/(2\eta) + \nu_{t+1}/2$; when $\mu_{t+1} = \nu_{t+1} = 0$ the trade collapses to pure harvesting $u_t^\star = -v_t/2$. Feedback makes the liquidation path endogenous: the same strategic trades that supply liquidity against future imbalances can also shift the rule that generates those imbalances, unlike standard predatory-trading and order-anticipation models in which the order imbalance is taken as given. The decomposition also clarifies when the attacker may rationally trade in the same direction as forced flow. Such procyclical trading is never selected by the static harvesting benchmark, since it worsens the current payoff. It can be optimal only when continuation value dominates the harvesting motive, either because inventory considerations shift trading toward earlier dates or because moving the current price changes future requirements and enlarges future forced flow enough to compensate for the extra impact cost today. Corollary 4 in Internet Appendix A.6.4 formalizes this argument. Writing $\chi_{t+1} = \mu_{t+1} + \eta\nu_{t+1}$, with $v_t \neq 0$ the optimal trade runs with the forced order, $u_t^\star v_t > 0$, if and only if $\operatorname{sign}(\chi_{t+1}) = \operatorname{sign}(v_t)$ and $|\chi_{t+1}| > \eta|v_t|$; and the current payoff is negative, $\pi_t(s_t, u_t^\star) < 0$, exactly when $|\chi_{t+1}| > \eta|v_t|$.

5.3 How the rebalancing rule shapes the manipulation motive

The trigger motive ν_{t+1} from Proposition 3 factors into three primitives of the disclosed rule: the sensitivity of the sampled statistic to the current transaction price, the pass-through of the statistic into the next requirement, and the shadow value of tightening that requirement. Because it is their product, the motive switches off whenever any one is zero: a rule is locally non-manipulable where its schedule is flat, as inside a cap, or where its statistic does not respond to the transaction price. When both links are active, the current price moves next-period constrained demand by $dX_{t+1}/dP_t = -B_{t+1}\, g'(\Gamma_t)\, \partial_P \Gamma(y_t, P_{t-1}, P_t)$. Internet Appendix A.6.5 states the decomposition (Proposition 9) and its shutdown conditions (Corollary 5).

5.4 Endgame structure, implementation lags, and closed-form harvesting

The terminal round-trip constraint forces an unwind at the end of the horizon. This endgame discipline rules out indefinitely carrying an induced position and makes late-horizon behavior largely mechanical. Implementation lags further limit the set of dates on which a trade can affect within-horizon requirements. At the terminal trade date $t = T - 1$ the unique admissible trade is $u_{T-1}^\star = -x_{T-1}$. Therefore $V_{T-1}(s_{T-1}) = \pi_{T-1}(s_{T-1}, -x_{T-1})$. With an implementation lag $\ell \geq 0$, so that $M_{t+1} = g(\Gamma_{t-\ell})$, any date-$t$ trade with $t \geq T-1-\ell$ can no longer affect a within-horizon requirement or forced flow, so $\nu_{t+1} = 0$ and the continuation problem reduces to harvesting against the pre-

determined forced-flow sequence and unwinding inventory by T (in closed form by Proposition 4 when trade bounds are slack, and by Proposition 10 under binding bounds). The last date at which a trade can trigger additional within-horizon forced flow is $t_{\text{last}} = T - 2 - \ell$. Lemma 6 in Internet Appendix A.6.6 states this formally, and it captures a practical implication of implementation lags. Late trades can still move the measured statistic, but any induced change in posted requirements arrives too late to affect constrained positions within the remaining horizon. From the last trigger date onward, the attacker faces a fixed sequence of predictable imbalance and an obligatory unwind, linking the endgame to optimal execution with exogenous flow under convex impact costs and to the broader literature on predictable trades and anticipated shocks (Bessembinder et al., 2016). More generally, once the remaining forced-flow path $(v_\tau)_{\tau=t}^{T-1}$ is fixed, the continuation problem reduces to intermediation and inventory management. The manipulation motive drops out because within-horizon requirements, and thus within-horizon forced flows, no longer depend on the current price.

Proposition 4 (Closed-form continuation when future forced flow is predetermined)**.** *Fix a date t and suppose that, conditional on the state s_t, the remaining forced flows $(v_\tau)_{\tau=t}^{T-1}$ are predetermined at time t and do not depend on the continuation actions $(u_\tau)_{\tau=t}^{T-1}$. Assume the per-date bounds are slack at the optimum. Let $n = T - t$ denote the number of remaining trading dates and define the average forced flow over the remainder of the horizon,*

$$\bar{v}_t = \frac{1}{n}\sum_{\tau=t}^{T-1} v_\tau.$$

Then the unique maximizer of the continuation problem satisfies, for $\tau = t, \dots, T-1$,

$$u_\tau^\star = -\frac{1}{2}(v_\tau - \bar{v}_t) - \frac{x_t}{n}. \tag{51}$$

Moreover, the associated maximal continuation profit is

$$V_t^{\text{harv}} = \frac{\eta}{4}\sum_{\tau=t}^{T-1}(v_\tau - \bar{v}_t)^2 + \eta x_t \bar{v}_t - \frac{\eta}{n}x_t^2. \tag{52}$$

Proof. See Internet Appendix A.6.7. □

Proposition 4 shows that once the sector's flow is exogenous to the attacker, she is optimally intermediating a known sequence of imbalances while respecting the terminal unwind. The average level of forced flow does not by itself generate a timing opportunity; the timing rent comes from predictable variation around the average. This is captured by $\frac{\eta}{4}\sum_{\tau=t}^{T-1}(v_\tau - \bar{v}_t)^2$, which vanishes when v_τ is constant over the remaining dates, leaving only mechanical liquidation. The remaining terms, $\eta x_t \bar{v}_t - \frac{\eta}{n}x_t^2$, summarize the inventory tradeoff: carrying inventory aligned with the average imbalance reduces the net intermediation burden, but outstanding inventory becomes increasingly costly as n shrinks because it must be unwound over fewer remaining dates under quadratic impact.

Corollary 1 (Only dispersion generates timing rents)**.** *In the setting of Proposition 4,*

$$V_t^{\text{harv}} = \frac{\eta}{4}\sum_{\tau=t}^{T-1}(v_\tau - \bar{v}_t)^2 + \eta x_t \bar{v}_t - \frac{\eta}{n}x_t^2.$$

The only component of V_t^{harv} that depends on the temporal pattern of forced flow beyond its average is the dispersion term $\frac{\eta}{4}\sum_{\tau=t}^{T-1}(v_\tau - \bar{v}_t)^2$. In particular, if $(v_\tau)_{\tau=t}^{T-1}$ is constant over time, say $v_\tau = c$, then

$$u_\tau^\star = -\frac{x_t}{n} \quad \textit{for all } \tau, \qquad V_t^{\text{harv}} = \eta x_t c - \frac{\eta}{n} x_t^2.$$

If, in addition, $x_t = 0$, then $u_\tau^\star = 0$ and $V_t^{\text{harv}} = 0$.

Proof. See Internet Appendix A.6.8. □

Corollary 2 (A single future liquidation event)**.** *In the setting of Proposition 4, let $n = T - t$ and suppose $x_t = 0$ and forced flow is concentrated at the terminal trade date: $v_\tau = 0$ for $\tau = t, \ldots, T-2$ and $v_{T-1} = v < 0$. Then $\bar{v}_t = v/n$ and the optimal trades are*

$$u_\tau^\star = \frac{v}{2n} < 0 \textit{ for } \tau = t, \ldots, T-2, \qquad u_{T-1}^\star = -\frac{n-1}{2n}\, v > 0.$$

Moreover, the maximal continuation profit is

$$V_t^{\text{harv}} = \frac{\eta}{4}\sum_{\tau=t}^{T-1}(v_\tau - \bar{v}_t)^2 = \frac{\eta}{4}\frac{n-1}{n}v^2.$$

Proof. See Internet Appendix A.6.9. □

Holding the forced-flow path fixed, continuation value increases with its temporal dispersion. Design features that bunch rebalancing into a narrow window (e.g., closing auctions, benchmark setting dates, etc.) therefore raise dispersion and increase the rents available from subsequent intermediation. Corollary 2 illustrates the optimal response to a bunched liquidation: ahead of a forced sale the attacker sells in small increments, building a short position, and then supplies liquidity on the liquidation date by buying against the forced order, unwinding the short. In the full feedback model, early trades can change the forced-flow path by shifting future requirements; once a particular (v_τ) is induced, the remaining problem is exactly the harvesting-and-unwind continuation characterized above. This concentration is empirically relevant: the instruments that motivate the analysis rebalance predominantly at or near the daily close rather than continuously, so almost the entire mechanically induced order flow arrives in a single narrow window. Leveraged and inverse exchange-traded funds are the clearest case, and their end-of-day rebalancing generates predictable flow (Cheng and Madhavan, 2009); target-volatility funds and the volatility-controlled indices underlying structured products share the same once-a-day timing. By Corollary 2, this single-moment concentration maximizes the dispersion term and the rents available to an intermediating attacker. The design compounds the trigger channel as well: because the rule samples the closing price, the attacker can move a single price to reset the next requirement, an incentive closely related to the manipulation of closing prices (Hillion and Suominen, 2004). End-of-day rebalancing therefore favors attacks on two counts, by bunching the forced flow that is later harvested and by isolating the single price that must be moved to trigger it.

6 Multi-asset optimal attacks

We extend the characterization of the exploiting round trip to N assets with temporary cross-impact under total flow. When the mechanically rebalancing sector liquidates along a fixed portfolio direction b and the risk statistic is a scalar function of prices, the attacker uses one portfolio to move the statistic and a different portfolio to intermediate the induced liquidation. This separation has no analogue in the single-asset case, where one impact coefficient governs both triggering and harvesting. Cross-impact also reshapes the trade-off on each side: it changes the rents from intermediating forced flow and the marginal cost of moving the price features that feed the statistic.

6.1 The dynamic program

The program mirrors Section 5.1 with the scalar impact η replaced by a symmetric positive semidefinite cross-impact matrix A and x_t, u_t, v_t vectors in $\mathbb{R}^N$. Impose Assumption 5 with N assets on a finite horizon T, conditional on a deterministic unaffected-price path $(S_t)_{t=-1}^{T-1}$. The date-t state $s_t = (x_t, y_t, P_{t-1}, M_t, X_t, X_{t-1})$, the rule update $y_{t+1} = F(y_t, P_t)$, $\Gamma_t = \Gamma(y_t, P_{t-1}, P_t)$, $M_{t+1} = g(\Gamma_t)$, and the forced flow $v_t = X_t - X_{t-1}$ are exactly as in the single-asset case. The one new primitive is the liquidation direction: the fully constrained benchmark holds a fixed $b \in \mathbb{R}^N$ scaled by W/M_t,

$$X_t = \frac{W}{M_t} b, \tag{53}$$

with local sensitivity $-\partial X_t / \partial M_t = (W/M_t^2)b$. The attacker chooses $u_t \in \mathbb{R}^N$ subject to the per-date speed limit

$$\|u_t\|_1 \le \bar{u}, \tag{54}$$

and to terminal-feasibility, giving the admissible set

$$\mathcal{U}_t(x_t) = \left\{ u \in \mathbb{R}^N : \ \|u\|_1 \le \bar{u}, \ \|x_t + u\|_1 \le (T - 1 - t)\bar{u} \right\}. \tag{55}$$

Total flow is $q_t = u_t + v_t$, transaction prices satisfy

$$P_t = S_t + A(u_t + v_t), \tag{56}$$

and the date-t payoff

$$\pi_t(s_t, u_t) = -u_t^\top A(u_t + v_t) \tag{57}$$

splits into the impact penalty $-u_t^\top A u_t \le 0$ and the harvesting term $-u_t^\top A v_t$, which is positive when the attacker trades against v_t. The transition is

$$\mathcal{T}(s_t, u_t) = \big(x_t + u_t, \ F(y_t, P_t), \ P_t, \ g(\Gamma_t), \ (W/g(\Gamma_t))b, \ X_t\big), \tag{58}$$

and the Bellman recursion is

$$V_t(s) = \max_{u \in \mathcal{U}_t(x)} \left\{ \pi_t(s, u) + V_{t+1}(\mathcal{T}(s, u)) \right\}, \qquad V_T(s) = 0 \text{ if } x_T = 0, \ -\infty \text{ otherwise.} \tag{59}$$

6.2 Parallels to the single-asset case

The solution reproduces Section 5.1 term for term, so we state it compactly and relegate the derivations to Internet Appendix A.7. Myopic harvesting sets $u_t^{\text{myopic}} = -v_t/2$ with rent $\frac{1}{4}v_t^\top A v_t$ (Lemma 7). The interior optimum satisfies the Euler equation $A(2u_t^\star + v_t) = \mu_{t+1} + A\nu_{t+1}$, which decomposes into harvesting, inventory management, and triggering exactly as in Proposition 3 (Proposition 11); and ν_{t+1} unpacks through the same chain rule, $\nu_{t+1} = \partial_P V_{t+1} + (\partial_P F)^\top \nabla_y V_{t+1} + \Lambda_{t+1} g'(\Gamma_t)\partial_P \Gamma$, with Λ_{t+1} the shadow value of tightening the requirement (Proposition 12).

A new feature is that cross-impact reshapes all three motives: harvesting rewards only the part of v_t that moves execution prices, and ν_{t+1} matters only along directions in which trades move the prices the rule reads. These distinctions drive the geometry of the next subsection.

6.3 The geometry of multi-asset attacks

Two portfolios govern the multi-asset attack, and the central point is that they need not be the same: the portfolio the sector is forced to trade, and the portfolio the rule watches. Take a volatility-controlled note on a fixed $70/30$ equity-bond basket. The liquidation direction b is the mix the sector holds: by (53) the position is $X_t = (W/M_t)\, b$, so when the rule deleverages the desk sells the same proportions, here $b = (0.7, 0.3)$, namely 70 cents of equity and 30 cents of bonds per dollar. The marked portfolio d is the exposure the rule reads: if the disclosed rule sets leverage from the volatility of the equity leg alone, it marks $d = (1, 0)$, so only the equity price enters the statistic. Triggering pushes the marked portfolio d: churn the equity to inflate its measured volatility and force the note to deleverage. Harvesting trades against the forced flow along b: buy the $70/30$ basket the note is made to sell. The single-asset model is the knife-edge case $b = d$. The two results below identify the cheapest way to move the statistic, and reduce any optimal attack to these two portfolios.

Lemma 4 (Minimal-cost portfolios)**.** *Fix a date t and suppose the statistic is differentiable in the current transaction-price vector with gradient $j_t = \partial_P \Gamma(y_t, P_{t-1}, P_t) \in \mathbb{R}^N$. Under (56), an infinitesimal trade perturbation $\delta u \in \mathbb{R}^N$ induces $\delta P_t = A\,\delta u$ and therefore the first-order statistic change*

$$\delta\Gamma_t = j_t^\top \delta P_t = j_t^\top A\,\delta u.$$

Assume $j_t^\top A j_t > 0$. Among all perturbations that deliver a given marginal statistic move $\delta\Gamma_t = \xi$ (with ξ a prescribed scalar), a perturbation with minimal instantaneous impact cost $\delta u^\top A\,\delta u$ is

$$\delta u^\star = \frac{\xi}{j_t^\top A j_t}\, j_t, \qquad \min_{\{\delta u:\ j_t^\top A\,\delta u = \xi\}} \delta u^\top A\,\delta u = \frac{\xi^2}{j_t^\top A j_t}.$$

If $j_t^\top A j_t = 0$, then $A j_t = 0$ and hence $j_t^\top A\,\delta u = 0$ for all δu; the statistic is locally insensitive to trading at date t.

Proof. Minimize the quadratic impact cost $\delta u^\top A\,\delta u$ subject to $j_t^\top A\,\delta u = \xi$. The first-order condition gives $A\,\delta u \propto A j_t$, so the price-moving component of δu is proportional to j_t, and the exposure constraint fixes the scale. When $j_t^\top A j_t > 0$, $\delta u^\star = (\xi / j_t^\top A j_t)\, j_t$ with minimal cost $\xi^2/(j_t^\top A j_t)$. When $j_t^\top A j_t = 0$, $A j_t = 0$, so $j_t^\top A\,\delta u = 0$ for all δu. □

The rule reads the single exposure $j_t^\top P_t$, so the cheapest trigger aligns δu with j_t at cost $\xi^2/(j_t^\top A j_t)$; when the rule marks a portfolio d, $j_t \propto d$ and the cost of a trigger of size ξ is $\xi^2/(c^2\, d^\top A d)$, falling as $d^\top A d$ rises (here $j_t = c\, d$): the more the marked portfolio moves per unit traded, the cheaper it is to shift the statistic. Nothing forces j_t to align with the liquidation direction b. In the example the rule reads the equity leg (d) while the sector holds the whole basket (b), so the trigger and liquidation portfolios are generally distinct.

However many assets trade, an optimal attack needs only two portfolios, the one the sector liquidates and the one the rule marks, and cross-impact enters through only three numbers.

Theorem 2. *Suppose the margin rule computes its scalar statistic by marking a fixed portfolio direction $d \in \mathbb{R}^N$: there exist functions $\bar{F}$ and $\bar{\Gamma}$ such that*

$$y_{t+1} = \bar{F}(y_t, d^\top P_t), \qquad \Gamma_t = \bar{\Gamma}(y_t, d^\top P_{t-1}, d^\top P_t),$$

for all relevant t. Consider the multi-asset dynamic program in Section 6.1 under temporary total-flow impact (56) and constrained demand (53). Consider the relaxed problem in which the per-date trade bound (54) (and hence the admissible set (55)) is removed but the terminal round-trip constraint $x_T = 0$ is retained. Then the supremum value of the relaxed problem is unchanged when restricting attention to attacks $(u_t)_{t=0}^{T-1}$ that trade only the two economically relevant portfolios,

$$u_t \in \operatorname{span}\{b, d\} \qquad \text{for all } t.$$

In particular, if the supremum is attained then there exists an optimal attack $(u_t^\star)_{t=0}^{T-1}$ satisfying this span restriction. Moreover, once trades are restricted to $\operatorname{span}\{b, d\}$, cross-impact matters only through three scalar cost coefficients:

$$b^\top A b, \qquad b^\top A d = d^\top A b, \qquad d^\top A d.$$

These numbers summarize, respectively, the execution-cost intensity of trading the liquidation portfolio b, the cross-impact interaction between the two portfolios, and the execution-cost intensity of trading the marked portfolio d. When $N = 1$ this reduction collapses to the single-asset model.

Proof. See Internet Appendix A.7.5. □

In the example, $b^\top A b$ prices trading the $70/30$ basket to harvest, $d^\top A d$ prices churning the equity to trigger, and $b^\top A d$ is their interaction, nonzero because the basket contains the equity the rule reads. A trade in any other direction is never worth using: whatever it does to the forced flow or the marked exposure, some combination of b and d does the same at no greater impact cost.

7 Volatility-controlled indices in structured products

Volatility-controlled (also called risk-control, volatility-managed, or target-volatility) indices supply a clean application for the results in this paper. They are embedded in retail structured products and index-linked annuities, so the mapping from sampled transaction prices into mandated reallocations is a contractual rule rather than a discretionary strategy. The methodology is publicly disclosed:

a rolling return window (often 20 days) yields a realized-volatility statistic, which determines the next-day risky weight, subject to caps, floors, smoothing, and implementation lags. Replication is mechanical: hedge desks trade the implied exposure in the underlying instruments, usually liquid ETFs. When measured volatility rises, the rule deleverages; when it falls, it re-levers (Krein and Fernandez, 2012; Moreira and Muir, 2017). The rebalancing pressure this generates is detectable in the data: the predicted weight change forecasts returns on the underlying funds (Appendix B.7).

This setting yields a capacity bound that requires only public inputs. Theorem 1 delivers the largest linked notional for which the induced feedback admits no dynamic arbitrage, taking as inputs only the published methodology and a market-impact calibration in the underlying hedge instruments. No holdings, dealer inventories, or ownership data enter, so an outside observer, a sponsor, or a supervisor can compute the bound directly from public information.

7.1 Product types

Marketed designs share a common blueprint: the contract holds a risky reference portfolio alongside cash or a bond-like instrument, and a publicly disclosed volatility-control rule adjusts the risky weight to keep risk near a stated target. Three parties implement it: the index sponsor publishes the methodology and the resulting weights; the issuer or insurer sells a contract whose payoff references the index, inheriting the methodology mechanically; and a hedge desk replicates the liability in liquid instruments (ETFs, futures, swaps, cash). Each rebalancing date forces a mechanical hedge trade. Internet Appendix B maps published methodologies into the rule objects used here. We focus on two designs:

1. *Single underlying (Template A):* a single parent index with risky weight equal to a capped inverse-volatility function of a rolling volatility estimate.

2. *Multi-underlying, fixed basket (Template B):* a fixed multi-asset basket (often implemented with ETFs or futures), scaled relative to cash by a scalar volatility-control factor.

Some products add a signal-driven layer that varies the reference portfolio over time (for example, trend or rotation rules, or an optimization over regime-dependent caps).

7.2 Optimal attacks against pure target-volatility rules

We specialize the single-asset dynamic program of Section 5.1 to the interior region of a pure target-volatility methodology, where the exposure schedule is smooth and the price-to-requirement mapping differentiable; we abstract from caps, floors, and turnover buffers. The risky weight is the target volatility divided by a rolling realized-volatility estimate, evaluated with implementation lag ℓ:

$$w_{t+1} = \frac{\sigma^*}{\widehat{\sigma}_{t-\ell}}, \qquad M_{t+1} = \frac{1}{w_{t+1}} = \frac{\widehat{\sigma}_{t-\ell}}{\sigma^*} = \frac{\sqrt{\Gamma_{t-\ell}}}{\sigma^*}, \qquad s_t = \frac{\partial M_{t+1}}{\partial \Gamma_{t-\ell}} = \frac{1}{2\sigma^* \widehat{\sigma}_{t-\ell}}. \tag{60}$$

Replication implies the volatility-managed sector holds $X_{t+1} = W/M_{t+1}$, generating forced flow $v_{t+1} = X_{t+1} - X_t$. Prices follow temporary impact, $P_t = S_t + \eta(u_t + v_t)$, and the attacker earns the

payoff (41). At an interior optimum, Proposition 3 gives the decomposition

$$u_t^\star = -\frac{v_t}{2} + \frac{1}{2\eta}\mu_{t+1} + \frac{1}{2}\nu_{t+1}, \tag{61}$$

with (μ_{t+1}, ν_{t+1}) defined there: inventory management against $x_T = 0$ and the feedback motive of moving the current price into future requirements. Proposition 9 in Internet Appendix A.6.5 unpacks ν_{t+1} through the rule's chain rule from the schedule slope in (60), the rolling-window shift F, and a closed-form derivative of the statistic in P_t, so the optimal policy is computable directly from the disclosed methodology. For Template A, the statistic is rolling realized variance of log returns,

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right), \qquad \Gamma_t = \frac{a_{\text{ann}}}{m}\sum_{i=0}^{m-1} r_{t-i}^2. \tag{62}$$

A minimal Markov state for the window is $y_t = (r_{t-1}, \ldots, r_{t-m+1})$, updated by a deterministic shift once P_t pins down r_t. The hard m-day window mechanically changes $\widehat{\sigma}$ as old returns roll out, generating predictable future forced flow the attacker can anticipate.

7.2.1 A predictable flow reversal

A perturbation δr_j enters Γ_t for every window $\{j, \ldots, j+m-1\}$ containing date j: holding other returns fixed,

$$\delta\Gamma_t = \frac{a_{\text{ann}}}{m}\,\delta(r_j^2)\cdot \mathbf{1}\{t \in \{j, \ldots, j+m-1\}\}. \tag{63}$$

and the derivative drops to zero at $t = j+m$ when r_j rolls out. With implementation lag ℓ, the rule applies a block of tighter requirements over dates $t+1 \in \{j+\ell+1, \ldots, j+\ell+m\}$, generating a predictable entry-exit pattern in forced flow $v_{t+1} = X_{t+1} - X_t$: deleveraging when the block begins to bind, releveraging when it ends. This is backward-looking procyclicality of hard rolling windows; exponentially weighted estimators replace the discrete roll-off with gradual decay.

7.2.2 The fragility of low-volatility states

In the interior region $g(\Gamma) = \sqrt{\Gamma}/\sigma^*$, so the schedule slope $s_t = g'(\Gamma_{t-\ell})$ from (60) is steepest when measured variance is small. The sector holds $X_{t+1} = W/M_{t+1} = W\sigma^*/\sqrt{\Gamma_{t-\ell}}$ with demand sensitivity

$$B_{t+1} = \frac{W}{M_{t+1}^2} = \frac{W\sigma^{*2}}{\Gamma_{t-\ell}}, \qquad B_{t+1}s_t = \frac{W\sigma^*}{2\,\Gamma_{t-\ell}^{3/2}}.$$

The product $B_{t+1}s_t$ is the mechanical gain from statistic to forced flow in binding states. Its $\Gamma^{-3/2}$ scaling means the rule is locally most responsive in calm markets. The attacker moves Γ only by moving transaction prices, so the marginal effect on next-day forced flow is

$$\underbrace{B_{t+1}s_t}_{\text{pass-through into demand}} \times \underbrace{\partial_P \Gamma}_{\text{sensitivity of the measured input to prices}}.$$

For rolling-window variance, calm states are special because of the proportional move: a given perturbation to measured variance is a larger relative change when the baseline Γ is low, exactly where the pass-through $B_{t+1}s_t$ is steepest. Calm configurations therefore combine large baseline exposure with the steepest statistic-to-demand pass-through, and are the most manipulable. These are not crisis states; they are exactly where the methodology levers up most aggressively. This is distinct from the leverage cycle. In leverage-cycle models, low measured risk relaxes financing and balance sheets expand (Geanakoplos, 2010; Adrian and Shin, 2010). Here, even holding sector size fixed, the schedule and the estimator amplify fragility through the two extra channels above. That a disclosed risk constraint can distort volatility in general equilibrium is not new: Basak and Shapiro (2001) show that value-at-risk limits amplify volatility in bad states. Their constraint is evaluated against horizon wealth fixed at the initial date, so no transaction price feeds back into a later requirement, and the distortion arises where the constraint binds. We ask instead how large the constrained sector can be before the feedback it creates admits a profitable round trip. This is a rule-based version of the "paradox of financial instability" (Borio and Drehmann, 2009; Borio, 2011). In our admissibility and capacity calculations, binding constraints are therefore often pinned down by low-volatility, high-exposure states rather than by conventional stress episodes. The test is thus forward-looking for benign states: given the disclosed methodology and standard impact inputs, it quantifies how close low-volatility configurations are to violating admissibility, even when backward-looking risk measurement labels them safe.

7.2.3 Inventory-light volatility triggering

The attacker can move Γ while keeping net inventory near zero. For any calendar date j,

$$\frac{\partial \Gamma_t}{\partial P_j} = \frac{2a_{\text{ann}}}{m\,P_j}\Big(\mathbf{1}\{t-m+1 \le j \le t\}\, r_j - \mathbf{1}\{t-m \le j \le t-1\}\, r_{j+1}\Big), \tag{64}$$

where $r_j = \log(P_j/P_{j-1})$. A single transaction price P_j enters two adjacent returns with opposite signs: it raises r_j and lowers r_{j+1}. Because Γ_t aggregates squared returns, a short-lived two-day disturbance with returns $+a$ on day j and $-a$ on day $j+1$ leaves the price level nearly unchanged but adds

$$\delta\Gamma_t \approx \frac{a_{\text{ann}}}{m}a^2\,\mathbf{1}\{t \in \{j,\dots,j+m-1\}\} + \frac{a_{\text{ann}}}{m}a^2\,\mathbf{1}\{t \in \{j+1,\dots,j+m\}\}$$

to the statistic on every window containing the spike. The attacker pays execution cost to generate the burst but keeps net inventory small relative to gross trading, so continuation value can be raised by shifting Γ without warehousing a directional position.

7.2.4 The effect of rule parameters: lags, horizons and target volatilities

An implementation lag ℓ kills the feedback motive in the endgame: by Lemma 6, no date-t trade with $t \ge T-1-\ell$ affects a within-horizon requirement, so $\nu_{t+1} = 0$, the last triggering date is $t_{\text{last}} = T-2-\ell$, and the continuation reduces to intermediation against a predetermined forced-flow sequence (Proposition 4). Window length m enters (64) two ways. The $1/m$ scaling attenuates

the marginal effect of any single return on Γ, but a longer window delays roll-off and stretches the induced reversal. These forces shift optimal triggering earlier and raise the value of remaining time to harvest. Target volatility σ^* scales requirements down ($M_{t+1} = \widehat{\sigma}_{t-\ell}/\sigma^*$) and hence exposure up. Although the schedule slope $s_t = 1/(2\sigma^*\widehat{\sigma}_{t-\ell})$ itself declines in σ^*, the demand sensitivity $B_{t+1} = W\sigma^{*2}/\Gamma_{t-\ell}$ rises faster: holding linked notional W fixed, $B_{t+1}s_t = W\sigma^*/(2\Gamma_{t-\ell}^{3/2})$ is increasing in σ^*, so higher targets raise exposure and amplify the flow response to statistic perturbations.

7.3 Testing vulnerability

Fix a market state z, a linked notional W, and a disclosed methodology. By Theorem 1, the economy is free of feedback manipulation at $(z; W)$ if and only if the augmented symmetric cost matrix $\widehat{H}(z; W)$ is positive semidefinite on the round-trip subspace, yielding the statewise capacity

$$W_{\max}(z) = \sup\Big\{W \geq 0 : \ \widehat{H}(z; W) \text{ is positive semidefinite on round trips}\Big\}.$$

Given a stress set $\mathcal{Z}$ drawn from the state distribution, the natural conservative target is the lower tail of the population distribution of $W_{\max}(z)$.[9] The sample minimum is an unstable estimator of that tail: as an extreme-value statistic over $|\mathcal{Z}|$ draws it drifts downward as $|\mathcal{Z}|$ grows and is seed-sensitive. We therefore report the first-percentile capacity

$$W_{\max}^{p01}(\mathcal{Z}) = Q_{0.01}\big(\{W_{\max}(z) : z \in \mathcal{Z}\}\big),$$

which converges at the standard $|\mathcal{Z}|^{-1/2}$ rate and admits a transparent reading: 1% of stress states are at least this fragile. We use $|\mathcal{Z}| = 10{,}000$ simulated states per configuration; the sample minimum and an independent-seed robustness check are reported in Internet Appendix B.5, Table 8, which shows that, relative to its level, the first percentile is more stable across seeds than the minimum. We also report the vulnerability curve, $\Pr_{z\in\mathcal{Z}}[W > W_{\max}(z)]$, the fraction of states at which admissibility fails at scale W. We construct $\mathcal{Z}$ from simulated price paths rather than from holdings or order flow, matching the objective of an ex ante test that uses only public rule information and standard liquidity inputs. Internet Appendix B details the simulation design and the impact calibration.

We want to stress that, even with no strategic trader, the rolling-window rule generates predictable forced flow as old returns leave the window, and predictable flow can in principle be front-run at any scale; this is the size-independent transfer that we separate from the scale-dependent rise in aggregate execution cost in Section 7.4, and it is present in every mechanical rebalancer. Whether that transfer is actually captured is a separate empirical question, and the evidence is mixed: predictable trades can draw in liquidity rather than predation (Bessembinder et al., 2016; Brøgger, 2021). What $W_{\max}(z)$ measures is the largest scale at which a trader cannot profitably move the sampled statistic and then harvest the rebalancing that its own impact induces. Positive semidefiniteness of

[9] Theorem 1 is exact at any given W; summarizing it by a per-state threshold presumes the admissible set $\{W \geq 0 : \widehat{H}(z; W)$ p.s.d. on round trips$\}$ is an interval. This holds in the two-period example of Section 4, where the round-trip form is proportional to $2 - k$ and the gain $k = -Bs\eta J_{0,0}$ is proportional to W through $B = W/M_0^2$, so the restricted form is affine in W. At longer horizons it is not guaranteed, but we verify it numerically: across 1,031 states over both templates and eight scenarios at $T = 126$, $\lambda_{\min}^{\mathcal{R}_T}(\widehat{H}(z; W))$ crosses zero exactly once in W at every state, so the admissible set is $[0, W_{\max}(z)]$. Evaluating the screen at a fixed W is exact regardless of the interval property.

$\widehat{H}$ measures how large the rebalancing sector can grow before the disclosed rule itself becomes manipulable, taking the ordinary cost of being predictable as the benchmark rather than as part of the capacity.

7.3.1 Results

We implement the test for the two volatility-control templates (A and B) at attacker horizons of 3, 6, and 12 months ($T \in \{63, 126, 252\}$ trading days). For each configuration we draw $|\mathcal{Z}| = 10{,}000$ stress states (Internet Appendix B.4), compute $W_{\max}(z)$ in days of ADV (linked notional; footnote 2), and record a failure whenever $W > W_{\max}(z)$.

The sector is fragile at scales that are small relative to the hedge market. Table 1 reports baseline magnitudes at $T = 126$. The conservative capacity $W^{p01}_{\max}(\mathcal{Z})$ is 0.166 days of ADV for Template A and 0.141, 0.165, and 0.313 for Template B with $N = 2, 4, 8$. At a sector size of one day of volume, the no-manipulation condition fails in 83.4% of stress states for Template A and in 85.0% to 97.5% for Template B; at two days of volume, failure rates rise to between 97.6% and 100.0%.

These ADV magnitudes translate directly into fractions of the underlying ETF. Khomyn et al. (2024) report a median annualized turnover ratio of 2.17 in their sample of large, liquid U.S. ETFs, equivalent to a daily ADV of $2.17/252 = 0.86\%$ of AUM. The Template A conservative capacity is therefore about $0.166 \times 0.86\% = 0.14\%$ of the underlying ETF; a scale of $W = 2$ corresponds to about $2 \times 0.86\% = 1.72\%$. The volatility-managed complex is manipulable in more than 80% of stress states well before it reaches 2% of the underlying market. The liquidity denominator throughout is full-day volume. Because these rules rebalance at the close, where only a fraction of daily volume trades, the notional a bunched rebalance can absorb is smaller, and the admissible scale falls roughly in proportion to that closing-window fraction, so the magnitudes above are generous relative to a closing-window execution. Liquidity and target volatility move the bound; implementation lags, volatility dynamics, and window choices cluster close to the baseline. Less liquid scenarios shift the vulnerability curve upward at every W (Figures 2 and 3), and Table 2 details the target-volatility and liquidity comparative statics. We include a 1 bp impact scenario, representative of a major underlying ETF and the most-liquid environment in our stress range. Even there, the test fails in roughly 40% of stress states by $W \approx 6$ days of ADV; under the Khomyn et al. (2024) ratio that is about 5% of the underlying ETF. The result survives the deepest plausible liquidity at sector sizes much larger than the baseline but still small relative to the hedge market. Internet Appendix B.5 reports $W^{p01}_{\max}(\mathcal{Z})$ for every scenario and horizon (Table 7) and the corresponding vulnerability curves for $N \in \{4, 8\}$ (Figures 4a–4b).

Table 2 varies the target volatility σ^* and the impact calibration, expressed as the cost in bps of trading 1% of volume, $c_{1\%}$. Lowering the target to $\sigma^* = 10\%$ raises conservative capacity by 25% in Template A and 30% in Template B with $N = 2$ relative to baseline. Raising the target to $\sigma^* = 15\%$ reduces capacity by roughly 17% in Template A and 9% in Template B with $N = 2$. In the larger portfolios the comparative static reverses: raising the target pushes a large share of stress states onto the leverage cap (Internet Appendix B.5, Table 9, where cap incidence at $\sigma^* = 15\%$ exceeds 60% for $N = 4$ and 80% for $N = 8$), where the rule's local sensitivity is truncated and the state drops out of the binding set, so the conservative percentile rises rather than falls (Internet Appendix B.5,

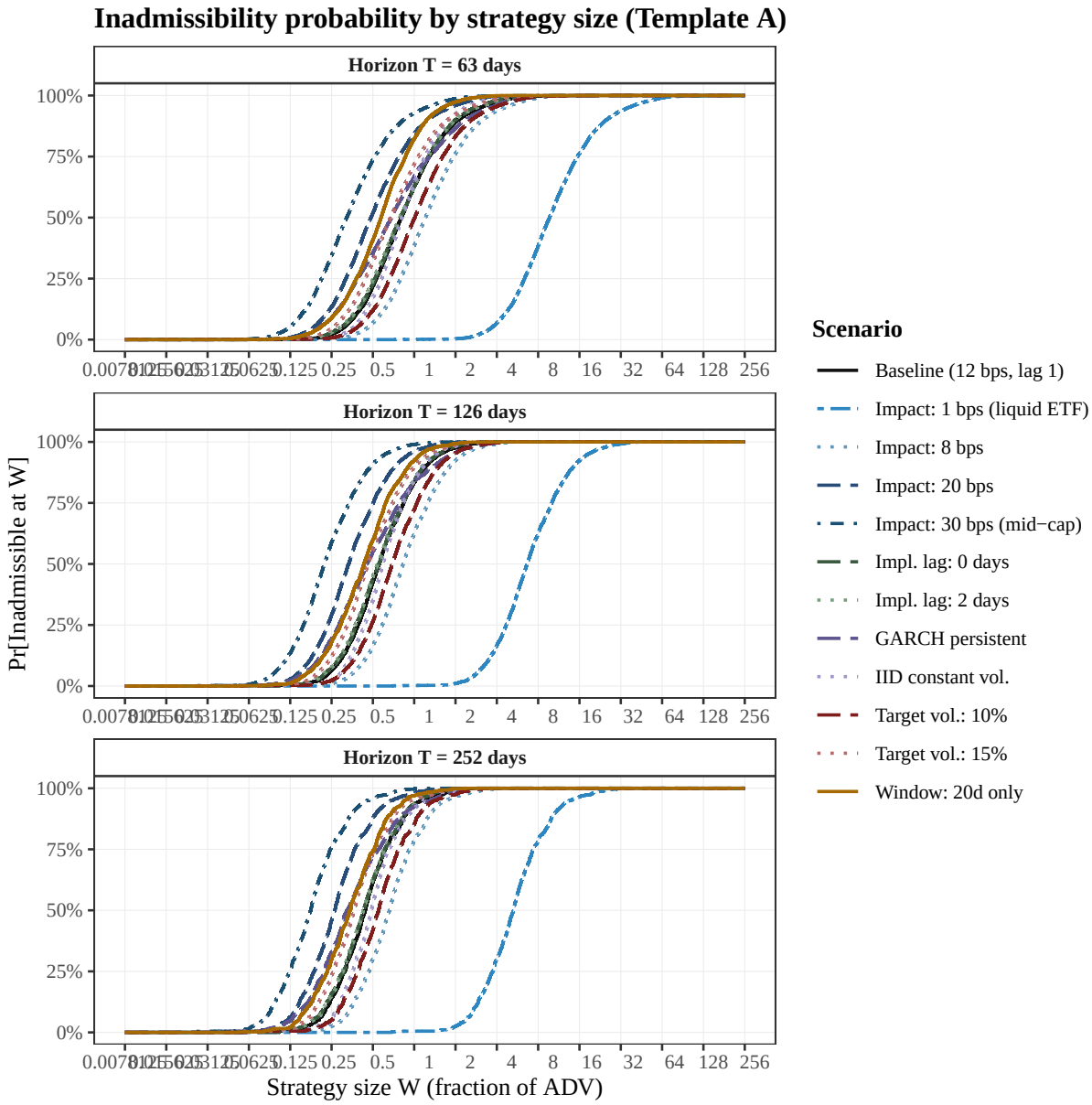


Figure 2 Vulnerability curves for Template A.

Panels correspond to $T \in \{63, 126, 252\}$. In each panel, the curve for a given scenario plots $\Pr_{z \in \mathcal{Z}}[W > W_{\max}(z)]$ across the $|\mathcal{Z}| = 10{,}000$ stress states as a function of sector scale W (in days of ADV). Different line types correspond to one-at-a-time deviations from the baseline configuration.

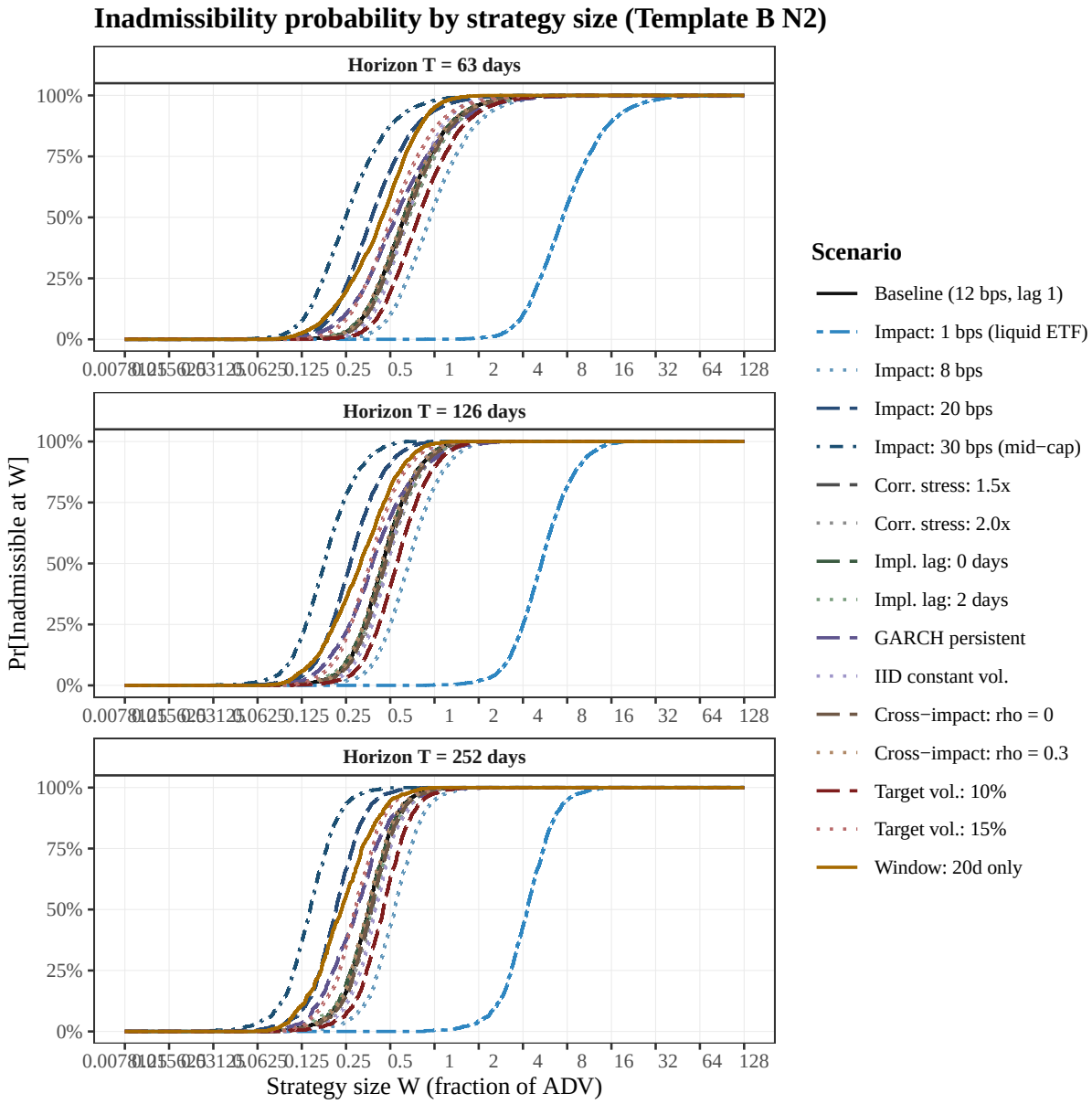


Figure 3 Vulnerability curves for Template B with $N = 2$.

Panels correspond to $T \in \{63, 126, 252\}$. In each panel, the curve for a given scenario plots $\Pr_{z \in \mathcal{Z}}[W > W_{\max}(z)]$ across the $|\mathcal{Z}| = 10{,}000$ stress states as a function of sector scale W (in days of ADV). Different line types correspond to one-at-a-time deviations from the baseline configuration.

		$W^{p01}_{\max}(\mathcal{Z})$	$W_{0.05}$	Median$(W_{\max})$	$\Pr[W_{\max} < 1]$	$\Pr[W_{\max} < 2]$
Template A	$N = 1$	0.166	0.235	0.556	83.4%	97.6%
Template B	$N = 2$	0.141	0.222	0.446	95.7%	100.0%
Template B	$N = 4$	0.165	0.241	0.446	97.5%	99.7%
Template B	$N = 8$	0.313	0.390	0.696	85.0%	97.6%

Table 1 Baseline admissibility and vulnerability at horizon $T = 126$.

$W^{p01}_{\max}(\mathcal{Z})$ is the conservative admissible scale, the 1st percentile of $W_{\max}(z)$ across the stress set $\mathcal{Z}$. $W_{0.05}$ is the 5th percentile of $W_{\max}(z)$ across $z \in \mathcal{Z}$, and "Median" is the median of $W_{\max}(z)$. Probabilities report the fraction of stress states with $W_{\max}(z) < W$ evaluated at $W = 1$ and $W = 2$, i.e., at one and two days of ADV. The stress set contains $|\mathcal{Z}| = 10{,}000$ simulated states per configuration.

Scenario	Template A ($N = 1$)	Template B ($N = 2$)
baseline	0.166 (1.00x)	0.141 (1.00x)
$\sigma^* = 10\%$	0.208 (1.25x)	0.183 (1.30x)
$\sigma^* = 15\%$	0.138 (0.83x)	0.129 (0.91x)
$c_{1\%} = 8$ bps	0.249 (1.50x)	0.212 (1.50x)
$c_{1\%} = 20$ bps	0.100 (0.60x)	0.085 (0.60x)

Table 2 Comparative statics at horizon $T = 126$.

Entries are the conservative admissible scale $W^{p01}_{\max}(\mathcal{Z})$ (1st percentile of $W_{\max}(z)$ across $\mathcal{Z}$), varying target volatility σ^* and liquidity $c_{1\%}$; parentheses give the ratio to the within-template baseline.

Table 7, Panels C and D). Liquidity shifts are close to proportional: improving impact from $c_{1\%} = 12$ bps (baseline) to 8 bps increases $W^{p01}_{\max}(\mathcal{Z})$ by 50%, whereas deteriorating to $c_{1\%} = 20$ bps reduces $W^{p01}_{\max}(\mathcal{Z})$ by 40%.

Table 3 highlights how capacity varies with horizon and selected feedback stresses. In Template A, the conservative percentile falls monotonically as the manipulation horizon lengthens (0.213, 0.166, and 0.131 at $T = 63$, 126, and 252): a longer horizon gives the attacker more dates on which to move the sampled statistic and harvest the induced rebalancing, so the largest safe scale shrinks. More persistent volatility dynamics lower the bound at every horizon (to 0.139, 0.100, and 0.084 across the three horizons). In Template B, diversification and cross-impact jointly matter for capacity: for $N = 8$ at $T = 126$, increasing cross-impact from $\rho = 0$ to $\rho = 0.3$ lowers $W^{p01}_{\max}(\mathcal{Z})$ from 0.388 to 0.286. Internet Appendix B.5 reports the full grid of horizons and scenarios and provides diagnostic plots for the cross-state distribution of $W_{\max}(z)$ and related quantities.

7.4 Profitability

The admissibility test flags a state as inadmissible when a profitable round trip exists there. The test uses the linearized dynamics, and at an inadmissible state the linearization also gives the trade in question: the worst round-trip direction (Theorem 1, Proposition 8). We check that this trade is profitable in the exact model, not only in the linearization. Scaling the direction to a small trade and running it through the nonlinear simulator, the round trip earns a strict profit at 100% of inadmissible states, under every impact calibration and at each scale $W \in \{1, 2, 5\}$ days of ADV.

The linearization only shows that a profitable round trip exists; it does not say how large the

Scenario	$T = 63$	$T = 126$	$T = 252$
Template A: baseline	0.213	0.166	0.131
Template A: $\sigma^* = 10\%$	0.267	0.208	0.163
Template A: $c_{1\%} = 20$ bps	0.128	0.100	0.078
Template A: GARCH persistent	0.139	0.100	0.084
Template B ($N = 2$): baseline	0.198	0.141	0.109
Template B ($N = 2$): $\sigma^* = 10\%$	0.254	0.183	0.137
Template B ($N = 2$): $c_{1\%} = 20$ bps	0.119	0.085	0.065
Template B ($N = 8$): baseline	0.349	0.313	0.278
Template B ($N = 8$): cross-impact $\rho = 0$	0.438	0.388	0.342
Template B ($N = 8$): cross-impact $\rho = 0.3$	0.319	0.286	0.251

Table 3 Horizon and feedback-stress summary for conservative admissible scale $W^{p01}_{\max}(\mathcal{Z})$.
Entries are $W^{p01}_{\max}(\mathcal{Z})$, the 1st percentile of $W_{\max}(z)$ across $\mathcal{Z}$, in ADV-normalized units. Selected scenarios are shown; Internet Appendix B.5 reports $W^{p01}_{\max}(\mathcal{Z})$ for all scenarios and horizons.

			Return on avg. capital		Return on peak capital			Excess-cost split	
W	Inadm.	Profitable	Median	p90	Median	p90	Sharpe (ann.)	Transfer	Agg. exec. cost
1	8,338	100%	11.0%	20.8%	4.1%	7.1%	2.36	60%	40%
2	9,760	100%	21.6%	38.5%	7.9%	13.7%	2.17	13%	87%
5	9,999	100%	60.8%	123.8%	22.9%	42.9%	1.41	1%	99%

Table 4 Profitability of the optimal attack at vulnerable states (Template A baseline, $T = 126$).
"Inadm." is the number of inadmissible stress states at scale W ($W_{\max}(z) < W$, out of $|\mathcal{Z}| = 10{,}000$). "Profitable" is the share admitting a strictly profitable round trip, certified at full coverage. The return, Sharpe ratio, and cost-split statistics use a random sample of 600 inadmissible states per scale, on which the single-asset dynamic program of Section 5 is solved on the realized path under a 1%-of-ADV daily trade cap. Returns are annualized, on average and on peak inventory; each pair reports the cross-state median and 90th percentile, and the Sharpe ratio is on average capital. "Excess-cost split" divides the sector's expected excess cost ΔC^{MR} in (25) into the transfer to the attacker and the change in aggregate execution cost, the latter a deadweight loss to the extent impact reflects the real cost of liquidity provision (Section 3.2).

profit is. To measure it, we solve the optimal-attack dynamic program of Section 5. The stress grid holds $|\mathcal{Z}| = 10{,}000$ states. A state is inadmissible at scale W when its capacity falls short of the sector size, $W_{\max}(z) < W$; at the remaining states the sector is within capacity, so no profitable round trip exists and there is nothing to attack. The number of inadmissible states is therefore a subset of the 10,000, and it grows with W because the sets are nested: 8,338 states at $W = 1$, 9,760 at $W = 2$, and 9,999 at $W = 5$ (the "Inadm." column of Table 4). At each scale we draw a random sample of 600 of these inadmissible states, and for each we replay its realized path under a 1%-of-ADV daily trade cap and impose the terminal condition $x_T = 0$, over a $T = 126$ horizon. We sample rather than solve at every inadmissible state because the dynamic program is costly to solve, and 600 draws estimate the reported medians and quantiles with ample precision.

Returns are annualized and shown two ways: per unit of average inventory carried over the round trip, and per unit of peak inventory. The median return on average capital is 11.0% at $W = 1$, 21.6% at $W = 2$, and 60.8% at $W = 5$, with the 90th-percentile at 124% ($W = 5$). On peak capital the medians are 4.1%, 7.9%, and 22.9%, about 40% as large, because the attack carries inventory

only briefly. The annualized Sharpe ratio runs from 2.36 at $W = 1$ to 1.41 at $W = 5$. The attack holds almost no inventory: mean peak inventory is 0.05 to 0.06 days of ADV even at $W = 5$, and mean inventory over the round trip is roughly half that. This is the spike-and-revert mechanism of Section 7.2.3. The magnitudes are robust to how impact decays over time: three propagator calibrations (transient impact with 21- and 63-day half-lives, and permanent impact) give median returns on average capital from 7 to 9% at $W = 1$ up to 32 to 43% at $W = 5$, and a profitable round trip still exists at 100% of inadmissible states under each. Internet Appendix B.6 reports the full grid and a representative optimal trade path.

The attack raises the mechanically rebalancing sector's execution cost by ΔC^{MR} in expectation, and this excess cost divides into two parts (Section 3.2): a transfer, equal to the attacker's profit, and the attack-induced rise in aggregate execution cost, a deadweight loss to the extent impact reflects the real cost of liquidity provision. The split is shown in equation (25): in expectation, the sector's excess cost minus the attacker's profit equals the rise in aggregate execution cost relative to no attack. The aggregate-execution-cost share grows with sector size, from 40% at one day of ADV to 87% at two and 99% at five, and the transfer share falls from 60% to 1% (last two columns of Table 4). Two distinct channels drive the shift. A small sector's excess cost is mostly transfer: the flow the attack adds is small at small scale, so the extra impact it creates is second-order, and the transfer to the attacker is the bulk of the excess cost. Leaning against the sector's predetermined forced flow does not by itself raise its cost; the sector's excess cost comes from the flow the attack adds. A large sector is harmed mainly through flow the attacker adds: a bigger sector lets the attacker inflate the rule's requirement and enlarge the forced flow, and that added flow pays quadratic impact. At scale, the excess cost is borne overwhelmingly as a higher aggregate execution cost rather than as a transfer to the attacker, a deadweight loss the rule imposes on its own participants to the extent impact reflects real liquidity-provision costs.

8 Conclusion

We provide a bound on how large a sector following a disclosed price-based rule can grow before its own rule-induced trading makes the market manipulable. Once today's transaction prices determine tomorrow's mandated trades, the classical no-manipulation conditions for market impact are no longer sufficient. Theorem 1 gives the sufficient condition as an eigenvalue test on the impact-cost matrix, enlarged to internalize rule-induced flow. It takes only the disclosed rule and a market-impact calibration, returns the largest sector size consistent with non-manipulability, and needs no holdings, dealer inventories, or ownership data, so a supervisor can run it using public data alone.

Applied to volatility-controlled indices in retail structured products and indexed annuities, we find that the safe scale is small: fractions of one day of ADV, as low as 0.14% of the underlying market. At one day of volume the condition fails in 83% to 98% of stress states, at two days in more than 97%. The optimal attack earns median annualized returns from 11.0% at one day of ADV to 60.8% at five, through a spike-and-revert strategy that inflates measured volatility while warehousing little inventory. The attacker captures a shrinking share of the harm, from 60% at one day of volume to 1% at five, the rest being an induced rise in aggregate execution cost, so enforcing the bound removes

not only the manipulator's rent but the induced execution cost as well, a social loss to the extent impact reflects real liquidity-provision costs. Fragility need not be greatest in volatile markets. Under inverse-volatility scaling the bound is often tightest at tranquil, high-exposure states. This rule-based paradox differs from leverage-cycle amplification: requirements tighten not because collateral values fall or funding worsens, but because a sampled risk statistic can itself be moved by trading, with no change in fundamentals or intermediary capital.

While we have applied it to volatility-managed portfolios, the test is much more general and covers any disclosed rule that maps past transaction prices into future binding demand: margin schedules, collateral haircuts, leverage limits, drawdown and value-at-risk triggers, and target-weight, momentum, or reversal mandates. For each it returns the largest safe scale, the share of states at which a given scale is already manipulable, and the trade that exploits it. To determine whether a market is viable, we have abstracted from modelling the manipulator and from detection and enforcement, which remain open questions for future work.

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Internet Appendix

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How Big Can Mechanical Rebalancing Get?

A Further theory results and proofs

This appendix develops the theory behind the capacity test and collects the proofs of the main-text results. We reduce the no-manipulation condition to a single eigenvalue (Appendix A.2) and turn it into a closed-form, conservative ceiling on the size of the constrained sector (Appendix A.3); Appendix A.5 shows the test is not an artifact of the linearization, and Appendices A.6 and A.7 collect the proofs. We assume requirements are set on the previous day's prices, $M_{t+1} = g(\Gamma_t)$, so a trade moves the constrained sector's feasibility only from the next day on and never feeds back on itself within a day (Lemma 5). Same-day updating $M_t = g(\Gamma_t)$ would instead let a trade move the contemporaneous prices, and the feedback could then blow up, arbitrarily small trades moving flows and prices without bound. We start by recording the local regularity conditions under which the linearization of Section 4 is valid, referenced there and used in Definition 4 below.

Assumption 6 (Local Regularity for Linearization). *Fix the reference configuration used for the linearization in Section 4. There exists an open neighborhood of the reference transaction-price path (equivalently, the wedge-price path since $\delta S = 0$) such that:*

(i) The statistic map $\Gamma(P)$ is differentiable at the reference path, so the Jacobian J used in the linearization exists; the schedule g is differentiable at the relevant statistic value, so $g'(\Gamma)$ is well defined. If the constraint has nondifferentiable pieces (caps, floors, turnover buffers, window max or min), the linearizations are taken at interior points where the active branch is locally fixed and the defining inequalities are slack; at a kink, replacing J and $g'(\Gamma)$ by a one-sided derivative or subgradient selection gives a conservative version of the screen.

(ii) The reference configuration is a binding steady state with zero baseline flow: at the reference, the binding requirement is constant across dates, the mechanically rebalancing sector's reference forced flow is zero, and the strategic trader's reference strategy is no trade. Deviations from the reference therefore coincide with levels: $q = u + v$ is total order flow and the execution wedge is $P - S = \mathcal{I}q$.

Definition 4 (Feedback manipulation and quasi-arbitrage). *Under Assumptions 1, 5, and 6, let $\mathcal{U}_T$ denote the deterministic admissible strategies of Definition 1 (each uniformly bounded by its own constant); the multi-asset case is identical, with Π and $\mathcal{U}_T$ replaced by their stacked N-asset analogues under Assumptions 3 and 4.*

(i) A feedback manipulation is an admissible round trip $u \in \mathcal{U}_T \cap \mathcal{R}_T$ with $\mathbb{E}[\Pi(u)] > 0$.

(ii) A feedback quasi-arbitrage is a sequence $\{u^n\}_{n\geq 1} \subset \mathcal{U}_T \cap \mathcal{R}_T$ with $\mathbb{E}[\Pi(u^n)] \to \infty$ and

$$\frac{\mathbb{E}[\Pi(u^n)]}{\sqrt{\mathbb{V}\mathrm{ar}(\Pi(u^n))}} \to \infty.$$

The notion is from Huberman and Stanzl (2004): scaling a positive-mean round trip generates a sequence with diverging expected profits and Sharpe ratios.

Lemma 5 (Invertibility of $I + \mathcal{K}$ under lagged updating). *Fix a horizon T and stack time-indexed paths in chronological order as column vectors $x = (x_0, \ldots, x_{T-1})^\top$. Suppose $M_{t+1} = g(\Gamma_t)$, and forced flow is the mechanically rebalancing sector's net trade $v_t = X_t - X_{t-1}$ with $X_t = X(M_t)$. Assume additionally that (i)*

the linearized impact operator $\mathcal{I}$ is such that its $T \times T$ matrix representation satisfies $\mathcal{I}_{t,s} = 0$ for all $s > t$, and (ii) Γ_t depends only on prices up to date t, so its Jacobian $J_{t,s} = 0$ for all $s > t$. Then $\mathcal{K}$ satisfies $\mathcal{K}_{t,s} = 0$ for all $s \geq t$, i.e., it is strictly lower triangular. Then $I + \mathcal{K}$ is invertible on $\mathbb{R}^T$, and

$$(I + \mathcal{K})^{-1} = \sum_{n=0}^{T-1} (-\mathcal{K})^n.$$

Proof. The lag operator L is strictly lower triangular, and D and $J\mathcal{I}$ are lower triangular. Hence $\mathcal{K} = BsDLJ\mathcal{I} = Bs\,(DL)(J\mathcal{I})$, a product of lower-triangular factors one of which is strictly lower triangular times the positive scalars B and s, is strictly lower triangular, and its T-th matrix power vanishes, $\mathcal{K}^T = 0$ (here and below the superscript on $\mathcal{K}$ denotes a matrix power, not a transpose). Then

$$(I + \mathcal{K}) \sum_{n=0}^{T-1} (-\mathcal{K})^n = I - (-\mathcal{K})^T = I,$$

so $I + \mathcal{K}$ is invertible with inverse the displayed finite sum (Neumann series; Sargent and Stachurski, 2022, Theorem 1.2.5). □

A.1 Permanent and transient impact

The impact kernel splits into a permanent level and a transient part,

$$G_\ell = G_\infty + g_\ell, \qquad G_\infty = \lim_{\ell \to \infty} G_\ell \geq 0, \qquad g_\ell \to 0.$$

The permanent level is a pure cost to any round trip, so viability is decided by the shape of g_ℓ. Aggregated over a large metaorder, the same g_ℓ delivers the square-root price impact seen in the data.

Proposition 5 (Permanent impact is never a source of manipulation)**.** *The permanent level adds $\frac{1}{2}\eta G_\infty \|u\|^2 \geq 0$ to the cost of every round trip $u \in \mathcal{R}_T$, so it can never fund a profitable one. Any manipulation must come from the transient part g_ℓ.*

Proof. Under the permanent level the execution price carries a wedge proportional to the position built so far,

$$P_t - S_t = \eta G_\infty \, x_t, \qquad x_t = \sum_{s \leq t} u_s.$$

The trader pays this wedge on every order, and the unaffected price contributes nothing over a round trip (Lemma 1), so the permanent part of its cost is $\eta G_\infty \sum_t u_t x_t$. Summing by parts,

$$\sum_{t=0}^{T-1} u_t x_t = \tfrac{1}{2}\big(x_{T-1}^2 + \|u\|^2\big) = \tfrac{1}{2}\|u\|^2,$$

since a round trip ends flat, $x_{T-1} = 0$. The permanent cost is therefore $\frac{1}{2}\eta G_\infty \|u\|^2 > 0$: the trader accumulates into its own rising price and unwinds into its own falling price, always on the wrong side. This is the linear permanent-impact benchmark of Huberman and Stanzl (2004). □

The round-trip cost is the quadratic form

$$\mathbb{E}\Big[\sum_t u_t P_t\Big] = \tfrac{1}{2}\,\eta\, u^\top (K + K^\top) u, \qquad (Ku)_t = \textstyle\sum_{s\le t} G_{t-s} u_s,$$

and a round trip profits when it is negative for some $u \in \mathcal{R}_T$ (Proposition 1). With two dates, $u = (a, -a)$ and kernel (G_0, G_1), the cost is $\eta(2G_0 - G_1)a^2$, negative iff $G_1 > 2G_0$: lagged impact above twice the instantaneous, which a decaying kernel ($G_1 \le G_0$) never produces. More generally, no round trip profits when g_ℓ is nonincreasing and convex, the discrete counterpart of the convex-decay condition of Alfonsi et al. (2012) and Gatheral et al. (2012). Geometric decay $g_\ell \propto \rho^\ell$ qualifies, so $G_\ell = c + (1-c)\rho^\ell$ is manipulation free for every c.

Proposition 6 (Square-root impact of large orders)**.** *With a power-law kernel $G_\ell = (1+\ell)^{-\gamma}$, $\gamma \in (0,1)$, an order of size Q executed at a constant rate q over $N = Q/q$ dates has peak price impact and average cost per share of order $Q^{1-\gamma}$. At $\gamma = \frac{1}{2}$ this is the square-root law; the one-period impact remains linear.*

Proof. Execute $u_s = q$ for $s < N$. The displacement $D_t = P_t - S_t$ builds through execution and decays after, peaking at $t = N-1$, where $D_{N-1} = \eta q \sum_{\ell=0}^{N-1} G_\ell$. For $\gamma \in (0,1)$ the partial sums grow as $\sum_{\ell=0}^{N-1}(1+\ell)^{-\gamma} \sim N^{1-\gamma}/(1-\gamma)$. Hence $D_{N-1} \sim \frac{\eta}{1-\gamma}\, q\, N^{1-\gamma}$, and with $N = Q/q$ at fixed q, $D_{\text{peak}}(Q) \sim \frac{\eta q^\gamma}{1-\gamma}\, Q^{1-\gamma}$. The average cost per share obeys the same power law. Since $(Ku)_t = q\,\Sigma_{t+1}$ with $\Sigma_k = \sum_{\ell<k} G_\ell \sim k^{1-\gamma}/(1-\gamma)$, the cost is $\eta\, u^\top K u = \eta q^2 \sum_{t<N} \Sigma_{t+1}$, and the same asymptotic gives $\sum_{t<N} \Sigma_{t+1} \sim N^{2-\gamma}/[(1-\gamma)(2-\gamma)]$, so

$$\frac{\eta\, u^\top K u}{Q} \sim \frac{\eta q^\gamma}{(1-\gamma)(2-\gamma)}\, Q^{1-\gamma}.$$

Both are of order $Q^{1-\gamma}$, equal to $\sqrt{Q}$ at $\gamma = \frac{1}{2}$, even though they have been generated by the one-period impact $f(q) = \eta q$, which is linear. □

That a linear propagator with decay reproduces concave, approximately square-root impact is well known in the propagator-model literature (Bouchaud et al., 2004; Donier et al., 2015).

A.2 Eigenvalue tests

Theorem 1 rules out manipulation when $u^\top \widehat{H} u \ge 0$ for every $u \in \mathcal{R}_T$. We compress this into a single number: the worst-case value over unit-norm round trips (Boyd and Vandenberghe, 2004, Section A.5.2).

Theorem 3. *Under the assumptions of Theorem 1 and with $\widehat{H}$ as in (34), define*

$$\lambda_{\min}^{\mathcal{R}_T}(\widehat{H}) = \inf\left\{ u^\top \widehat{H} u :\ u \in \mathcal{R}_T,\ \|u\|_2 = 1 \right\}.$$

Then $\widehat{H}$ is positive semidefinite on $\mathcal{R}_T$ if and only if $\lambda_{\min}^{\mathcal{R}_T}(\widehat{H}) \ge 0$. The same holds in the multi-asset model after replacing $\mathcal{R}_T$ by $\mathbf{R}_T$ and $\widehat{H}$ by $\widehat{\mathbf{H}}$.

Proof. Positive semidefiniteness on $\mathcal{R}_T$ is equivalent to $u^\top \widehat{H} u \ge 0$ for every unit-norm $u \in \mathcal{R}_T$, hence to $\lambda_{\min}^{\mathcal{R}_T}(\widehat{H}) \ge 0$. □

A.3 A closed-form capacity bound

The exact test of Theorem 3 requires inverting $I + \mathcal{K}$. Under temporary impact the test reduces to the most negative eigenvalue of $\mathcal{K} + \mathcal{K}^\top$, without requiring an inversion.

Proposition 7 (Closed-form capacity bound). *Suppose $\mathcal{I} = \eta\, I_T$ with $\eta > 0$ and $I + \mathcal{K}$ is invertible (Lemma 5). Then $\widehat{H}$ in (34) is positive semidefinite on all of $\mathbb{R}^T$ if and only if*

$$\lambda_{\min}\big(\mathcal{K} + \mathcal{K}^\top\big) \geq -2. \tag{65}$$

When (65) holds, $\widehat{H}$ is also positive semidefinite on $\mathcal{R}_T$, so by Theorem 1 the combined system is feedback manipulation free. Because $B = W/M_0^2$ is proportional to W, the feedback operator factors as $\mathcal{K} = W\,\mathcal{K}_1$ with $\mathcal{K}_1$ independent of W, and (65) holds for every

$$W \leq W_{\text{loop}} = \frac{2}{\big|\lambda_{\min}\big(\mathcal{K}_1 + \mathcal{K}_1^\top\big)\big|}, \tag{66}$$

with $W_{\text{loop}} = +\infty$ when $\mathcal{K}_1 + \mathcal{K}_1^\top$ is positive semidefinite. Because the condition is necessary and sufficient on $\mathbb{R}^T$, W_{loop} is the exact capacity of the unrestricted test, and the full-space admissible set is $[0, W_{\text{loop}}]$. It is a closed-form floor for the statewise capacity of Section 7.3, $W_{\text{loop}} \leq W_{\max}(z)$, and needs one symmetric eigenvalue rather than a search over the restricted eigenvalue.

Proof. Because $I + \mathcal{K}$ is invertible, $q = (I + \mathcal{K})^{-1}u$ is a bijection of $\mathbb{R}^T$; write $u = (I + \mathcal{K})q$. Since $\widehat{H} = \widetilde{\mathcal{I}} + \widetilde{\mathcal{I}}^\top$, the scalar $u^\top \widehat{H} u$ equals $2\, u^\top \widetilde{\mathcal{I}} u$, and $\widetilde{\mathcal{I}} = \eta(I + \mathcal{K})^{-1}$ gives

$$\tfrac{1}{2}\, u^\top \widehat{H} u = \eta\, u^\top (I + \mathcal{K})^{-1} u = \eta\, q^\top (I + \mathcal{K})^\top q = \eta\big(\|q\|^2 + q^\top \mathcal{K}^\top q\big).$$

As $q^\top \mathcal{K}^\top q$ is a scalar it equals $\frac{1}{2}\, q^\top (\mathcal{K} + \mathcal{K}^\top) q$, so under (65)

$$\tfrac{1}{2}\, u^\top \widehat{H} u = \eta\big(\|q\|^2 + \tfrac{1}{2}\, q^\top (\mathcal{K} + \mathcal{K}^\top) q\big) \geq \eta\big(1 + \tfrac{1}{2}\lambda_{\min}(\mathcal{K} + \mathcal{K}^\top)\big)\|q\|^2 \geq 0.$$

Because q ranges over all of $\mathbb{R}^T$ as u does, the inequality is not only sufficient: $\widehat{H}$ is positive semidefinite on $\mathbb{R}^T$ if and only if $I + \frac{1}{2}(\mathcal{K} + \mathcal{K}^\top)$ is p.s.d., which is (65), and positive semidefiniteness on $\mathbb{R}^T$ implies it on $\mathcal{R}_T$. The term $\eta\|q\|^2$ is the trader's own impact cost, always nonnegative. The term $\frac{1}{2}\eta\, q^\top (\mathcal{K} + \mathcal{K}^\top) q$ is the feedback contribution to cost; it is negative, and the round trip gains from the forced flow in directions along which that flow leans in the trader's favor. A profitable round trip requires this contribution to be negative and larger in magnitude than the impact cost, $q^\top (\mathcal{K} + \mathcal{K}^\top) q < -2\|q\|^2$ for some q, the negation of (65). For the scaling, $\mathcal{K} = Bs\, DLJ\mathcal{I}$ in (31) with $B = W/M_0^2$, so $\mathcal{K} = W\,\mathcal{K}_1$; the eigenvalues of $\mathcal{K} + \mathcal{K}^\top = W(\mathcal{K}_1 + \mathcal{K}_1^\top)$ scale linearly in $W \geq 0$, and (65) is equivalent to $W\,\big|\lambda_{\min}(\mathcal{K}_1 + \mathcal{K}_1^\top)\big| \leq 2$. □

The threshold -2 is a harvesting break-even. Myopic harvesting recovers half of any forced flow it leans against (Lemma 3), so a trigger pays for itself only once the induced flow reaches twice the trade that generates it. Condition (65) is the finite-horizon form of the two-date no-manipulation condition $k \leq 2$, under which the augmented cost matrix $\widehat{H} = \widetilde{\mathcal{I}} + \widetilde{\mathcal{I}}^\top$ is positive semidefinite.

Computing $W_{\max}(z)$ inverts $I + W\mathcal{K}_1$ and searches over W; because $\mathcal{K} = W\mathcal{K}_1$ enters linearly, W factors out and the bound needs only the single W-independent eigenvalue $\lambda_{\min}(\mathcal{K}_1 + \mathcal{K}_1^\top)$, with no inversion and no search. It holds only for temporary impact, $\mathcal{I} = \eta I_T$, where the trader's own cost is the same in every direction. Under a propagator that cost depends on the direction of attack, the closed form no longer applies, and we use the exact test $W_{\max}(z)$.

Table 5 compares the closed-form bound $W_{\text{loop}}(z)$ of (66) with the exact capacity $W_{\max}(z)$ on the calibrated stress set of Internet Appendix B.4. By Proposition 7, W_{loop} is the exact capacity of the full-space test, so the gap $W_{\max} - W_{\text{loop}}$ is the slack from restricting attention to round trips. On this set that gap is negligible: the cross-state median ratio $W_{\text{loop}}/W_{\max}$ is one and the two conservative first-percentile capacities coincide. The round-trip restriction is nearly irrelevant because the rolling-window forced flow reverts within the horizon, so the worst full-space direction is close to a round trip. The only visible conservatism is at the shortest horizon, where the worst state is about ten percent below the exact bound.

Scenario	T	median	p05	min	$W_{\max}^{p01}$	W_{loop}^{p01}
Baseline	63	1.000	0.993	0.895	0.213	0.213
Baseline	126	1.000	1.000	0.990	0.166	0.166
Baseline	252	1.000	1.000	0.996	0.131	0.131
$c_{1\%} = 20$ bps	126	1.000	1.000	0.990	0.100	0.100
$\sigma^* = 10\%$	126	1.000	1.000	0.990	0.208	0.208
Window 20 only	126	1.000	1.000	0.987	0.122	0.122
GARCH persistent	126	1.000	0.999	0.985	0.100	0.100

Table 5 The closed-form capacity bound against the exact capacity.

For each stress state z we compute the exact statewise capacity $W_{\max}(z)$ from the restricted-eigenvalue test of Theorem 3 and the capacity bound $W_{\text{loop}}(z) = 2/|\lambda_{\min}(\mathcal{K}_1(z)+\mathcal{K}_1(z)^\top)|$ of Proposition 7, on the same per-state operator $\mathcal{K}_1$. Columns report the cross-state median, fifth percentile, and minimum of $W_{\text{loop}}/W_{\max}$, and the conservative first-percentile capacities $W_{\max}^{p01}$, W_{loop}^{p01} in days of ADV. The stress set has 10,000 states per cell, matching the main-table stress sets, so $W_{\max}^{p01}$ reproduces the capacity reported there; $W_{\text{loop}} \le W_{\max}$ at every state up to the search tolerance.

A.4 Date-varying sensitivities

The slope $s = g'(\Gamma_0)$ and the demand sensitivity $B = W/M_0^2$ are local objects, evaluated at the binding reference. When the binding requirement is identical across dates they are constants and the feedback operator $\mathcal{K}$ in the main text is exact. Along a realized path, however, the requirement M_t and the lagged statistic that sets it vary by date, so the quantitative implementation evaluates the sensitivities date by date, replacing the scalars by the diagonal operators

$$\mathbf{S} = \operatorname{diag}(s_t), \quad s_t = g'(\Gamma_{t-1-\ell}), \qquad \mathbf{B} = \operatorname{diag}\big(W/M_t^2\big),$$

so that $\delta X = -\mathbf{B}\,\delta M$ and the forced-flow response is $v = -D\,\mathbf{B}\,\mathbf{S}\,L^{\ell+1}\,J\,\delta\Delta P$, with feedback operator $\mathcal{K} = D\,\mathbf{B}\,\mathbf{S}\,L^{\ell+1}\,J\,\mathcal{I}$; here $\ell \ge 0$ is the implementation lag and the diagonal entry $s_t = g'(\Gamma_{t-1-\ell})$ is indexed by the requirement date t (under the baseline timing $\ell = 0$ the lag power reduces to the

single L of the main text, and the slope of Internet Appendix B.3, indexed there by the trigger date, is the same object shifted by one date). The constant-B, constant-s expressions in (29)–(31) are the special case of a flat binding reference. Evaluating B at the date-t requirement $M_t = \widehat{\sigma}_{t-1-\ell}/\sigma^*$ produces the $1/\widehat{\sigma}^3$ scaling that drives the calm-market fragility.

A.5 Robustness to nonlinearity

Theorem 1 is derived in a model linearized around a binding configuration. This appendix shows that, wherever the capacity test flags a state as unsafe, a genuinely profitable manipulation exists in the exact nonlinear model. We linearize the rule primitives (Γ, g, X); nonlinearity of the price-impact is a separate question.

Proposition 8 (A failed test implies a real profitable round trip)**.** *Fix a binding reference state (z, W) at which the rule primitives (Γ, g, X) are locally smooth (no cap or floor switching) and $I + \mathcal{K}(z; W)$ is invertible. Suppose the linearized test fails: some admissible round trip $\bar{U}$ has $\bar{U}^\top \widehat{H}(z; W)\, \bar{U} < 0$. Then, for all small enough $\varepsilon > 0$, the scaled round trip $\pm\varepsilon\bar{U}$ is admissible in the exact nonlinear model and, for at least one sign, earns strictly positive expected profit, at least*

$$-\tfrac{1}{2}\,\varepsilon^2\,\bar{U}^\top \widehat{H}(z; W)\,\bar{U} + o(\varepsilon^2) > 0,$$

with equality for the direction $\bar{U}$ itself when the reference has zero baseline flow. The multi-asset statement is identical with $\widehat{\mathbf{H}}$ in place of $\widehat{H}$.

Proof. By the implicit function theorem, local smoothness and invertibility of $I + \mathcal{K}$ give a unique smooth nonlinear flow path $q(U)$ near the baseline q^0, with $q(\pm\varepsilon\bar{U}) = q^0 \pm \varepsilon(I + \mathcal{K})^{-1}\bar{U} + o(\varepsilon)$. The unaffected price contributes zero in expectation (Lemmas 1 and 2), so $\mathbb{E}[\Pi(\pm\varepsilon\bar{U})] = \mp\varepsilon\,\bar{U}^\top\mathcal{I}q^0 - \varepsilon^2\,\bar{U}^\top\mathcal{I}(I + \mathcal{K})^{-1}\bar{U} + o(\varepsilon^2)$. Averaging over the two signs cancels the linear term, and $\bar{U}^\top\mathcal{I}(I + \mathcal{K})^{-1}\bar{U} = \frac{1}{2}\bar{U}^\top\widehat{H}\bar{U}$ by (34) leaves the displayed quadratic, which is positive; the larger sign is at least the average, and the linear term vanishes when $q^0 = 0$. □

A.5.1 Numerical validation and full-coverage existence

We verify Proposition 8 on two fronts: the leading-order accuracy of $\widehat{H}$ as the Hessian of realized profit, and the existence of a profitable round trip at every inadmissible state.

Leading-order accuracy. At a random sample of eight vulnerable baseline states ($T = 126$, $W_{\max}(z) < 1$), we scale round trips to a peak daily trade of 10^{-5} of ADV and compare the simulator's sign-symmetrized profit to the operator prediction $-\frac{1}{2}\varepsilon^2\bar{U}^\top\widehat{H}\bar{U}$ (Table 6). At the six states with ordinary curvature the ratio of realized to predicted profit is close to one, 0.91 to 1.11. At the two most fragile states ($W_{\max} \approx 0.11$ and 0.13) the curvature $\widehat{H}$ is about a thousand times larger, so even a trade this small already induces enough forced flow to hit the rule's nonlinearities: the leverage cap binds and the response saturates. The linear prediction ignores that saturation, so it overstates the profit and the ratio falls toward zero.

Full-coverage existence. The existence claim needs only the sign of realized profit. For each z with $W_{\max}(z) < W$ we take the critical eigenvector $\bar{U}$ of $\widehat{H}(z; W)$ and evaluate the exact nonlinear sign-symmetrized profit of $\pm\varepsilon\bar{U}$ over a ladder of scales ε (from 10^{-4} to 10^{-1} of the ADV cap). Across four calibrations (Template A and three impact propagators) and three scales $W \in \{1, 2, 5\}$, a strictly profitable round trip exists at all 117,470 inadmissible states, with no exceptions, and $\lambda_{\min}^{\mathcal{R}_T}(\widehat{H}) < 0$ at every one. The optimal-attack program of Section 7.4 then quantifies how large those profits are (Tables 4 and 10).

$W_{\max}(z)$	0.11	0.13	0.35	0.39	0.39	0.66	0.67	0.91
realized / predicted	0.00	0.00	1.11	0.91	0.94	0.99	1.02	0.99
optimal attack profitable	yes	yes	yes	yes	yes	yes	yes	yes

Table 6 Numerical validation of Proposition 8.

Ratio of the simulator's realized (sign-symmetrized) round-trip profit to the linearized prediction $-\frac{1}{2}\varepsilon^2\bar{U}^\top\widehat{H}\bar{U}$, at a random sample of eight vulnerable baseline states ($T = 126$, $W_{\max}(z) < 1$), for round trips scaled to a peak daily trade of 10^{-5} of ADV (median over five random mean-zero directions per state). At the two most fragile states the same trade size exceeds the local linearization radius and the realized response saturates (see text). The last row reports whether the optimal attack of Section 7.4 earns strictly positive profit at the state ($W = 1$).

As a round trip is scaled up, expected profit grows quadratically but the random price component only linearly, so the linearized Sharpe ratio diverges (the quasi-arbitrage of Definition 4). However, in the exact model, the realized Sharpe ratio is bounded: flat at small scale, then declining once the leverage cap and convex impact costs bite – consistent with the finite cross-state values in Section 7.4.

A.6 Proofs for the single-asset model

A.6.1 Stochastic dynamic program

Sections 5.1–6.1 condition on a deterministic unaffected-price path. The program extends to stochastic prices without changing its structure. If the unaffected price is driven by a Markov state Z_t that is $\mathcal{F}_{t-1}$-measurable, augment the state to $\tilde{s}_t = (s_t, Z_t)$; the Bellman recursion and the within-period first-order conditions of Propositions 3 and 11 carry over unchanged, with the shadow values μ_{t+1} and ν_{t+1} replaced by conditional expectations of the next-period marginal values given $\tilde{s}_t$. The Euler equations remain $\eta(2u_t^\star + v_t) = \mu_{t+1} + \eta\nu_{t+1}$ in the single-asset case and $A(2u_t^\star + v_t) = \mu_{t+1} + A\nu_{t+1}$ in the multi-asset case.

A.6.2 Proof of Proposition 3

Proof. Fix s_t and set $\mu_{t+1} = \partial_x V_{t+1}(s_{t+1}^\star)$, $\nu_{t+1} = \frac{d}{dP_t}V_{t+1}(s_{t+1}^\star)$, where $s_{t+1}^\star = \mathcal{T}(s_t, u_t^\star)$ and $\frac{d}{dP_t}$ is the total derivative through all components of s_{t+1} depending on P_t. By (41), $\partial_u \pi_t(s_t, u) = -\eta(2u + v_t)$, and u enters $\mathcal{T}(s_t, u)$ only through $x_{t+1} = x_t + u$ and $P_t = S_t + \eta(u + v_t)$. Therefore, by the chain rule,

$$\frac{d}{du}V_{t+1}(\mathcal{T}(s_t, u)) = \partial_x V_{t+1}(s_{t+1}) \cdot \frac{dx_{t+1}}{du} + \frac{d}{dP_t}V_{t+1}(s_{t+1}) \cdot \frac{dP_t}{du} = \mu_{t+1} + \eta\nu_{t+1},$$

evaluated at $u = u_t^\star$. At an interior optimum,

$$0 = \frac{d}{du}\Big[\pi_t(s_t, u) + V_{t+1}(\mathcal{T}(s_t, u))\Big]_{u=u_t^\star} = -\eta(2u_t^\star + v_t) + \mu_{t+1} + \eta\nu_{t+1},$$

which gives (48) and hence (49). Finally, (50) follows from Lemma 3 and the identity

$$u_t^\star + \tfrac{v_t}{2} = \frac{1}{2\eta}\big(\mu_{t+1} + \eta\nu_{t+1}\big),$$

which is obtained by rearranging (48). □

A.6.3 Trigger dates versus harvest dates

Corollary 3 (Trigger dates versus harvest dates)**.** *Fix a date t with $v_t = 0$. Lemma 3 implies $\pi_t(s_t, u) \le 0$ for all u. Assume $0 \in \mathcal{U}_t(x_t)$ (equivalently, $|x_t| \le (T-1-t)\bar{u}$), so that $u = 0$ is feasible. Then any optimal action with $u_t^\star \neq 0$ must be chosen for its effect on continuation value in (45), since it cannot raise the current-period wedge payoff. If, in addition, the conditions of Proposition 3 hold at $(s_t, u_t^\star)$, then the decomposition (49) reduces to*

$$u_t^\star = \frac{1}{2\eta}\mu_{t+1} + \frac{1}{2}\nu_{t+1},$$

so any nonzero trade is driven by the inventory and/or rule-manipulation motives. Conversely, at any interior optimum with $\mu_{t+1} = 0$ and $\nu_{t+1} = 0$, the optimal trade reduces to pure harvesting, $u_t^\star = -v_t/2$.

Proof. Since $0 \in \mathcal{U}_t(x_t)$, $u = 0$ is admissible. If $v_t = 0$, Lemma 3 gives $\pi_t(s_t, u) \le 0 = \pi_t(s_t, 0)$, so optimality requires

$$\pi_t(s_t, u_t^\star) + V_{t+1}(\mathcal{T}(s_t, u_t^\star)) \ge V_{t+1}(\mathcal{T}(s_t, 0)),$$

i.e., $u_t^\star \neq 0$ only if the continuation term outweighs the contemporaneous loss $\pi_t(s_t, u_t^\star) \le 0$. Under Proposition 3, setting $v_t = 0$ in (49) gives the reduction; setting $\mu_{t+1} = \nu_{t+1} = 0$ in addition gives $u_t^\star = -v_t/2$. □

A.6.4 Procyclical trading with forced flow

Corollary 4. *Fix $t \in \{0, \dots, T-1\}$ and suppose the conditions of Proposition 3 hold at $(s_t, u_t^\star)$. Define*

$$\chi_{t+1} = \mu_{t+1} + \eta\nu_{t+1}.$$

(i) If $v_t \neq 0$, then $u_t^\star v_t > 0$ if and only if $\operatorname{sign}(\chi_{t+1}) = \operatorname{sign}(v_t)$ and $|\chi_{t+1}| > \eta|v_t|$.

(ii) The one-period payoff at the optimal action satisfies $\pi_t(s_t, u_t^\star) < 0$ if and only if $|\chi_{t+1}| > \eta|v_t|$.

Proof. From (49), $u_t^\star = -v_t/2 + \chi_{t+1}/(2\eta)$. If $v_t \neq 0$, then $u_t^\star v_t > 0$ is equivalent to

$$\left(-\frac{v_t}{2} + \frac{\chi_{t+1}}{2\eta}\right) v_t > 0 \quad \Longleftrightarrow \quad \chi_{t+1} v_t > \eta v_t^2,$$

which holds if and only if $\operatorname{sign}(\chi_{t+1}) = \operatorname{sign}(v_t)$ and $|\chi_{t+1}| > \eta|v_t|$. For (ii), the payoff identity (50)

implies

$$\pi_t(s_t, u_t^\star) = \frac{\eta}{4} v_t^2 - \frac{1}{4\eta} \chi_{t+1}^2,$$

which is negative if and only if $|\chi_{t+1}| > \eta |v_t|$. □

A.6.5 The rebalancing rule and the manipulation motive

Proposition 3 identifies ν_{t+1} as the continuation-value gain from moving the current transaction price through the rule update. The following result expresses ν_{t+1} in terms of the rule primitives (F, Γ, g) and derivatives of the value function, making transparent which parts of the rule generate a manipulation incentive.

Proposition 9 (Decomposition of ν_{t+1})**.** *Fix $t \in \{0, \dots, T-1\}$ and consider any state-action pair (s_t, u_t). Let P_t be given by (40) and let $s_{t+1} = \mathcal{T}(s_t, u_t)$. Assume V_{t+1} is differentiable at s_{t+1}, F is differentiable in its price argument at (y_t, P_t), and g and Γ are differentiable at $\Gamma_t = \Gamma(y_t, P_{t-1}, P_t)$ and in P_t at (y_t, P_{t-1}, P_t), respectively. Write $B_{t+1} = W/M_{t+1}^2$ and define*

$$\Lambda_{t+1} = \partial_M V_{t+1}(s_{t+1}) - B_{t+1}\, \partial_X V_{t+1}(s_{t+1}), \tag{67}$$

the shadow value of tightening the requirement when the mechanically rebalancing sector remains on $X_{t+1} = W/M_{t+1}$. Then $\nu_{t+1} = \frac{d}{dP_t} V_{t+1}(s_{t+1})$ satisfies

$$\nu_{t+1} = \partial_P V_{t+1}(s_{t+1}) + \big(\nabla_y V_{t+1}(s_{t+1})\big) \cdot \partial_P F(y_t, P_t) + \Lambda_{t+1}\, g'(\Gamma_t)\, \partial_P \Gamma(y_t, P_{t-1}, P_t). \tag{68}$$

Proof. View V_{t+1} as a differentiable function of the next state

$$s_{t+1} = (x_{t+1}, y_{t+1}, P_t, M_{t+1}, X_{t+1}, X_t).$$

When differentiating with respect to P_t, the component $x_{t+1} = x_t + u_t$ is fixed. The components that depend on P_t are the stored price P_t itself, $y_{t+1} = F(y_t, P_t)$, and (M_{t+1}, X_{t+1}), where

$$M_{t+1} = g(\Gamma_t), \qquad \Gamma_t = \Gamma(y_t, P_{t-1}, P_t), \qquad X_{t+1} = \frac{W}{M_{t+1}}.$$

By the chain rule,

$$\begin{aligned} \nu_{t+1} &= \frac{d}{dP_t} V_{t+1}(s_{t+1}) \\ &= \partial_P V_{t+1}(s_{t+1}) + \big(\nabla_y V_{t+1}(s_{t+1})\big) \cdot \frac{\partial y_{t+1}}{\partial P_t} + \partial_M V_{t+1}(s_{t+1}) \cdot \frac{dM_{t+1}}{dP_t} + \partial_X V_{t+1}(s_{t+1}) \cdot \frac{dX_{t+1}}{dP_t}. \end{aligned}$$

Since $y_{t+1} = F(y_t, P_t)$, we have $\frac{\partial y_{t+1}}{\partial P_t} = \partial_P F(y_t, P_t)$. Moreover,

$$\frac{dM_{t+1}}{dP_t} = g'(\Gamma_t)\, \partial_P \Gamma(y_t, P_{t-1}, P_t).$$

Finally, $X_{t+1} = W/M_{t+1}$ implies

$$\frac{dX_{t+1}}{dP_t} = -\frac{W}{M_{t+1}^2}\frac{dM_{t+1}}{dP_t} = -B_{t+1}\frac{dM_{t+1}}{dP_t}.$$

Substituting the last two into the expansion for ν_{t+1} and grouping the (∂_M, ∂_X) terms yields

$$\nu_{t+1} = \partial_P V_{t+1}(s_{t+1}) + \big(\nabla_y V_{t+1}(s_{t+1})\big) \cdot \partial_P F(y_t, P_t) + \left(\partial_M V_{t+1}(s_{t+1}) - B_{t+1}\partial_X V_{t+1}(s_{t+1})\right)\frac{dM_{t+1}}{dP_t},$$

which is (68) after inserting the definition (67). □

The first two terms in (68) reflect the fact that P_t enters the next state directly: it is stored as P_t and it updates the internal memory variable $y_{t+1} = F(y_t, P_t)$; with a rolling window or a lag, the rule-induced incentives for requirement dates beyond $t+1$ run through these memory terms. The third term isolates the contemporaneous requirement-update channel. Along $X_{t+1} = W/M_{t+1}$, a marginal change dM_{t+1} induces $dX_{t+1} = -B_{t+1}dM_{t+1}$, so the change in continuation value induced by tightening is $dV_{t+1} = \Lambda_{t+1}dM_{t+1}$. The rule-induced incentive to move P_t therefore factors into the statistic sensitivity $\partial_P\Gamma(y_t, P_{t-1}, P_t)$, the local pass-through $g'(\Gamma_t)$, and the shadow value Λ_{t+1} of tightening next-period feasibility.

Corollary 5. *Under the conditions of Proposition 9, the requirement-update component of ν_{t+1} in (68) is*

$$\nu_{t+1}^M = \Lambda_{t+1}\, g'(\Gamma_t)\, \partial_P\Gamma(y_t, P_{t-1}, P_t).$$

Accordingly, the requirement-update channel is locally inactive whenever $g'(\Gamma_t) = 0$, $\partial_P\Gamma(y_t, P_{t-1}, P_t) = 0$, or $\Lambda_{t+1} = 0$. When it is active, its sign is

$$\operatorname{sign}(\nu_{t+1}^M) = \operatorname{sign}\big(\Lambda_{t+1}g'(\Gamma_t)\partial_P\Gamma(y_t, P_{t-1}, P_t)\big).$$

Proof. By (68) the margin-update component is $\nu_{t+1}^M = \Lambda_{t+1}g'(\Gamma_t)\partial_P\Gamma(y_t, P_{t-1}, P_t)$. Being a product of three scalar factors, it vanishes whenever any one of $g'(\Gamma_t)$, $\partial_P\Gamma(y_t, P_{t-1}, P_t)$, or Λ_{t+1} is zero. The sign claim follows directly from the same product representation. □

Manipulation through the requirement update requires both links in the feedback chain to be operative: the current price must move the statistic, $\partial_P\Gamma \neq 0$, and the statistic must move the requirement, $g'(\Gamma_t) \neq 0$. If either link is flat at date t, marginal changes in P_t do not change M_{t+1} and thus do not change X_{t+1} under $X_{t+1} = W/M_{t+1}$; the contemporaneous trigger motive is then locally inactive. With a rolling window the stored price still affects later requirements through the first two terms of (68). The motive vanishes at every date under a global shutdown: $g' = 0$, or Γ insensitive to transaction prices in all its arguments, which requires more than $\partial_P\Gamma = 0$ because the price also enters through P_{t-1} and the memory y_t. The attacker then only harvests predetermined flow and manages inventory. When both links are active, the induced sensitivity of next-period constrained demand to the current price is

$$\frac{dX_{t+1}}{dP_t} = -B_{t+1}\, g'(\Gamma_t)\, \partial_P\Gamma(y_t, P_{t-1}, P_t),$$

and Λ_{t+1} converts the resulting marginal tightening into continuation value.

A.6.6 Last trigger date under implementation lags

Lemma 6. *Suppose requirements are applied with an implementation lag $\ell \geq 0$, so that $M_{t+1} = g(\Gamma_{t-\ell})$. Then for any date $t \geq T-1-\ell$, the action u_t cannot affect any requirements that arrive by the final trade date, namely $(M_{t+1}, \ldots, M_{T-1})$, and hence cannot affect any within-horizon forced flows $(v_{t+1}, \ldots, v_{T-1})$. In particular, from such dates onward the only action-affected dependence of the continuation value on the state is through inventory, so $\nu_{t+1} = 0$ whenever the derivative in Proposition 3 exists.*

Therefore, from date t onward the continuation problem reduces to harvesting against the predetermined forced-flow sequence and unwinding inventory by T, solved in closed form by Proposition 4 when trade bounds are slack (and by Proposition 10 under binding bounds). Equivalently, the last date at which a trade can trigger additional within-horizon forced flow is $t_{\text{last}} = T-2-\ell$.

Proof. For $t \geq T-1-\ell$ and $\tau \in \{t+1, \ldots, T-1\}$, $M_\tau = g(\Gamma_{\tau-1-\ell})$ with $\tau-1-\ell \leq t-1$, so (M_τ, X_τ) depends only on prices up to $t-1$. The within-horizon forced flows $(v_{t+1}, \ldots, v_{T-1})$ are therefore predetermined at t, so $V_{t+1}(s_{t+1})$ is locally constant in P_t and $\nu_{t+1} = 0$. For the last-trigger-date claim: if $t \leq T-2-\ell$, then $t+\ell+1 \leq T-1$ and Γ_t enters $M_{t+\ell+1} = g(\Gamma_t)$ within the horizon; if $t \geq T-1-\ell$, the first part shows that no within-horizon requirement depends on Γ_t. □

A.6.7 Proof of Proposition 4

Proof. When future forced flows are exogenous, the continuation objective from date t is

$$\max_{(u_\tau)_{\tau=t}^{T-1}} \; -\eta \sum_{\tau=t}^{T-1} \left(u_\tau^2 + u_\tau v_\tau\right) \qquad \text{s.t.} \qquad \sum_{\tau=t}^{T-1} u_\tau = -x_t,$$

(equivalent to $x_T = 0$ given $x_{t+1} = x_t + u_t$). The objective is strictly concave quadratic on an affine set, so the maximizer is unique. With Lagrange multiplier λ on the inventory constraint, the Lagrangian is

$$\mathcal{L} = -\eta \sum_{\tau=t}^{T-1} \left(u_\tau^2 + u_\tau v_\tau\right) + \lambda \left(\sum_{\tau=t}^{T-1} u_\tau + x_t \right).$$

The first-order condition for each τ is

$$-2\eta u_\tau - \eta v_\tau + \lambda = 0 \qquad \Longrightarrow \qquad u_\tau = -\frac{v_\tau}{2} + \frac{\lambda}{2\eta}.$$

Summing over $\tau = t, \ldots, T-1$ and imposing $\sum_{\tau=t}^{T-1} u_\tau = -x_t$ yields

$$-\frac{1}{2} \sum_{\tau=t}^{T-1} v_\tau + \frac{n\lambda}{2\eta} = -x_t \qquad \Longrightarrow \qquad \frac{\lambda}{2\eta} = \frac{\bar{v}_t}{2} - \frac{x_t}{n},$$

which gives (51).

To obtain the value, use Lemma 3 date by date:

$$-\eta u_\tau(u_\tau + v_\tau) = -\eta\Big(u_\tau + \tfrac{v_\tau}{2}\Big)^2 + \frac{\eta}{4}v_\tau^2.$$

Under (51),

$$u_\tau^\star + \tfrac{v_\tau}{2} = \frac{\bar v_t}{2} - \frac{x_t}{n},$$

which is constant across τ. Therefore

$$V_t^{\text{harv}} = \sum_{\tau=t}^{T-1}\left[\frac{\eta}{4}v_\tau^2 - \eta\left(\frac{\bar v_t}{2} - \frac{x_t}{n}\right)^2\right] = \frac{\eta}{4}\sum_{\tau=t}^{T-1} v_\tau^2 - \eta n\left(\frac{\bar v_t}{2} - \frac{x_t}{n}\right)^2.$$

Expanding the last square and using $\sum_{\tau=t}^{T-1}(v_\tau - \bar v_t)^2 = \sum_{\tau=t}^{T-1} v_\tau^2 - n\bar v_t^2$ yields (52). □

A.6.8 Proof of Corollary 1

Proof. The first equation is exactly (52). If $v_\tau = c$, then $\bar v_t = c$ and $\sum_{\tau=t}^{T-1}(v_\tau - \bar v_t)^2 = 0$, so (52) gives $V_t^{\text{harv}} = \eta x_t c - \frac{\eta}{n}x_t^2$. Equation (51) yields $u_\tau^\star = -\frac{1}{2}(c - c) - \frac{x_t}{n} = -\frac{x_t}{n}$ for all τ. Finally, if $x_t = 0$ then $u_\tau^\star = 0$ and $V_t^{\text{harv}} = 0$. □

A.6.9 Proof of Corollary 2

Proof. Under the stated forced-flow path, $\bar v_t = \frac{1}{n}\sum_{\tau=t}^{T-1} v_\tau = v/n$. Applying (51) with $x_t = 0$ gives, for $\tau = t, \dots, T-2$,

$$u_\tau^\star = -\frac{1}{2}\left(0 - \frac{v}{n}\right) = \frac{v}{2n},$$

and for $\tau = T-1$,

$$u_{T-1}^\star = -\frac{1}{2}\left(v - \frac{v}{n}\right) = -\frac{n-1}{2n}v.$$

Since $v < 0$, the terminal trade is positive. Finally, with $x_t = 0$, Corollary 1 gives $V_t^{\text{harv}} = \frac{\eta}{4}\sum_{\tau=t}^{T-1}(v_\tau - \bar v_t)^2$. Here there are $n-1$ terms equal to $(0 - v/n)^2 = v^2/n^2$ and one term equal to $(v - v/n)^2 = v^2(n-1)^2/n^2$, so

$$\sum_{\tau=t}^{T-1}(v_\tau - \bar v_t)^2 = \frac{v^2}{n^2}\big((n-1) + (n-1)^2\big) = \frac{n-1}{n}v^2,$$

which yields the stated value. □

A.6.10 Harvesting with per-date trade bounds

This appendix completes the optimal-attack characterization of Section 5 when the per-date trade bound $|u_\tau| \le \bar u$ binds.

Proposition 10 (Harvesting with per-date trade bounds). *Fix a date t and suppose that, conditional on s_t, the remaining forced flows $(v_\tau)_{\tau=t}^{T-1}$ are predetermined at time t and do not depend on the continuation trades*

$(u_\tau)_{\tau=t}^{T-1}$. *Consider the continuation problem under temporary impact with per-date bounds* $|u_\tau| \le \bar{u}$*:*

$$\max_{(u_\tau)_{\tau=t}^{T-1}} \; -\eta \sum_{\tau=t}^{T-1} \big(u_\tau^2 + u_\tau v_\tau\big) \qquad \text{s.t.} \qquad \sum_{\tau=t}^{T-1} u_\tau = -x_t, \qquad |u_\tau| \le \bar{u} \;\; \forall \tau.$$

Assume feasibility, $|x_t| \le (T-t)\bar{u}$*. Then the problem has a unique maximizer* $(u_\tau^\star)_{\tau=t}^{T-1}$*. Moreover, there exists a scalar* $c_t \in \mathbb{R}$ *such that, for* $\tau = t, \dots, T-1$*,*

$$u_\tau^\star = \mathrm{clip}_{[-\bar{u},\bar{u}]}\Big(-\frac{v_\tau}{2} + c_t\Big), \tag{69}$$

where $\mathrm{clip}_{[-\bar{u},\bar{u}]}(z) = \min\{\bar{u}, \max\{-\bar{u}, z\}\}$ *and* c_t *can be chosen so that* $\sum_{\tau=t}^{T-1} u_\tau^\star = -x_t$*. If at least one bound is slack at the optimizer (so* $|u_\tau^\star| < \bar{u}$ *for some* τ*), then* c_t *is unique. If the bounds are slack for all* τ*, then* (69) *reduces to* (51)*.*

Proof. The objective is a strictly concave quadratic in $(u_\tau)_{\tau=t}^{T-1}$ and the constraints define a nonempty compact convex set under feasibility, so a unique maximizer exists. Introduce a multiplier λ on the inventory constraint and multipliers $\alpha_\tau, \beta_\tau \ge 0$ on the lower and upper bounds. The Lagrangian is

$$\mathcal{L} = -\eta \sum_{\tau=t}^{T-1} \big(u_\tau^2 + u_\tau v_\tau\big) + \lambda \left(\sum_{\tau=t}^{T-1} u_\tau + x_t\right) + \sum_{\tau=t}^{T-1} \alpha_\tau (u_\tau + \bar{u}) + \sum_{\tau=t}^{T-1} \beta_\tau (\bar{u} - u_\tau).$$

The KKT conditions imply, for each τ,

$$-2\eta u_\tau - \eta v_\tau + \lambda + \alpha_\tau - \beta_\tau = 0, \qquad \alpha_\tau(u_\tau + \bar{u}) = 0, \qquad \beta_\tau(\bar{u} - u_\tau) = 0,$$

together with $\alpha_\tau, \beta_\tau \ge 0$, $-\bar{u} \le u_\tau \le \bar{u}$, and the equality constraint $\sum_{\tau=t}^{T-1} u_\tau = -x_t$. These conditions are equivalent to the pointwise form

$$u_\tau = \mathrm{clip}_{[-\bar{u},\bar{u}]}\left(-\frac{v_\tau}{2} + \frac{\lambda}{2\eta}\right).$$

Setting $c_t = \lambda/(2\eta)$ yields (69).

It remains to justify that c_t can be chosen to satisfy the inventory identity. Define the continuous, nondecreasing function

$$\Phi(c) = \sum_{\tau=t}^{T-1} \mathrm{clip}_{[-\bar{u},\bar{u}]}\Big(-\frac{v_\tau}{2} + c\Big).$$

Feasibility implies $-x_t \in [-(T-t)\bar{u}, (T-t)\bar{u}]$, while $\Phi(c) \to -(T-t)\bar{u}$ as $c \to -\infty$ and $\Phi(c) \to (T-t)\bar{u}$ as $c \to +\infty$. Hence there exists c_t with $\Phi(c_t) = -x_t$, and the corresponding $(u_\tau^\star)$ satisfies the equality constraint. If $|u_{\bar{\tau}}^\star| < \bar{u}$ for some $\bar{\tau}$, then in a neighborhood of c_t the term $\mathrm{clip}_{[-\bar{u},\bar{u}]}(-v_{\bar{\tau}}/2 + c)$ is locally equal to $-v_{\bar{\tau}}/2 + c$ and thus varies strictly with c. Therefore Φ is strictly increasing at c_t, which implies that the equation $\Phi(c) = -x_t$ has a unique solution; hence c_t is unique. If all bounds bind, then Φ is locally flat at c_t and the multiplier (hence c_t) need not be unique, even though $(u_\tau^\star)$ is unique by strict concavity.

Finally, if no bound binds at the optimizer, then $\mathrm{clip}_{[-\bar{u},\bar{u}]}$ is the identity and stationarity gives

$u_\tau^\star = -v_\tau/2 + c_t$ with a common offset c_t. Imposing $\sum_{\tau=t}^{T-1} u_\tau^\star = -x_t$ yields $c_t = \bar{v}_t/2 - x_t/(T-t)$, which recovers (51). □

A.7 Proofs for the multi-asset model

A.7.1 Pseudoinverse and projections when A is p.s.d.

Let $A \in \mathbb{R}^{N\times N}$ be symmetric and positive semidefinite. Denote by $A^\dagger$ the Moore-Penrose pseudoinverse and define

$$\Pi_A = A^\dagger A = AA^\dagger,$$

the orthogonal projector onto $\mathrm{Range}(A)$; see Boyd and Vandenberghe (2004, Appendix A.5.4).

We use two standard facts about the p.s.d. pseudoinverse (Boyd and Vandenberghe, 2004, Example 4.5): (i) for all $z \in \mathbb{R}^N$, $z^\top Az = (\Pi_A z)^\top A(\Pi_A z)$; and (ii) for any $w \in \mathrm{Range}(A)$, the solutions of $Az = w$ are $\{A^\dagger w + n : \ n \in \ker(A)\}$, and every solution satisfies $z^\top Az = w^\top A^\dagger w$.

A.7.2 Myopic harvesting in multiple assets

Lemma 7 (Myopic harvesting in multiple assets)**.** *Fix a date t and a state s_t with predetermined forced flow $v_t = X_t - X_{t-1}$. Under (57), the one-period payoff satisfies*

$$\pi_t(s_t, u) = -\Big(u + \tfrac{v_t}{2}\Big)^\top A\Big(u + \tfrac{v_t}{2}\Big) + \frac{1}{4} v_t^\top A v_t, \tag{70}$$

concave in u, with unconstrained maximizers $u_t^{\mathrm{myopic}} = -v_t/2 + n$ for $n \in \ker A$ and maximal rent $\frac{1}{4} v_t^\top A v_t$. If A is positive definite, the maximizer is unique and equals $-v_t/2$.

Proof. Using (57) and symmetry of A, $\pi_t(s_t, u) = -u^\top Au - u^\top Av_t$. Completing the square yields (70), and concavity follows from positive semidefiniteness of A. The square term vanishes when $A(u + v_t/2) = 0$, i.e. $u = -v_t/2 + n$ with $An = 0$; positive definiteness forces $n = 0$. □

A.7.3 Separation of harvesting, inventory, and triggering

Proposition 11. *Fix $t \in \{0, \dots, T-1\}$ and a state s_t. Suppose the optimizer $u_t^\star$ in (59) is interior, $u_t^\star \in \operatorname{int} \mathcal{U}_t(x_t)$, and assume that V_{t+1} is differentiable at the continuation state $s_{t+1}^\star = \mathcal{T}(s_t, u_t^\star)$ and that $\mathcal{T}(s_t, u)$ is differentiable at $u_t^\star$ (in particular, F, Γ, and g are differentiable at the points they are evaluated at, as in Proposition 12). Define*

$$\mu_{t+1} = \nabla_x V_{t+1}(s_{t+1}^\star) \in \mathbb{R}^N, \qquad \nu_{t+1} = \frac{d}{dP_t} V_{t+1}(s_{t+1}^\star) \in \mathbb{R}^N, \qquad P_t = S_t + A(u_t^\star + v_t),$$

where $\frac{d}{dP_t}$ denotes the total derivative through all components of s_{t+1} that depend on P_t under the transition map. Then $u_t^\star$ satisfies the Euler equation

$$A(2u_t^\star + v_t) = \mu_{t+1} + A\,\nu_{t+1}. \tag{71}$$

If A is positive definite, (71) uniquely determines the interior optimizer:

$$2u_t^\star = -v_t + A^{-1}\mu_{t+1} + \nu_{t+1}.$$

If A is only positive semidefinite, the objective and the rule respond only to the price-moving component of trades. Trading in an impact-neutral direction does not move the price under (56) and does not generate cashflows, so such directions are economically irrelevant for harvesting and for triggering. Let $A^\dagger$ denote the Moore–Penrose pseudoinverse and write $\Pi_A = A^\dagger A$ for the operator that extracts the price-moving component of a trade. Then (71) pins down $\Pi_A u_t^\star$, and the canonical choice that sets the impact-neutral component to zero satisfies

$$\Pi_A u_t^\star = -\frac{1}{2}\,\Pi_A v_t + \frac{1}{2}A^\dagger \mu_{t+1} + \frac{1}{2}\,\Pi_A \nu_{t+1}. \tag{72}$$

If A is positive definite, then $A^\dagger = A^{-1}$ and $\Pi_A = I$, so (72) reduces to the previous expression. Moreover, the resulting current-period wedge payoff can be written as

$$\pi_t(s_t, u_t^\star) = \frac{1}{4}v_t^\top A v_t - \Big(u_t^\star + \tfrac{v_t}{2}\Big)^\top A\Big(u_t^\star + \tfrac{v_t}{2}\Big) = \frac{1}{4}v_t^\top A v_t - \frac{1}{4}\big(\mu_{t+1} + A\nu_{t+1}\big)^\top A^\dagger\big(\mu_{t+1} + A\nu_{t+1}\big). \tag{73}$$

Proof. Fix t, s_t, and an interior maximizer $u_t^\star$ in (59), and write $\Psi(u) = \pi_t(s_t, u) + V_{t+1}(\mathcal{T}(s_t, u))$. By (57) and symmetry of A, $\nabla_u \pi_t(s_t, u) = -A(2u + v_t)$. The control enters $\mathcal{T}$ only through $x_{t+1} = x_t + u$ and $P_t = S_t + A(u + v_t)$, so by the chain rule

$$\nabla_u V_{t+1}(\mathcal{T}(s_t, u)) = \nabla_x V_{t+1}(s_{t+1}) \cdot \frac{\partial(x_t + u)}{\partial u} + \frac{d}{dP_t}V_{t+1}(s_{t+1}) \cdot \frac{\partial P_t}{\partial u} = \mu_{t+1} + A\nu_{t+1},$$

where $\mu_{t+1} = \nabla_x V_{t+1}(s_{t+1})$ and $\nu_{t+1} = \frac{d}{dP_t}V_{t+1}(s_{t+1})$. At an interior optimum, $\nabla_u \Psi(u_t^\star) = 0$, hence

$$-A(2u_t^\star + v_t) + \mu_{t+1} + A\nu_{t+1} = 0,$$

which is (71).

Since A may be singular, (71) pins down only the price-moving component $\Pi_A u_t^\star$ of the optimizer. Multiplying (71) by $A^\dagger$ directly,

$$\Pi_A(2u_t^\star + v_t) = A^\dagger \mu_{t+1} + \Pi_A \nu_{t+1}.$$

Using linearity and $\Pi_A u_t^\star = u_t^\star$, we have

$$\Pi_A(2u_t^\star + v_t) = 2u_t^\star + \Pi_A v_t,$$

hence

$$2u_t^\star = -\Pi_A v_t + A^\dagger \mu_{t+1} + \Pi_A \nu_{t+1},$$

which is (72).

By Lemma 7,

$$\pi_t(s_t, u_t^\star) = \frac{1}{4}v_t^\top A v_t - \Big(u_t^\star + \tfrac{v_t}{2}\Big)^\top A\Big(u_t^\star + \tfrac{v_t}{2}\Big) = \frac{1}{4}v_t^\top A v_t - \frac{1}{4}(2u_t^\star + v_t)^\top A(2u_t^\star + v_t).$$

Using (71), $A(2u_t^\star + v_t) = \mu_{t+1} + A\nu_{t+1} \in \mathrm{Range}(A)$. By fact (ii) of Appendix A.7.1,

$$(2u_t^\star + v_t)^\top A(2u_t^\star + v_t) = (\mu_{t+1} + A\nu_{t+1})^\top A^\dagger(\mu_{t+1} + A\nu_{t+1}).$$

Substituting yields (73). □

A.7.4 Chain-rule decomposition of the manipulation motive

Proposition 12 (Chain-rule decomposition of ν_{t+1})**.** *Fix $t \in \{0, \dots, T-1\}$ and consider any state-action pair (s_t, u_t). Let P_t be given by (56) and let $s_{t+1} = \mathcal{T}(s_t, u_t)$. Assume V_{t+1} is differentiable at s_{t+1}, F is differentiable in its price argument at (y_t, P_t), and the maps g and Γ are differentiable at $\Gamma_t = \Gamma(y_t, P_{t-1}, P_t)$ and in P_t at (y_t, P_{t-1}, P_t), respectively. Define*

$$\Lambda_{t+1} = \partial_M V_{t+1}(s_{t+1}) - B_{t+1}^\top \nabla_X V_{t+1}(s_{t+1}), \tag{74}$$

the shadow value of tightening the requirement along $X_{t+1} = (W/M_{t+1})b$, where $B_{t+1} = (W/M_{t+1}^2)\, b \in \mathbb{R}^N$ is the vector demand sensitivity $-\partial X_{t+1}/\partial M_{t+1}$. Let

$$\nu_{t+1} = \frac{d}{dP_t} V_{t+1}(s_{t+1}) \in \mathbb{R}^N$$

denote the total derivative of the continuation value with respect to the current price P_t through the transition map. Then

$$\nu_{t+1} = \partial_P V_{t+1}(s_{t+1}) + \big(\partial_P F(y_t, P_t)\big)^\top \nabla_y V_{t+1}(s_{t+1}) + \Lambda_{t+1}\, g'(\Gamma_t)\, \partial_P \Gamma(y_t, P_{t-1}, P_t). \tag{75}$$

Proof. Fix (s_t, u_t) and let $s_{t+1} = \mathcal{T}(s_t, u_t)$. In the transition map, P_t enters the next state directly as the lagged price component, and it also affects $(y_{t+1}, M_{t+1}, X_{t+1})$ via $y_{t+1} = F(y_t, P_t)$, $M_{t+1} = g(\Gamma_t)$, and $\Gamma_t = \Gamma(y_t, P_{t-1}, P_t)$. Let $\nu_{t+1} = \frac{d}{dP_t} V_{t+1}(s_{t+1})$. By the multivariate chain rule,

$$\nu_{t+1} = \partial_P V_{t+1}(s_{t+1}) + \big(\partial_P F(y_t, P_t)\big)^\top \nabla_y V_{t+1}(s_{t+1}) + \partial_M V_{t+1}(s_{t+1})\, \frac{\partial M_{t+1}}{\partial P_t} + \nabla_X V_{t+1}(s_{t+1})^\top \frac{\partial X_{t+1}}{\partial P_t}.$$

Since $M_{t+1} = g(\Gamma_t)$ with $\Gamma_t = \Gamma(y_t, P_{t-1}, P_t)$,

$$\frac{\partial M_{t+1}}{\partial P_t} = g'(\Gamma_t)\, \partial_P \Gamma(y_t, P_{t-1}, P_t).$$

Also, $X_{t+1} = (W/M_{t+1})b$, so $\frac{\partial X_{t+1}}{\partial M_{t+1}} = -B_{t+1}$ and therefore

$$\frac{\partial X_{t+1}}{\partial P_t} = -B_{t+1}\, g'(\Gamma_t)\, \partial_P \Gamma(y_t, P_{t-1}, P_t)^\top.$$

Substituting implies

$$\nabla_X V_{t+1}(s_{t+1})^\top \frac{\partial X_{t+1}}{\partial P_t} = -\big(B_{t+1}^\top \nabla_X V_{t+1}(s_{t+1})\big)\, g'(\Gamma_t)\, \partial_P \Gamma(y_t, P_{t-1}, P_t),$$

so grouping the $g'(\Gamma_t)\partial_P\Gamma$ terms yields (75) with Λ_{t+1} defined in (74). □

A.7.5 Proof of Theorem 2

Assume the rule is portfolio marked in direction d as in the theorem statement. Let $\mathcal{W} = \mathrm{span}\{b, d\}$ and write $B = [b\, d] \in \mathbb{R}^{N\times 2}$. Define the 2×2 Gram matrix $G = B^\top AB$ and let $G^\dagger$ be its Moore-Penrose pseudoinverse.

For any $u \in \mathbb{R}^N$ define $\Pi_{\mathcal{W}}u$ as the minimizer of

$$\min_{w\in\mathcal{W}} (u-w)^\top A(u-w).$$

Writing $w = Bc$, this is the convex quadratic problem

$$\min_{c\in\mathbb{R}^2} (u-Bc)^\top A(u-Bc).$$

A standard pseudoinverse solution (Boyd and Vandenberghe, 2004, Appendix A.5.4 and Example 4.5) yields the canonical minimizer

$$c^\star = G^\dagger B^\top Au, \qquad \Pi_{\mathcal{W}}u = Bc^\star = B\,G^\dagger B^\top Au,$$

so $\Pi_{\mathcal{W}}u \in \mathcal{W}$. By optimality, the residual $u - \Pi_{\mathcal{W}}u$ is A-orthogonal to $\mathcal{W}$, i.e.

$$w^\top A(u-\Pi_{\mathcal{W}}u) = 0 \qquad \text{for all } w \in \mathcal{W}. \tag{76}$$

In particular, taking $w = b$ and $w = d$ gives

$$b^\top A\,\Pi_{\mathcal{W}}u = b^\top Au, \qquad d^\top A\,\Pi_{\mathcal{W}}u = d^\top Au. \tag{77}$$

Moreover, (76) implies the Pythagorean identity

$$u^\top Au = (\Pi_{\mathcal{W}}u)^\top A(\Pi_{\mathcal{W}}u) + (u-\Pi_{\mathcal{W}}u)^\top A(u-\Pi_{\mathcal{W}}u), \tag{78}$$

so the projection weakly decreases the A-norm:

$$(\Pi_{\mathcal{W}}u)^\top A(\Pi_{\mathcal{W}}u) \le u^\top Au. \tag{79}$$

Fix any admissible round trip $(u_t)_{t=0}^{T-1}$ for the relaxed problem (terminal constraint $x_T = 0$, no per-date bounds) and set $\tilde{u}_t = \Pi_{\mathcal{W}}u_t$. Linearity of $\Pi_{\mathcal{W}}$ gives $\sum_t \tilde{u}_t = 0$, so $(\tilde{u}_t)$ is also a round trip. Let $z_t = d^\top P_t$. We show by induction on t that z_t coincides under (u_t) and $(\tilde{u}_t)$. The base case is v_0, which is pinned by initial conditions. Given $(z_0,\dots,z_{t-1})$ identical, portfolio marking implies (y_t, M_t, X_t) and $v_t = X_t - X_{t-1}$ coincide; then (56) gives $z_t = d^\top S_t + d^\top A(u_t + v_t)$, and (77) with $u = u_t$ gives $d^\top A\tilde{u}_t = d^\top Au_t$, so z_t matches. Hence requirements and forced flows coincide under both strategies.

Compare profits period by period: $\pi_t(u) = -u^\top Au - u^\top Av_t$. Since $v_t \in \mathrm{span}\{b\}$, the interac-

tion term depends on u only through $b^\top Au$, preserved by (77) under replacement $u_t \to \tilde{u}_t$. By (79), $\tilde{u}_t^\top A\tilde{u}_t \le u_t^\top Au_t$. Therefore $\pi_t(\tilde{u}_t) \ge \pi_t(u_t)$ at each t, so summing gives weakly higher total wedge profits. The supremum is therefore unchanged under the span restriction, and an optimizer (if attained) can be chosen in $\mathcal{W}$. Once $u_t \in \mathcal{W}$ and $v_t \in \operatorname{span}\{b\}$, all quadratic wedge terms depend on A only through the Gram matrix $G = B^\top AB$ with entries $b^\top Ab$, $b^\top Ad$, $d^\top Ad$.

A.7.6 KKT conditions under the ℓ_1 per-date trade bound

We record the first-order conditions for (59) when the ℓ_1 constraints bind. Fix t, s_t, and assume V_{t+1} is differentiable at the continuation state. Let

$$G_t(u) = \pi_t(s_t, u) + V_{t+1}(\mathcal{T}(s_t, u)).$$

Its gradient is

$$\nabla_u G_t(u) = -A(2u + v_t) + \mu_{t+1}(u) + A\nu_{t+1}(u),$$

where $\mu_{t+1}(u) = \nabla_x V_{t+1}(\mathcal{T}(s_t, u))$ and $\nu_{t+1}(u) = \nabla_{P_t} V_{t+1}(\mathcal{T}(s_t, u))$. The feasible set is the intersection of two ℓ_1 balls,

$$\|u\|_1 \le \bar{u}, \qquad \|x_t + u\|_1 \le (T - 1 - t)\bar{u}.$$

A (possibly) binding optimum $u_t^\star$ admits multipliers $\lambda_t, \kappa_t \ge 0$ and subgradients $\zeta_t \in \partial \|u_t^\star\|_1$, $\xi_t \in \partial \|x_t + u_t^\star\|_1$ such that

$$0 \in \nabla_u G_t(u_t^\star) - \lambda_t \zeta_t - \kappa_t \xi_t, \tag{80}$$

$$\lambda_t \left(\|u_t^\star\|_1 - \bar{u}\right) = 0, \qquad \kappa_t \left(\|x_t + u_t^\star\|_1 - (T - 1 - t)\bar{u}\right) = 0. \tag{81}$$

Here $\partial \|z\|_1$ is the standard ℓ_1 subdifferential: componentwise, $\zeta_i = \operatorname{sign}(z_i)$ if $z_i \ne 0$ and $\zeta_i \in [-1, 1]$ if $z_i = 0$. Equation (80) reduces to the interior Euler equation (71) when both constraints are slack.

B Applications

Volatility-controlled (risk-control, target-volatility) indices publish a deterministic exposure rule: given recent realized returns, they set the next-day risky exposure, typically subject to caps, floors, and implementation lags. When these indices underlie structured products and index-linked annuities, dealer replication turns the published rule into mechanical hedging flow in the underlying. This is exactly the environment the linearized admissibility condition in Theorem 1 is meant to evaluate.

B.1 Institutional setting

Volatility-controlled indices are widely used as underlyings for retail structured products and insurance wrappers. In these contracts the payoff is indexed to the published index level, so the issuer or insurer must replicate the index exposure to hedge its liabilities. The market is relatively large: U.S. retail annuity sales were on the order of hundreds of billions of dollars in 2024, and fixed indexed annuities (FIAs) are a major and growing segment (LIMRA, 2025; American Academy of Actuaries, 2026). Index providers explicitly design and license risk-control indices for product embedding, including structured products and indexed investment products (MSCI, 2020, 2024; S&P Dow Jones Indices, 2025).

B.1.1 From rulebooks to trading flow

Three parties map the disclosed rulebook into trading flow relevant for our admissibility test.

1. *Index sponsor and calculation agent.* The sponsor specifies the methodology (sampling convention, volatility estimator, exposure rule, caps/floors, fees, and any lags) and publishes the official index level. These documents pin down the feedback map from prices to next-day exposure (MSCI, 2024; S&P Dow Jones Indices, 2025; BlackRock, 2023).

2. *Issuer/insurer.* The issuer sells a contract whose payoff references the index level (structured note) or credits interest using an index-linked option (FIA/RILA). The liability is therefore indexed to the rule (American Academy of Actuaries, 2026).

3. *Dealer replication and hedge instruments.* Replication is implemented in futures, ETFs, swaps, and cash. When the index rebalances, replication requires trading these hedge instruments. Conditional on the published rule, the direction and timing of the flow are mechanical.

B.1.2 Single-asset versus multi-asset indices

Most marketed designs share a common architecture: a reference portfolio and a cash account, with a volatility-control constraint that scales exposure to the reference portfolio to target a stated risk level (MSCI, 2024; S&P Dow Jones Indices, 2025, 2021; Bloomberg Index Services Limited, 2025). They differ mainly in the construction of the reference portfolio.

- *Single-underlying (Template A).* The reference portfolio is a single parent index. The risk-control rule varies the weight on the parent index versus cash as an inverse-volatility function of a rolling volatility estimator (MSCI, 2024; S&P Dow Jones Indices, 2025).

- *Multi-underlying, fixed basket (Template B).* The reference portfolio is a fixed multi-asset basket (often equity and rates, implemented with ETFs or futures), and the rule scales the basket versus cash to hit a volatility target (Bloomberg Index Services Limited, 2025).
- *Multi-underlying, signal-driven basket (Template C).* The reference portfolio is itself rule-based and time-varying (e.g., rotation, momentum/trends), and a volatility-control rule then scales the resulting portfolio versus cash (J.P. Morgan, 2015; Goldman Sachs Group, 2021; S&P Dow Jones Indices, 2024b; BNP Paribas Indices, 2025; UBS Investment Bank, 2025). A marketed instance is the BlackRock Adaptive U.S. Equity Index Series, which sets weights by an optimization subject to regime-dependent exposure caps and an adaptive variance budget, and whose volatility control responds to the index's own realized volatility (BlackRock, 2023).

We focus on Templates A and B.

B.2 Formulae and timing conventions

Our baseline formulas come from public index methodology documents and representative offering materials. Methodologies specify the estimator, caps and floors, turnover buffers, fees, the cash component, and the calculation day; disclosure documents restate the same mechanics and make the lag between volatility measurement and exposure implementation explicit (MSCI, 2024; S&P Dow Jones Indices, 2021; BlackRock, 2023; Royal Bank of Canada, 2021; Morgan Stanley, 2025).

B.2.1 Template A: single-underlying daily clipped target-vol

The single-underlying template allocates between a parent index (e.g., MSCI World Index) and cash. Let B_t be the parent index level used for the rule, sampled once per business day, and let $r_t = \log(B_t/B_{t-1})$ denote the daily log return used for volatility estimation. An annualized realized volatility estimate is constructed as a function of recent returns. For example, MSCI computes a short-horizon realized volatility and a long-horizon realized volatility using equally weighted daily gross total returns and sets Parent Index volatility as the maximum of the two (with $N = 20$ and $N = 60$ trading days) (MSCI, 2024). S&P documents an analogous structure, allowing simple-weighted or exponentially weighted volatility and using the maximum of short- and long-term estimates (S&P Dow Jones Indices, 2024a, 2021). Given a target volatility σ^*, the exposure to the parent index is set as a ratio of target to estimated volatility, subject to a maximum leverage cap $L_{\max}$,

$$L_t = \min\left\{L_{\max},\ \frac{\sigma^*}{\widehat{\sigma}_{t-\ell}}\right\},$$

with the residual weight $1 - L_t$ allocated to a cash component (and, when $L_t > 1$, borrowing costs applied to the leveraged portion). The lag ℓ is explicit in some rulebooks and disclosures. For instance, MSCI determines index leverage for an effective date using volatility estimated two trading days before the effective date and applies a turnover buffer so leverage is updated only when changes exceed a threshold (MSCI, 2024). Representative offering documents for risk-control underliers also describe a two-day lag between leverage-factor calculation and implementation and disclose the leverage cap (e.g., 150%) (Morgan Stanley, 2025; Royal Bank of Canada, 2021).

B.2.2 Template B: multi-underlying risk-control

In the multi-underlying template, the reference portfolio is a basket of tradable constituents, and the rule scales the basket versus cash to track a volatility target. The basket composition is fixed and only the overall risk budget varies. Formally, let $P_t \in \mathbb{R}^N$ denote the vector of constituent price levels used for index calculation (sampled once per business day), and let $d \in \mathbb{R}^N$ be fixed risky-basket weights. The reference-basket return is $r_t^{\text{ref}} = d^\top r_t$ where r_t are constituent returns. The rule computes a realized volatility estimate $\widehat{\sigma}_t$ from the history of r^{ref} (or an equivalent basket-level statistic) and sets a scalar leverage factor L_t as above, subject to caps/floors and a lag. The resulting risky weights are $w_t = L_t d$, and the cash weight is the residual $1 - \mathbf{1}^\top w_t$. A concrete industry example is the Bloomberg US EQ:FI 60:40 volatility-target index series, which scales a fixed 60/40 equity-fixed-income basket against an overnight cash index to a stated volatility target (for example 5%), subject to a 150% maximum exposure and a one-business-day lag (Bloomberg Index Services Limited, 2025). The basket weights are held fixed and only the overall risk budget varies; index levels are computed from constituent closing prices, and the specification documents the realized-volatility estimator, the exposure cap, and an excess-return calculation relative to the cash index (Bloomberg Index Services Limited, 2025).

B.2.3 Timing conventions

Three timing details matter for the no-arbitrage and stress-test computations.

1. Methodologies commonly use end-of-day closing prices to compute index levels and the statistics entering the rebalancing rule (BlackRock, 2023).

2. Many designs implement leverage and weights with a one- or two-business-day lag. This lag is explicit in methodology documents (MSCI, 2024; Morgan Stanley, 2025).

3. Caps, floors, turnover buffers, rounding, and exchange-holiday conventions create kinks and discrete updates (MSCI, 2024; BlackRock, 2023).

B.3 Derivatives and construction of the admissibility test

An index methodology gives a deterministic rule from recent returns to next-day risky weight, including caps, floors, and any implementation lag. We translate it into the objects used in the paper.

B.3.1 (Γ, g) and $s = g'(\Gamma)$

For a standard target-volatility rule,

$$w_{t+1} = \text{clip}\left(w_{\min},\, w_{\max},\, \frac{\sigma^*}{\widehat{\sigma}_{t-\ell}}\right), \qquad M_{t+1} = \frac{1}{w_{t+1}}, \tag{82}$$

with $\widehat{\sigma}_t = \sqrt{\Gamma_t}$. Many methodologies add a turnover buffer: the weight is held fixed unless the raw update exceeds a threshold τ,

$$w_{t+1}^{\text{raw}} = \text{clip}\left(w_{\min},\, w_{\max},\, \frac{\sigma^*}{\widehat{\sigma}_{t-\ell}}\right), \qquad w_{t+1} = \begin{cases} w_t, & \text{if } |w_{t+1}^{\text{raw}} - w_t| \le \tau, \\ w_{t+1}^{\text{raw}}, & \text{if } |w_{t+1}^{\text{raw}} - w_t| > \tau, \end{cases} \qquad M_{t+1} = \frac{1}{w_{t+1}}.$$

When the buffer binds (first case), exposure is locally flat in $\widehat{\sigma}_{t-\ell}$, so $s_t = 0$. On the interior region (cap/floor inactive and buffer inactive),

$$M_{t+1} = \frac{\widehat{\sigma}_{t-\ell}}{\sigma^*} = \frac{\sqrt{\Gamma_{t-\ell}}}{\sigma^*}, \qquad s_t = \frac{\partial M_{t+1}}{\partial \Gamma_{t-\ell}} = \frac{1}{2\sigma^* \widehat{\sigma}_{t-\ell}}. \tag{83}$$

B.3.2 J for Template A

Template A uses a single price series P_t and a rolling realized-variance statistic. Define log returns and the statistic by

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right), \qquad \Gamma_t = \frac{a_{\text{ann}}}{m} \sum_{i=0}^{m-1} r_{t-i}^2, \tag{84}$$

with window length m and annualization factor $a_{\text{ann}} > 0$. Some rulebooks compute realized variance at multiple horizons (e.g., 20 and 60 days) and set Γ_t equal to the maximum across horizons. For a finite set $\mathcal{M}$, define

$$\Gamma_t^{(m)} = \frac{a_{\text{ann}}}{m} \sum_{i=0}^{m-1} r_{t-i}^2, \qquad m \in \mathcal{M}, \qquad \Gamma_t = \max_{m \in \mathcal{M}} \Gamma_t^{(m)}.$$

Away from ties, let m_t^* be the unique maximizing window and take the Jacobian row at date t from that active window. At ties we choose the smallest window. Differentiating (84) gives, for any date j,

$$\frac{\partial \Gamma_t}{\partial P_j} = \frac{2a_{\text{ann}}}{m\, P_j} \Big(\mathbf{1}\{t - m + 1 \le j \le t\}\, r_j - \mathbf{1}\{t - m \le j \le t - 1\}\, r_{j+1} \Big). \tag{85}$$

On a T-date horizon, writing $\delta\Gamma = J\, \delta\Delta P$, the matrix J is the $T \times T$ matrix with (t, j) entry given by (85) evaluated at the reference state. Only prices inside the rolling window affect Γ_t.

B.3.3 J for Template B

Template B scales exposure to a fixed risky basket against cash. Let $P_t \in \mathbb{R}^N$ collect the constituent prices used by the methodology, and define constituent log returns

$$r_t^k = \log\left(\frac{P_t^k}{P_{t-1}^k}\right), \qquad k = 1, \ldots, N. \tag{86}$$

Fix basket weights $d \in \mathbb{R}^N$ and write the basket return as $\bar{r}_t = d^\top r_t$. Realized-variance is

$$\Gamma_t = \frac{a_{\text{ann}}}{m} \sum_{i=0}^{m-1} \bar{r}_{t-i}^2. \tag{87}$$

As in Template A, some rulebooks take the maximum across horizons. For implementation we break ties in favor of the smallest window. Differentiating (87) with respect to a constituent price P_j^k yields

$$\frac{\partial \Gamma_t}{\partial P_j^k} = \frac{2a_{\text{ann}}}{m\, P_j^k}\, d_k \Big(\mathbf{1}\{t-m+1 \le j \le t\}\, \bar{r}_j - \mathbf{1}\{t-m \le j \le t-1\}\, \bar{r}_{j+1} \Big). \tag{88}$$

Stack deviations over dates and assets as

$$\delta\Delta P = \big((\delta\Delta P_0)^\top, \dots, (\delta\Delta P_{T-1})^\top \big)^\top \in \mathbb{R}^{NT}.$$

Then (88) defines J such that $\delta\Gamma = J\, \delta\Delta P$, as in (27).

B.3.4 $\mathcal{K}$ and $\widehat{H}$

Given J and s, requirement deviations satisfy

$$\delta M = s\, L^{\ell+1}\, \delta\Gamma = s\, L^{\ell+1}\, J\, \delta\Delta P, \tag{89}$$

where L is the one-step lag operator and ℓ is the disclosed implementation lag, so the total shift is $\ell+1$. Forced flow comes from the change in constrained demand. Assume the mechanically rebalancing sector trades along a fixed liquidation direction $b \in \mathbb{R}^N$ (for the single-underlying template, $N = 1$ and $b = 1$), so $X_t = X(M_t)\, b$ with local sensitivity $B = -X'(M_0) > 0$ as in (28). Stacking over dates,

$$\delta X = -B\, (I_T \otimes b)\, \delta M, \qquad v = (D \otimes I_N)\, \delta X = -Bs\, (D \otimes I_N)(I_T \otimes b)\, L^{\ell+1}\, J\, \delta\Delta P, \tag{90}$$

where D is the first-difference operator on the T-date horizon (applied blockwise via $D \otimes I_N$). Wedge prices are generated by total flow: $\delta\Delta P = \mathcal{I} q$, with $q = u + v$. Substituting (90) yields

$$q = u - \mathcal{K} q, \qquad \mathcal{K} = Bs\, (D \otimes I_N)(I_T \otimes b)\, L^{\ell+1} J \mathcal{I}. \tag{91}$$

Consistent with the date-varying sensitivities noted in Section 4.2, the implementation evaluates the local sensitivities at each date rather than at a single reference: the scalar slope s is replaced by $\mathbf{S} = \operatorname{diag}(s_t)$ with s_t as in (83), and the scalar demand sensitivity B by $\mathbf{B} = \operatorname{diag}(W/M_t^2)$. The constant-$s$, constant-$B$ form displayed above is the flat-reference special case in which M_t and $\widehat{\sigma}_{t-1-\ell}$ are constant across dates. If $I + \mathcal{K}$ is invertible, then

$$q = (I + \mathcal{K})^{-1} u, \qquad \delta\Delta P = \mathcal{I}(I + \mathcal{K})^{-1} u. \tag{92}$$

Define $\widetilde{\mathcal{I}} = \mathcal{I}(I + \mathcal{K})^{-1}$. Theorem 1 applies to

$$\widehat{H} = \widetilde{\mathcal{I}} + \widetilde{\mathcal{I}}^{\top} = \mathcal{I}(I + \mathcal{K})^{-1} + \Big((I + \mathcal{K})^{-1}\Big)^{\top} \mathcal{I}^{\top}, \tag{93}$$

and the manipulation-free condition is that $\widehat{H}$ is positive semidefinite on the round-trip subspace.

B.4 Stress-test design and impact calibration

The stress test takes two inputs at each state z: the rule-induced price-to-trade sensitivity, computed from the published volatility-index methodology, and a market-impact estimate. We hold the per-unit impact constant across states; the price-to-trade sensitivity varies with z through the rule.

We construct $\mathcal{Z}$ from simulated return paths. For Template A we simulate a single daily log-return process. For Template B we simulate an N-vector of daily log returns with a realistic correlation structure. The baseline generator is multivariate GARCH(1,1):

$$r_t^k = \sqrt{h_t^k}\, \varepsilon_t^k, \qquad h_{t+1}^k = \omega_k + \alpha\, (r_t^k)^2 + \beta\, h_t^k, \qquad k = 1, \dots, N,$$

with $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta < 1$. Innovations $\varepsilon_t \in \mathbb{R}^N$ are i.i.d. over t with mean zero, unit variances, and a specified correlation matrix; the baseline uses a multivariate Student-t distribution with ν degrees of freedom, scaled to unit variance componentwise. For correlation stress in Template B, we multiply off-diagonal correlations by $\kappa \geq 1$, cap absolute correlations at 0.99, and project to the nearest valid correlation matrix before simulation.

We simulate $n_{\text{paths}} = 50$ independent paths of length 2500 business days and discard the first 500 days as burn-in. From each path we sample 200 evaluation dates uniformly at random (without replacement), so $|\mathcal{Z}| = 10{,}000$ states per template and scenario. Innovations are Student-t with $\nu = 8$. We consider $(\alpha, \beta) \in \{(0, 0), (0.05, 0.90), (0.05, 0.94)\}$. With $a_{\text{ann}} = 252$, we set $\omega_k = (1 - \alpha - \beta)\, \sigma_{\text{ann},k}^2 / a_{\text{ann}}$ so that unconditional annualized volatility equals $\sigma_{\text{ann},k}$. For Template A we set $\sigma_{\text{ann}} = 0.20$. For Template B we consider $N \in \{2, 4, 8\}$ with baseline off-diagonal correlation 0.2 (equicorrelation for $N \in \{4, 8\}$) and stress factors $\kappa \in \{1.0, 1.5, 2.0\}$. Annualized volatilities and fixed weights (d, b) are:

$$\begin{aligned}
N = 2: &\quad \sigma_{\text{ann}} = (0.20, 0.10), \quad d = b = (0.6, 0.4), \\
N = 4: &\quad \sigma_{\text{ann}} = (0.20, 0.15, 0.12, 0.10), \quad d = b = \tfrac{1}{4}\,\mathbf{1}, \\
N = 8: &\quad \sigma_{\text{ann}} = (0.22, 0.20, 0.18, 0.16, 0.14, 0.12, 0.10, 0.08), \quad d = b = \tfrac{1}{8}\,\mathbf{1}.
\end{aligned}$$

We calibrate $\mathcal{I}$ from one-day execution-cost benchmarks expressed in basis points per 1% of ADV. Two complementary references guide the range. Kyle and Obizhaeva (2016) report total costs of 10.71 bps (linear) and 14.16 bps (square-root) for a 1% ADV order (Table V), motivating 12 bps per 1% ADV as a baseline and $[8, 20]$ bps as a conservative range. Frazzini et al. (2018) report mean market impact of 8.90 bps for large-cap trades and 18.95 bps for small-cap trades (Table II, Panel A), supporting the upper end of the range in stressed scenarios. We implement these magnitudes as a local linear impact law, which is the object required by the theory: in the single-asset case, $\mathcal{I}$ is set so that trading 1% of ADV generates a one-day cost of $c_{1\%} \in \{8, 12, 20\}$ bps, supplemented by two book-end scenarios,

$c_{1\%} = 1$ bp (a very deep underlying ETF) and $c_{1\%} = 30$ bps (a mid-cap name), which appear in the vulnerability figures and in the liquidity discussion of Sections 1 and 7.3.1. The linear and square-root cost models in Kyle and Obizhaeva (2016) bracket the 12 bps baseline at exactly 1% of ADV, the size at which daily trades are capped in the profitability exercise of Internet Appendix B.6, so the calibration matches the level of execution costs where the strategies trade rather than extrapolating a small-trade slope. Below the anchor a concave law implies a larger price response per unit of flow than the linear law, so the linear generator, if anything, understates how easily small trades move the statistic the rule reads. The capacity grid also includes transient-propagator variants of $\mathcal{I}$ (exponentially decaying impact with 21- and 63-day half-lives, and permanent impact); these select the states used for the propagator robustness of the profitability exercise in Internet Appendix B.6. In the multi-asset case we use a common $c_{1\%}$ across constituents. Cross-impact is added as a separate sensitivity through the symmetric parameterization

$$\mathcal{I}_{ij} = \rho \sqrt{\mathcal{I}_{ii}\mathcal{I}_{jj}}\, \mathrm{corr}_{ij} \qquad (i \neq j),$$

with $\rho \in [0, 0.3]$ and baseline $\rho = 0.2$; the diagonal case $\rho = 0$ and the upper end $\rho = 0.3$ are reported as one-at-a-time deviations (cross-impact tightens capacity, so the diagonal case is the most permissive in the range). In computation we enforce that stressed correlation matrices are valid by projection to the nearest positive semidefinite correlation matrix. When adding cross-impact terms, we symmetrize the resulting impact matrix and apply a diagonal adjustment if needed to ensure it is positive semidefinite. These numerical safeguards address knife-edge cases and do not change the analytical objects used by the theorem.

B.5 Additional results for the volatility-control stress tests

We report the conservative admissible scale $W^{p01}_{\max}(\mathcal{Z})$ for every scenario and horizon, together with the vulnerability curves, the seed-robustness check, and the cap-regime incidence for the volatility-control stress tests.

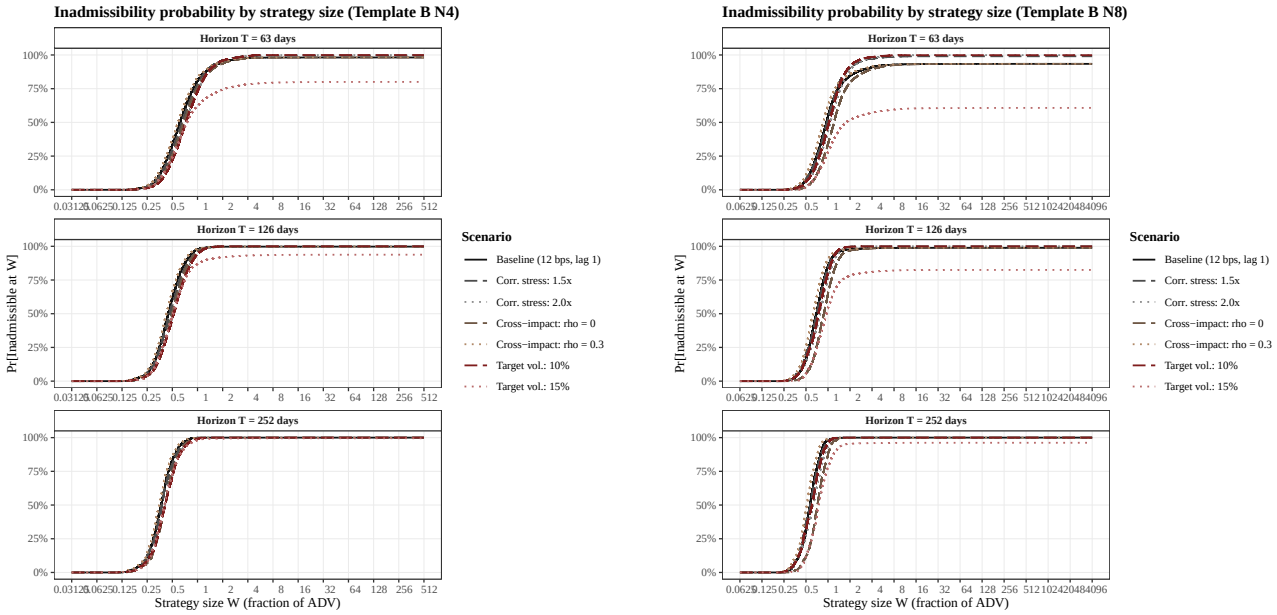
Scenario	$T = 63$	$T = 126$	$T = 252$
Panel A: Template A ($N = 1$)			
baseline	0.213	0.166	0.131
$\sigma^* = 10\%$	0.267	0.208	0.163
$\sigma^* = 15\%$	0.178	0.138	0.108
$c_{1\%} = 1$ bp (liquid ETF)	2.561	1.995	1.569
$c_{1\%} = 8$ bps	0.320	0.249	0.196
$c_{1\%} = 20$ bps	0.128	0.100	0.078
$c_{1\%} = 30$ bps (mid-cap)	0.085	0.066	0.052
iid const vol	0.262	0.204	0.166
GARCH persistent	0.139	0.100	0.084
lag $\ell = 0$	0.191	0.159	0.131
lag $\ell = 2$	0.189	0.141	0.103
window 20 only	0.139	0.122	0.088
Panel B: Template B, $N = 2$			
baseline	0.198	0.141	0.109
$\sigma^* = 10\%$	0.254	0.183	0.137
$\sigma^* = 15\%$	0.156	0.129	0.111
$c_{1\%} = 1$ bp (liquid ETF)	2.371	1.694	1.311
$c_{1\%} = 8$ bps	0.296	0.212	0.164
$c_{1\%} = 20$ bps	0.119	0.085	0.065
$c_{1\%} = 30$ bps (mid-cap)	0.079	0.056	0.044
iid const vol	0.231	0.173	0.160
GARCH persistent	0.130	0.106	0.085
lag $\ell = 0$	0.197	0.150	0.128
lag $\ell = 2$	0.178	0.130	0.107
window 20 only	0.103	0.093	0.080
cross-impact $\rho = 0$	0.204	0.145	0.113
cross-impact $\rho = 0.3$	0.194	0.139	0.108
corr stress $\kappa = 1.5$	0.193	0.152	0.123
corr stress $\kappa = 2.0$	0.205	0.154	0.125
Panel C: Template B, $N = 4$			
baseline	0.200	0.165	0.150
$\sigma^* = 10\%$	0.239	0.183	0.160
$\sigma^* = 15\%$	0.242	0.197	0.156
cross-impact $\rho = 0$	0.223	0.184	0.166
cross-impact $\rho = 0.3$	0.190	0.157	0.142
corr stress $\kappa = 1.5$	0.219	0.184	0.160
corr stress $\kappa = 2.0$	0.228	0.192	0.143
Panel D: Template B, $N = 8$			
baseline	0.349	0.313	0.278
$\sigma^* = 10\%$	0.351	0.294	0.264
$\sigma^* = 15\%$	0.466	0.416	0.382
cross-impact $\rho = 0$	0.438	0.388	0.342
cross-impact $\rho = 0.3$	0.319	0.286	0.251
corr stress $\kappa = 1.5$	0.349	0.314	0.288
corr stress $\kappa = 2.0$	0.353	0.286	0.264

Table 7 Conservative admissible scale $W_{\max}^{p01}(\mathcal{Z})$ by scenario and horizon.
Entries are $W_{\max}^{p01}(\mathcal{Z})$, the 1st percentile of $W_{\max}(z)$ across $\mathcal{Z}$, in ADV-normalized units ($|\mathcal{Z}| = 10{,}000$). Panel A is the single-asset Template A; Panels B–D are Template B with $N \in \{2, 4, 8\}$ hedge assets.

Template	T	$W^{p01}_{\max}(\mathcal{Z})$ seed 1	seed 2	seed 3	range	Sample minimum $W_{\max}(\mathcal{Z})$ seed 1	seed 2	seed 3	range
Template A ($N = 1$)	63	0.213	0.204	0.242	0.038	0.061	0.080	0.123	0.063
	126	0.166	0.157	0.200	0.043	0.057	0.075	0.117	0.059
	252	0.131	0.132	0.160	0.030	0.057	0.075	0.117	0.059
Template B ($N = 2$)	63	0.198	0.172	0.190	0.025	0.072	0.072	0.086	0.013
	126	0.141	0.124	0.149	0.025	0.072	0.072	0.082	0.010
	252	0.109	0.104	0.125	0.020	0.072	0.072	0.082	0.010
Template B ($N = 4$)	63	0.200	0.239	0.230	0.039	0.134	0.180	0.154	0.046
	126	0.165	0.217	0.198	0.052	0.129	0.166	0.154	0.037
	252	0.150	0.190	0.184	0.040	0.129	0.165	0.154	0.037
Template B ($N = 8$)	63	0.349	0.347	0.342	0.007	0.265	0.234	0.230	0.035
	126	0.313	0.304	0.288	0.025	0.262	0.234	0.227	0.035
	252	0.278	0.286	0.258	0.028	0.262	0.234	0.227	0.035

Table 8 Seed robustness of the first-percentile capacity and the sample minimum (baseline scenario). Both statistics recomputed under three independent simulation seeds, each at $|\mathcal{Z}| = 10{,}000$. The left block reports $W^{p01}_{\max}(\mathcal{Z})$, the right block the sample minimum $W_{\max}(\mathcal{Z})$, under seeds 1, 2, and 3, with the across-seed range in each block's final column. The seed-1 columns reproduce the values used in the main tables.

B.5.1 Vulnerability curves



(a) Template B, $N = 4$ **(b)** Template B, $N = 8$

Figure 4 Vulnerability curves for Template B.
Panels correspond to horizons $T \in \{63, 126, 252\}$; lines correspond to the scenarios implemented for each N.

B.5.2 Cap-regime incidence at the first responsive date

Table 9 summarizes cap-regime incidence at the first responsive date for Template B in the baseline scenario (which has $\sigma^* = 12.5\%$) and in the two target-volatility scenarios $\sigma^* \in \{10\%, 15\%\}$. For each (N, σ^*, T), the table reports the number of stress states with regime `cap` at the first responsive date, the share of the $|\mathcal{Z}| = 10{,}000$ stress set, and the conditional failure probability at $W_{\text{med}} = \text{median}_{z \in \mathcal{Z}} W_{\max}(z)$.

N	Scenario	T	$\#\{z : \text{cap}\}$	share of $\mathcal{Z}$	Pr[fail | cap] at W_{med}
2	baseline	63	16	0.16%	93.8%
2	baseline	126	16	0.16%	87.5%
2	baseline	252	16	0.16%	50.0%
2	$\sigma^* = 10\%$	63	0	0.00%	–
2	$\sigma^* = 10\%$	126	0	0.00%	–
2	$\sigma^* = 10\%$	252	0	0.00%	–
2	$\sigma^* = 15\%$	63	486	4.86%	84.2%
2	$\sigma^* = 15\%$	126	486	4.86%	65.0%
2	$\sigma^* = 15\%$	252	486	4.86%	60.3%
4	baseline	63	2647	26.47%	67.9%
4	baseline	126	2647	26.47%	61.2%
4	baseline	252	2647	26.47%	51.3%
4	$\sigma^* = 10\%$	63	234	2.34%	90.2%
4	$\sigma^* = 10\%$	126	234	2.34%	71.4%
4	$\sigma^* = 10\%$	252	234	2.34%	59.0%
4	$\sigma^* = 15\%$	63	6517	65.17%	52.1%
4	$\sigma^* = 15\%$	126	6517	65.17%	52.0%
4	$\sigma^* = 15\%$	252	6517	65.17%	50.4%
8	baseline	63	4504	45.04%	59.0%
8	baseline	126	4504	45.04%	54.7%
8	baseline	252	4504	45.04%	52.7%
8	$\sigma^* = 10\%$	63	620	6.20%	78.9%
8	$\sigma^* = 10\%$	126	620	6.20%	72.6%
8	$\sigma^* = 10\%$	252	620	6.20%	60.6%
8	$\sigma^* = 15\%$	63	8220	82.20%	49.0%
8	$\sigma^* = 15\%$	126	8220	82.20%	50.0%
8	$\sigma^* = 15\%$	252	8220	82.20%	49.9%

Table 9 Incidence of cap-regime at the first responsive date for Template B.

The "share of $\mathcal{Z}$" column reports $\#\{z : \text{cap}\}/|\mathcal{Z}|$ in percent. The last column, Pr[fail | cap], is a failure rate *conditional* on the cap regime; a dash marks cells with no cap-regime states, for which this conditional probability is undefined (it is not zero). Because the cap regime is classified at the first responsive date, which does not depend on the horizon, and the stress states are shared across horizons, the incidence is identical across T within each scenario.

B.6 Profitability of the optimal attack

Existence holds at full coverage for every calibration: across the four calibrations and the three scales $W \in \{1, 2, 5\}$, a strictly profitable round trip exists at 100% of the inadmissible stress states (the eigenvector-witness check of Internet Appendix A.5.1; $\lambda_{\min}^{\mathcal{R}_T} < 0$ at every one). The tables below

quantify the magnitude of the gains. For each calibration and sector scale W, we solve the single-asset optimal-attack dynamic program of Section 5 on a random sample of 600 inadmissible states drawn from $\{z : W_{\max}(z) < W\}$ in the corresponding capacity run, on each state's realized path, subject to a daily trade cap of 1% of ADV, terminal round trip $x_T = 0$, and horizon $T = 126$. Panel A of Table 10 reports return on *average* capital (annualized profit per unit of mean inventory over the round trip), the measure used in the main text, and Panel B return on *peak* capital (annualized profit per unit of maximum inventory warehoused). In both, "Inadm." is the number of inadmissible states at scale W (out of $|\mathcal{Z}| = 10{,}000$) and "Profitable" is the full-coverage share admitting a profitable round trip (100%); "Mean/Median/p90" are cross-state moments of the annualized return over the sampled attacks; the Sharpe ratio is annualized and treats each state as one independent T-day round trip; the final column is the corresponding inventory in days of ADV.

Panel A: return on average capital								
Scenario	W	Inadm.	Profitable	Mean	Median	p90	Sharpe	Avg inv.
Baseline (12 bps)	1	8,338	100%	12.0%	11.0%	20.8%	2.36	0.019
	2	9,760	100%	22.4%	21.6%	38.5%	2.17	0.021
	5	9,999	100%	72.4%	60.8%	123.8%	1.41	0.024
Prop. exp. $h = 21$d	1	9,596	100%	11.7%	8.7%	23.9%	1.74	0.020
	2	9,974	100%	22.2%	16.0%	50.6%	1.48	0.021
	5	10,000	100%	44.3%	37.4%	93.3%	1.84	0.021
Prop. exp. $h = 63$d	1	9,803	100%	11.8%	9.2%	24.3%	1.81	0.026
	2	10,000	100%	20.0%	15.7%	41.3%	1.72	0.023
	5	10,000	100%	48.9%	42.8%	99.3%	1.88	0.021
Prop. permanent	1	10,000	100%	8.3%	6.6%	17.9%	1.81	0.046
	2	10,000	100%	17.0%	12.9%	35.0%	1.61	0.040
	5	10,000	100%	46.6%	31.6%	105.0%	1.27	0.029
Panel B: return on peak capital								
Scenario	W	Inadm.	Profitable	Mean	Median	p90	Sharpe	Peak inv.
Baseline (12 bps)	1	8,338	100%	4.3%	4.1%	7.1%	2.80	0.049
	2	9,760	100%	8.2%	7.9%	13.7%	2.33	0.054
	5	9,999	100%	26.2%	22.9%	42.9%	1.47	0.060
Prop. exp. $h = 21$d	1	9,596	100%	3.8%	3.4%	7.1%	2.33	0.055
	2	9,974	100%	7.3%	5.7%	15.9%	1.68	0.053
	5	10,000	100%	18.3%	16.0%	36.7%	1.91	0.048
Prop. exp. $h = 63$d	1	9,803	100%	4.0%	3.5%	7.8%	2.01	0.068
	2	10,000	100%	7.3%	5.9%	14.4%	1.82	0.058
	5	10,000	100%	21.3%	18.8%	42.4%	1.98	0.047
Prop. permanent	1	10,000	100%	3.7%	3.2%	7.6%	1.97	0.099
	2	10,000	100%	7.4%	5.9%	15.2%	1.83	0.085
	5	10,000	100%	20.6%	15.5%	42.5%	1.40	0.063

Table 10 Profitability of the optimal attack, across impact calibrations and sector scales ($T = 126$).
Panel A reports return on *average* capital (annualized profit per unit of mean inventory over the round trip), the measure used in the main text; Panel B return on *peak* capital (per unit of the maximum inventory warehoused). Baseline is Template A with $c_{1\%} = 12$ bps and lag $\ell = 1$; the propagator rows replace purely temporary impact with exponentially decaying transient impact (half-lives of 21 and 63 trading days) and with permanent impact. Both panels use the same states and design; "Avg inv." and "Peak inv." are the corresponding mean inventories in days of ADV.

B.6.1 Incidence of the attack's excess cost

The excess cost on the mechanically rebalancing sector, ΔC^{MR}, splits via (25) into the transfer to the attacker and the change in aggregate execution cost. Table 11 reports this split, as shares of the expected excess cost, for the temporary-impact baseline – the case the Section 3.2 decomposition is derived for. Under the temporary-impact baseline the transfer share shrinks rapidly with sector size (from 60% at one day of ADV to 1% at five) and the aggregate-execution-cost share reaches 99%: a small sector is exploited mainly by trading against forced flow the rule would have generated anyway, so most of its excess cost is a pure transfer, while at scale the attacker amplifies that flow and the added execution cost dominates. One caveat: the decomposition is in expectation, and although the attacker profits at every inadmissible state, the attack *raises* the rebalancing sector's own execution cost ($\Delta C^{MR} > 0$) at only 41 to 84% of these states (last column of Table 11); at the rest the attacker harvests forced flow without raising the sector's cost, so the shares are informative in aggregate rather than state by state. Under the impact-propagator calibrations the picture differs: the attacker's offsetting trades against the predictable forced flow are partly liquidity-provision, which lowers the realized impact cost relative to the forced flow alone, so ΔC^{MR} is then predominantly a transfer (the aggregate-execution-cost component small, and at the shortest half-life marginally negative, i.e. the attack slightly lowers aggregate execution cost). The clean transfer-versus-execution-cost reading of Section 3.2 is specific to temporary impact, so we do not tabulate the split for the propagators.

Calibration	W	Transfer	Agg. exec. cost	Positive
Temporary (baseline)	1	60%	40%	41%
	2	13%	87%	65%
	5	1%	99%	84%

Table 11 Incidence of the attack-induced excess cost, temporary-impact baseline ($T = 126$).
At each scale W, the optimal attack of Section 5 is solved at the same random sample of 600 inadmissible states as in Table 10, and all sampled states enter the expectations. "Transfer" and "Agg. exec. cost" are the shares of the expected attack-induced excess cost ΔC^{MR} in (25) captured by the attacker and borne as a rise in aggregate execution cost, respectively (the latter a deadweight loss to the extent impact reflects real liquidity-provision costs); they sum to 100% and are ratios of expected components across stress states. "Positive" is the share of states at which the attack raises the sector's execution cost above its no-attack benchmark ($\Delta C^{MR} > 0$). We report the split only for the temporary-impact baseline, the case the Section 3.2 decomposition is derived for.

B.7 The footprint of volatility rebalancing

We test whether the rebalancing implied by a disclosed volatility-control rule predicts returns on its underlying funds. The rule is the rolling-window target-volatility rule of equations (82) and (84), with a 20-trading-day variance window ($m = 20$, $a_{\text{ann}} = 252$), target $\sigma^* = 5\%$, bounds $w_{\min} = 0$ and $w_{\max} = 1.5$, under the baseline update timing $M_{t+1} = g(\Gamma_t)$ (that is, $\ell = 0$ in equation (82)); its implied trade is the weight change $\Delta w_{t+1} = w_{t+1} - w_t$, fixed by volatility through day t (equation (85)) and executed the next day. The "lag" varied in Panel B is the horizon at which this predicted rebalance is matched to fund returns; the headline uses one trading day. The treated funds are the ETFs

that published volatility-control indices trade: the constituents named in the multi-asset designs of Appendix B.1, and the most liquid fund tracking each single-asset parent index; each is matched to control funds in the same asset class that no such index references. Returns are daily total returns from CRSP for 2007 to 2024, and the factors are the Fama-French five factors and the VIX. Each day we standardize Δw within fund and form a long-short portfolio, long the top third and short the bottom third of the cross-section by Δw.

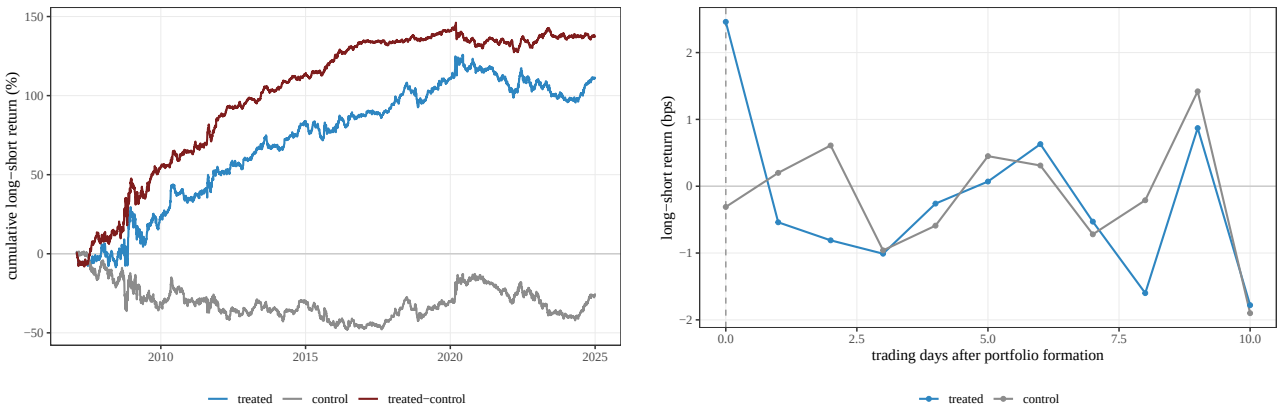
The long-short portfolio earns 6.3% a year on the treated funds and nothing on the controls (Panel A of Table 12); the treated-minus-control difference is 7.7% a year, and 7.4% after the Fama-French five factors and VIX innovations. It is 6.4% under a median split, 6.1% under quartiles, 4.7% within asset class, and 6.7% dropping the 2008, 2009, and 2020 crises, and is 6.7% and 8.0% in the more and less liquid funds, respectively. Figure 5b shows the effect as about two and a half basis points on the formation day that reverse within days, and Figure 5a shows the difference compounding from 2007 to 2024 while the control stays flat.

Varying one parameter at a time, the difference changes little across target volatilities from 5% to 15%, the range disclosed indices use, or across leverage caps from 1.0 to no cap (Panel B of Table 12 and Figure 6). It is present only with the one day lag. The same predicted rebalance matched to returns two, three, or five days later predicts nothing. We therefore exclude we are measuring some source of risk-premia, since a risk premium would be earned at any horizon, whereas the footprint of the rule's own trade is earned only as the trade is executed, the day after the rebalance is fixed; so the concentration at one day attributes the return to the mechanical rebalancing rather than to compensation for risk.

Panel A: returns and factor-adjusted alphas			
Portfolio	Mean	FF5 α	FF5 $+\Delta$VIX α
Cross-asset sort			
Treated (rule underlyings)	6.3 (2.1)	6.4 (2.1)	5.2 (1.7)
Control (matched non-rule)	-1.4 (-0.5)	-1.5 (-0.5)	-2.2 (-0.8)
Treated $-$ Control	7.7 (3.4)	7.8 (3.6)	7.4 (3.4)
Within asset class			
Treated $-$ Control	4.7 (2.0)	4.8 (2.0)	4.9 (2.0)
Panel B: sensitivity to the rule's parameters			
	Treated	Control	Treated $-$ Control
Target volatility σ^*			
5%	6.3	-1.4	7.7 (3.4)
7.5%	6.9	-1.5	8.4 (3.6)
10%	6.3	-1.0	7.2 (3.0)
12.5%	5.1	-1.1	6.2 (2.4)
15%	5.0	-1.7	6.5 (2.6)
Signal lag (trading days)			
1	6.3	-1.4	7.7 (3.4)
2	-1.6	1.0	-2.6 (-1.1)
3	-2.5	0.2	-2.7 (-1.2)
5	-0.8	-3.3	2.5 (1.2)
Leverage cap $w_{\max}$			
1.0	6.9	-1.5	8.4 (3.6)
1.25	6.9	-1.5	8.4 (3.6)
1.5	6.3	-1.4	7.7 (3.4)
2.0	6.7	-1.4	8.0 (3.5)
None	5.5	-2.7	8.1 (3.5)

Table 12 The footprint of volatility rebalancing: ETF returns and parameter sensitivity.

Daily long-short portfolios on the predicted rebalancing Δw of the 20-day target-volatility rule: long the top third and short the bottom third of the cross-section by Δw (signal standardized within fund). "Treated" are ETFs underlying disclosed volatility-control indices; "Control" are matched same-class funds that no index references. Panel A reports mean returns and factor-adjusted alphas ("FF5" adjusts for the Fama-French five factors; "FF5 $+\Delta$VIX" adds daily VIX innovations, reporting the intercept as annualized alpha), with a within-asset-class block that sorts above versus below the median within each class. Panel B reports the treated-minus-control long-short as one parameter of the rule is varied at a time around the headline calibration (target $\sigma^* = 5\%$, lag $\ell = 1$, cap $w_{\max} = 1.5$). Entries are mean returns in percent per year; Newey-West t-statistics (5 lags) in parentheses. The treated and control columns are each leg's mean over its own valid days, and treated-minus-control is the mean of the daily difference on their common days; the two coincide except at $\sigma^* = 15\%$, where the legs' valid days differ on nine of about 4,500. Entries are rounded to one decimal, so columns need not net exactly. Daily, 2007 to 2024.



(a) Cumulative long-short portfolios returns

(b) Impact and reversal in event time

Figure 5 The predicted-pressure long-short portfolios.

(a) Cumulative return to the daily long-short portfolios on the 20-day rule's predicted rebalancing Δw, for treated ETFs, matched controls, and their difference. (b) Raw long-short portfolios return by trading day relative to portfolio formation, for treated and control funds. Daily, 2007 to 2024.

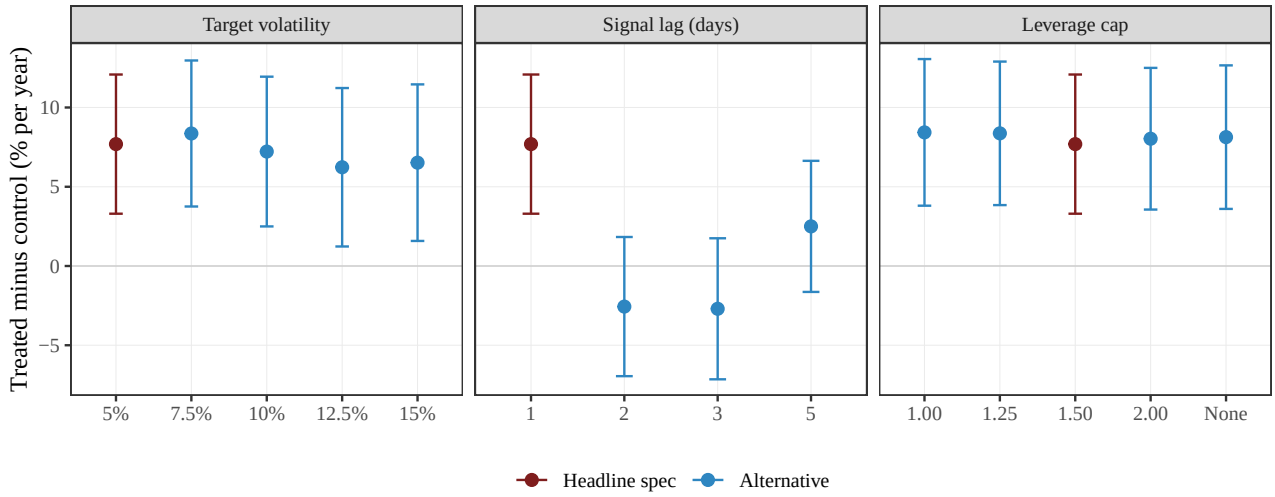


Figure 6 Sensitivity of the predicted-pressure long-short portfolios.

Treated-minus-control long-short portfolios, in percent per year, with 95% Newey-West confidence intervals, as the 20-day target-volatility rule varies one parameter at a time: the target volatility σ^*, the lag matching the predicted rebalance to returns, and the leverage cap $w_{\max}$. The headline calibration ($\sigma^* = 5\%$, lag 1, cap 1.5) is in dark red. Daily, 2007 to 2024.

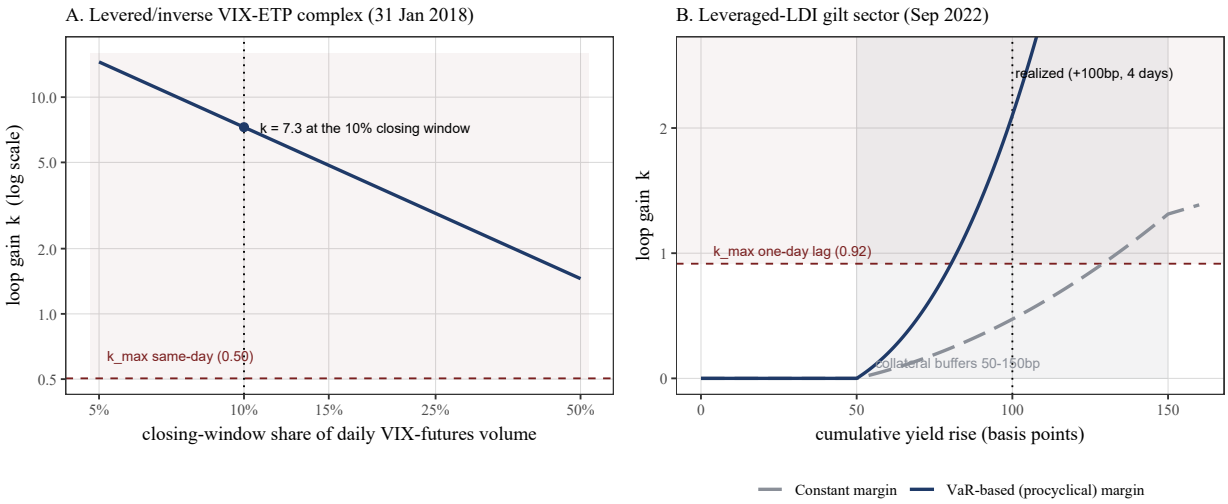


Figure 7 Ex-ante fragility of two episodes.

The figure plots the feedback gain k from pre-event public inputs against the one-month admissibility threshold (dashed), at each episode's timing: same-day $k_{\max} = 0.50$ for the market-on-close VIX rebalance (panel A) and one-day-lag $k_{\max} = 0.92$ for the next-day LDI margin (panel B). *Panel A*: the L&I VIX-ETP complex on 31 January 2018. *Panel B*: the leveraged-LDI gilt sector; k against the cumulative yield rise, with collateral buffers uniform on 50 to 150bp (shaded). The solid line includes the VaR-based margin ($m = \sigma/\sigma_0$); the long-dashed line holds margin per unit of exposure constant.

C Ex-ante fragility: two empirical examples

The capacity test needs only a disclosed rule and a market-impact calibration, so it can be run before any stress episode, from public information alone. We use that property to check the test against two sectors that later failed: the levered-and-inverse (L&I) VIX exchange-traded-product (ETP) complex of February 2018, and the leveraged liability-driven investment (LDI) gilt sector of September 2022, the two episodes from the introduction. For each, we form the screen from inputs public before the event and ask whether it would have flagged the sector in advance. At a single state the admissibility condition reduces to a scalar feedback gain, $k = \lambda G/\mathrm{ADV}$, where G is the sector's rebalancing gain (the forced flow generated per unit move in the price its rule reads), ADV is the volume that flow trades against, and λ is the impact per 1% of ADV (Appendix C.1). The gain k measures the strength of the feedback from prices to forced flow and back. At a one-month horizon the screen fails, and a profitable round trip exists, once k exceeds a threshold set by the rule's timing: $k_{\max} = 0.50$ when the requirement responds to the same day's price, as in a market-on-close rebalance, and 0.92 when it responds with a one-day lag, as in next-day-settled margin. The two episodes take these cases in turn: the VIX complex rebalances at the close (same-day), and the LDI sector meets margin the next day (one-day lag). The full model behind each screen is in Appendix C.1, which defines the gain G and, for the LDI sector, the breached share $F(\Delta y)$ and the margin multiplier $m(\Delta y)$ that appear below.

February 2018: the L&I volatility complex. These products deliver a constant daily multiple β of the return on VIX futures: inverse ($\beta = -1$) and levered-long ($\beta = +2$). Holding the multiple fixed

forces a daily rebalance. A product with assets A trades $\beta(\beta-1)A\,r$ in VIX futures after a futures return r (Cheng and Madhavan, 2009), so the flow is predictable and moves futures prices (Todorov, 2024; Brøgger, 2021).[10] Because $\beta(\beta-1)>0$ for both signs, inverse and levered-long products buy together when volatility rises, with aggregate gain $G=\sum_i \beta_i(\beta_i-1)A_i$, the counterpart of Bs in Section 4 (Appendix C.1). From CRSP and Compustat, $G\approx$ \$6.6bn at end-January 2018, against \$3.2bn of daily front-plus-second-month futures volume. The rebalance happens at the close, so its forced flow trades against only the volume in the closing window, not the full day's volume. We therefore compute k as a function of the closing-window share, the fraction of the day's volume that trades while the products rebalance: the smaller this share, the less liquidity absorbs the flow and the higher is k. In U.S. equities the closing auction accounts for about 7.5% of daily volume, and the last half hour including the auction for about 23.6% (Bogousslavsky and Muravyev, 2023). If 10% of the day's volume trades in the closing window, $k=7.3$; if instead half of it does, k falls to 1.5. Across the whole range, from a narrow window up to half the day, k stays above the same-day threshold (Figure 7, panel A). Half the day is more than twice the observed half-hour share, so the upper end of this range is far more conservative than the data require. The screen flags the volatility complex as vulnerable at prevailing conditions, before the event. On 5 February the front future rose 113%, the mandated \$7 to \$8bn close purchase met a market a fraction of that size, and the products lost most of their value overnight (Sushko and Turner, 2018; Augustin et al., 2021).

September 2022: leveraged LDI. Leveraged LDI funds hold long-duration gilts against a collateral buffer, under a convex value-at-risk margin, the model's price-based requirement $M_t=g(\Gamma_t)$. A fund sells nothing while the cumulative yield rise Δy stays inside its buffer, and delevers through variation margin once it is breached. The sector's forced selling then grows faster than the shock, for two reasons. First, funds hold different buffers, so as Δy grows a larger share have exhausted theirs and are selling; this breached share $F(\Delta y)$ rises from zero. Second, value-at-risk margin rises with volatility, so each breached fund must post more and sells more, scaling its flow by $m(\Delta y)=\sigma(\Delta y)/\sigma_0\geq 1$ (Brunnermeier and Pedersen, 2009; Adrian and Shin, 2010; Gârleanu and Pedersen, 2011). Both the number of sellers and each one's flow rise with the shock, so total selling accelerates. With $G\approx$ £250bn, constructed as LDI gilt sales per unit gilt-price move from Pinter et al. (2024) and detailed in Appendix C.1, £34bn of daily Long Gilt futures volume, and buffers centred on the roughly 100bp typical pre-crisis LDI buffer (Wilkins, 2024), a full breach is $W_0\approx 7$ days of ADV (Appendix C.1).

In the calm of 2021 the sector looks safe: $k\approx 0.2$, far below the bound. But k does not stay at 0.2; as Δy grows the accelerating selling drives it up, until it crosses the threshold past which the selling becomes self-reinforcing. Along the volatility path realized in 2022, that crossing comes at $+81$bp (Figure 7, panel B). Holding each fund's margin per unit of exposure fixed, so it sells only because its gilts have lost value, the ordinary force behind deleveraging, the crossing sits at $+130$bp, beyond the realized move in 2022. If we let value-at-risk margin instead rise with volatility, as in practice,

[10] The exploiting round trip harvests this flow by trading *against* it, the $u_t=-v_t/2$ motive of Section 5; Todorov (2024) finds that leverage rebalancing dominates the price effect and that trading against it is profitable. This differs from front-running the flow: Brøgger (2021) finds no excess profit from trading *ahead* of the predictable rebalancing, and that larger, more predictable flows draw in liquidity and carry smaller price impact.

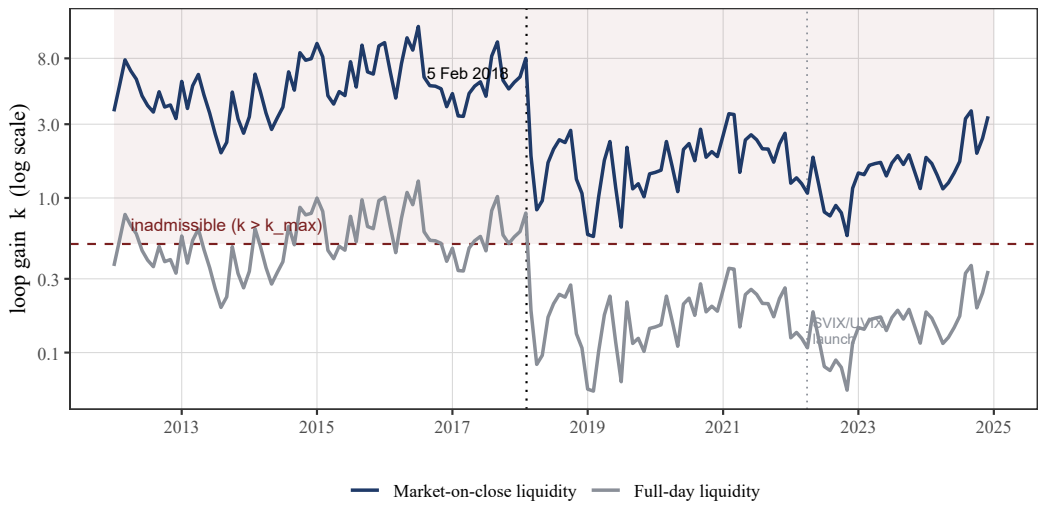


Figure 8 The volatility complex: 2012 to 2024.

The figure plots the monthly feedback gain k of the levered-and-inverse VIX-ETP complex from public data: $k = \lambda G(t)/\mathrm{ADV}(t)$, with $G = \sum_i \beta_i(\beta_i - 1)A_i$ from CRSP and Compustat, front-plus-second-month VIX-futures dollar volume for ADV, and λ the price impact. The dashed line is the same-day threshold $k_{\max} = 0.50$. The closing-window reading (blue) uses a tenth of the day's volume; the full-day reading (grey) uses the whole day. The dotted black line marks 5 February 2018, after which the complex collapses (XIV terminated, SVXY and UVXY leverages halved); the dotted grey line marks the relaunched SVIX and UVIX (from 2022).

the threshold is $+81$bp. The gap is the requirement responding to volatility itself. Loss-driven selling would have needed a larger move than came, so it is the volatility-scaled margin that carries the crossing inside the realized $+100$bp.

This calculation uses the crisis path, so it only confirms the fragility after the fact; the more important claim is that the risk was visible in advance. In Appendix C.1 we fit a fat-tailed model to gilt yields up to the eve of the episode and simulate paths forward.[11] The gain crosses the threshold with probability about 1% within a trading month and 7% within a quarter. Notice that the sector entered September at twice its calm-market volatility. In the actual event, thirty-year gilt yields rose over 100bp in four days and LDI funds sold $£25.3$bn of gilts in par terms over the following five weeks (Pinter et al., 2024), and the Bank of England intervened (Chen and Kemp, 2023).

Both examples use only pre-event public information to show sector vulnerability. The leverage gain of the volatility complex is on even in tranquil markets. In the time series, the gain stays above the threshold in every month of the sample, dropping sharply only when the complex collapses after the February 2018 event (Figure 8). The collateral gain of the LDI sector is instead masked by the buffers, so the sector clears a static reading and is flagged only when the volatility input is stressed. Notice that neither loop required a manipulator; both were triggered by market moves alone. However, Sections 5 and 6 show that the feedback need not wait for a shock: a strategic trader could have set it off deliberately.

[11] Specifically, the asymmetric GARCH(1,1) of Glosten et al. (1993) with skew-t innovations.

C.1 Feedback gains for the 2018 and 2022 episodes

The two episodes of this appendix are applications of the closed-form capacity bound of Appendix A.3. The test reads a scalar feedback gain

$$k = \lambda \frac{G}{\mathrm{ADV}}, \qquad \lambda = \frac{c_{1\%}}{100}, \tag{94}$$

where G is the sector's forced-flow gain (the forced flow induced per unit move in the price the rule reads), ADV is the volume that flow trades against, and λ is the impact per unit of ADV. The state is admissible when $k \le k_{\max}$, with $k_{\max}$ the finite-horizon threshold of Proposition 7 (0.916 at the one-day-lag $T = 21$ horizon, 0.504 same-day); the VIX market-on-close rebalance is same-day and the LDI margin is next-day-settled, so the two episodes use 0.504 and 0.916 respectively. Both thresholds are the single-channel value of the capacity bound (66), with $k = \lambda G/\mathrm{ADV}$ playing the role of the sector scale W and $k_{\max}$ evaluated numerically on the canonical rolling-window operator at horizon $T = 21$, rather than obtained in closed form. The two thresholds differ because of when the rule forces rebalancing. With next-day settlement the trade follows that price, so the feedback stays finite at any gain (Lemma 5). With a same-day rebalance the trade and the price are set together, and the closing price explodes at $k = 1$. Kyle and Obizhaeva (2016) make the cost of trading a fixed fraction of daily volume proportional to the asset's volatility. Writing the cost of trading 1% of ADV as $c_{1\%}$ basis points,

$$c_{1\%} = \kappa\, \sigma_{\mathrm{daily}}, \qquad \sigma_{\mathrm{daily}} = \frac{\sigma_{\mathrm{ann}}}{\sqrt{252}}, \tag{95}$$

with σ_{daily} in basis points and $\kappa = 0.095$ fixed so that an asset at $\sigma_{\mathrm{ann}} = 20\%$ carries $c_{1\%} = 12$ bps, the baseline of Section 7 and the impact calibration of Appendix B.4. Each instrument's own realized volatility then pins its impact: VIX front-month futures at 58% give $c_{1\%} = 35$ bps; the Long Gilt future gives 3 bps in the calm of 2021 (4.8%) and 13 bps at the crisis volatility (21%).

February 2018: the leverage-rebalancing gain. A daily-reset exchange-traded product with constant leverage β holds exposure βA against net assets A. After an underlying return r its exposure is $\beta A(1+r)$ and its net assets $A(1+\beta r)$, so restoring leverage β requires target exposure $\beta A(1+\beta r)$ and the closing rebalancing trade

$$\Delta = \beta A(1+\beta r) - \beta A(1+r) = \beta(\beta-1)\, A\, r \tag{96}$$

(Cheng and Madhavan, 2009). Since $\beta(\beta-1) > 0$ for both inverse ($\beta = -1 \Rightarrow 2$) and levered-long ($\beta = +2 \Rightarrow 2$) products, the whole complex trades in the direction of r, buying VIX futures when volatility rises. Aggregating gives the forced-flow gain

$$G = \sum_i \beta_i(\beta_i - 1)\, A_i, \tag{97}$$

with the VIX-futures level the price the rule reads and the leverage target the schedule, so G is the sensitivity of mandated flow to that price, the counterpart of $B\,s$. With $G = \$6.6$bn against $\$3.2$bn of front-plus-second-month futures volume on 31 January 2018, (94) gives $k = 0.73$ against the full

day and $k = 7.3$ against a closing window that clears a tenth of the day, both above the same-day threshold $k_{\max} = 0.504$.

September 2022: the convex collateral gain. A leveraged liability-driven fund holds long-duration gilts at leverage β against a collateral buffer. A cumulative yield rise Δy is a gilt loss; once a fund's buffer is breached it delevers through variation margin, shedding $(\beta - 1)e$ of exposure per unit gilt return, with e its equity. Two features make the aggregate response convex. First, buffers differ across funds. Wilkins (2024) reports that pre-crisis LDI buffers typically covered around 100bp, but does not report their dispersion; we therefore assume buffers uniform on $[50, 150]$ bp, centred on that figure. With this spread the breached share of the sector is

$$F(\Delta y) = \min\Big(\max\Big(\tfrac{\Delta y - 50}{100}, 0 \Big), 1 \Big), \tag{98}$$

rising from zero at 50 bp to a full breach at 150 bp. Second, the initial margin is set by value-at-risk, so it rises with measured volatility, multiplying the requirement by

$$m(\Delta y) = \frac{\sigma(\Delta y)}{\sigma_0} \geq 1 \tag{99}$$

(Brunnermeier and Pedersen, 2009; Adrian and Shin, 2010; Gârleanu and Pedersen, 2011), with σ_0 the calm-state volatility. This is the price-based requirement $M_{t+1} = g(\Gamma_t)$ with g a value-at-risk rule. The full-breach forced-flow gain of the leveraged-LDI gilt sector is $G_{\text{LDI}} \approx £250$bn. We construct it as forced sales per unit gilt-price move: Pinter et al. (2024) report £25.3bn of LDI gilt sales and fire-sale price discounts on the order of 10%, so $G_{\text{LDI}} \approx 25/0.10$. Their discount estimates run from 6.9% to 16.7%, which places G_{LDI} between £152bn and £368bn. This gives $W_0 = G_{\text{LDI}}/\text{ADV} \approx 7.4$ days against £34bn of daily Long Gilt futures volume, the 2021 average for the most active contract in our futures data, so the gain at a yield rise Δy is $W_0\, F(\Delta y)\, m(\Delta y)$ days of ADV. Because impact rises with volatility too ($\sigma(\Delta y) = \sigma_0\, m(\Delta y)$ in (95)), the feedback gain is doubly convex,

$$k(\Delta y) = \frac{c_{1\%}\big(\sigma_0\, m(\Delta y)\big)}{100}\, W_0\, F(\Delta y)\, m(\Delta y). \tag{100}$$

With m rising to the realized crisis volatility at +100 bp, $k(\Delta y)$ crosses $k_{\max} = 0.916$ at $\Delta y = +81$ bp under the procyclical margin, against +130 bp when margin per unit of exposure is held fixed; gilt yields rose +100 bp in four days.

An ex-ante probability. The +81 bp crossing evaluates $k(\Delta y)$ along the realized volatility path, so it is ex post: $m(\Delta y)$ ramps to the realized crisis level. For an ex-ante reading we fit a GJR-GARCH(1,1) (Glosten et al., 1993) with skew-t innovations to daily changes in the Bank of England ten-year nominal gilt yield through 22 September 2022, simulate yield paths forward from that day's fitted conditional volatility, and evaluate the feedback gain along each path with m and the impact set by the path's own simulated volatility rather than the realized crisis level. The bound is breached with probability about 1% within a trading month and 7% within a quarter (roughly 0.3 to 2% and 3 to 11% across GARCH specifications, widening with the innovation tail). It is non-negligible because the

sector entered September with volatility already about twice its 2021 level, so a first crossing needs only a $+100$ bp move; and it is a lower bound, the simulation being exogenous and omitting the price feedback from the funds' own forced sales.